

SEVEN OPEN UNIT EQUILATERAL TRIANGLES DO NOT COVER A REGULAR UNIT HEXAGON

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ABSTRACT. We prove that the regular hexagon of side length one cannot be covered by seven open equilateral triangles of side length one. The center and six vertices canonically label the seven covering triangles, and an exhaustive classification routes every hypothetical cover to a trace-length, area-loss, or finite equilateral-enclosure contradiction. The body gives the exact geometric and set-theoretic definitions and the compact interfaces used by these three mechanisms. Appendices A–F solve the associated optimization and feasibility problems. The zero-gap nine-point terminal culminates in an exact polynomial positivity certificate over $\mathbb{Q}(\sqrt{3})$; numerical scans are not proof dependencies.

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1. INTRODUCTION AND PROOF OUTLINE

1.1. The theorem and the canonical labeling. Set $O := 0 \in \mathbb{R}^2$. With indices in $\mathbb{Z}/6\mathbb{Z}$, choose $V_0 \in \mathbb{R}^2$, let $\text{Rot}_{O,\theta}$ denote ordinary rotation through θ about O , and define

$$\|V_0 - O\| = 1, \quad V_i := \text{Rot}_{O, i\pi/3}(V_0).$$

Put

$$H := \text{conv}\{V_0, \dots, V_5\}, \quad e_{i,i+1} := [V_i, V_{i+1}], \quad r_i := [O, V_i],$$

$$M_i := \frac{O + V_i}{2},$$

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and

$$S := \partial H \cup \bigcup_{i=0}^5 r_i.$$

Then

$$\mathcal{H}^1(\partial H) = 6, \quad \mathcal{H}^1\left(\bigcup_{i=0}^5 r_i\right) = 6, \quad \mathcal{H}^1(S) = 12.$$

Theorem 1.1 (Main theorem). *The hexagon H cannot be covered by seven open equilateral triangles of side length one.*

Corollary 1.2 (Soifer's Hexagon Problem). *For every $L > 1$, the regular hexagon $H_L := O + L(H - O)$ cannot be covered by seven closed unit equilateral triangles.*

The seven points $O, V_0, \dots, V_5$ are pairwise at distance at least one, whereas any two points in an open unit equilateral triangle are at distance strictly less than one. Hence the seven covering triangles admit a unique labeling

$$U_C, U_0, \dots, U_5$$

such that

$$O \in U_C, \quad V_i \in U_i.$$

We call U_C the *C triangle* and U_i the *V triangle* at V_i . Set

$$T_C = \overline{U_C}, \quad T_i = \overline{U_i}.$$

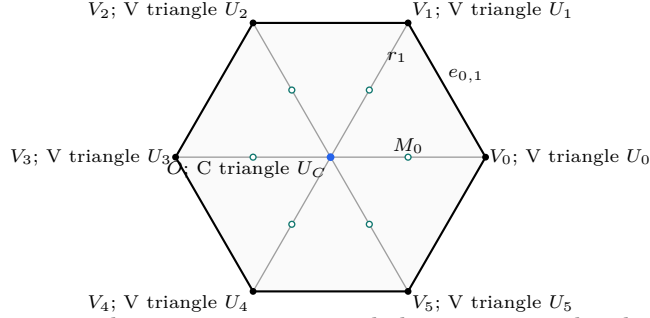


FIGURE 1. The center, vertices, radial segments, and radial midpoints of H .

1.2. Finite classifications. Define the closed C-triangle edge count by

$$N_E(T_C) := \left| \left\{ i \in \mathbb{Z}/6\mathbb{Z} : \mathcal{H}^1(T_C \cap e_{i,i+1}) > 0 \right\} \right|.$$

The closed classifications are

$$T_C \text{ has class CE0} \iff N_E(T_C) = 0,$$

$$T_C \text{ has class CE1} \iff N_E(T_C) = 1,$$

$$T_C \text{ has class CE2} \iff N_E(T_C) = 2.$$

Proposition 2.3 proves that these are the only possibilities.

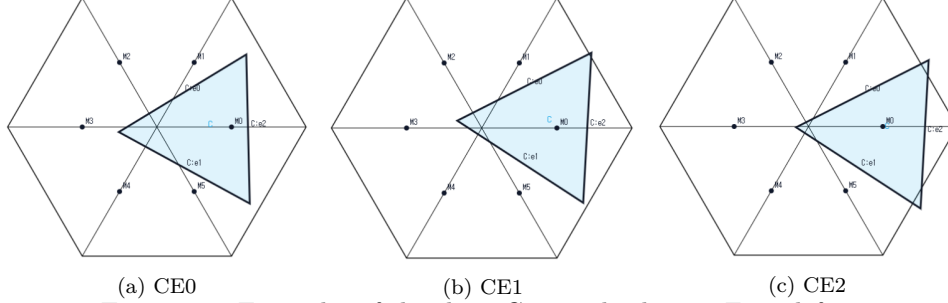


FIGURE 2. Examples of the three C triangle classes. From left to right, the C triangle has positive-length trace on zero, one, and two edges of ∂H ; point contacts do not count. The screenshots are illustrative and not to scale; no proof depends on visual inspection.

For the V triangle T_i , put

$$o(T_i) := |\text{vert}(T_i) \setminus H|, \quad n(T_i) := |\{j \in \{i-1, i+1\} : \mathcal{H}^1(T_i \cap r_j) > 0\}|,$$

with indices modulo 6. The raw pair $(3, 0)$ says that all three vertices of T_i lie outside H ; it does not say that T_i is disjoint from H , since $V_i \in \text{int}(T_i) \cap H$. Proposition 2.5 replaces every such triangle by a same-orientation translate with exactly the same trace in H , still containing V_i in its interior, and with at most two vertices outside H . We make this exact-trace normalization before defining any reach. The normalized exhaustive types used below are

$(o(T_i), n(T_i))$	type
$(1, 0), (2, 0)$	Vd0
$(1, 1)$	Vd1
$(1, 2)$	Vd2
$(2, 1)$	T3-like.

Define the edge and radial parametrizations

$$X_i(s) := V_i + s(V_{i+1} - V_i), \quad R_i(s) := V_i + s(O - V_i) \quad (0 \leq s \leq 1).$$

The actual maximal reaches are defined on the closed V triangle T_i by

$$\begin{aligned} A_i &:= \max\{s \in [0, 1] : V_i + s(V_{i-1} - V_i) \in T_i\}, \\ B_i &:= \max\{s \in [0, 1] : X_i(s) \in T_i\}, \\ C_i &:= \max\{s \in [0, 1] : R_i(s) \in T_i\}. \end{aligned}$$

Equivalently, in terms of the original open V triangle,

$$\begin{aligned} A_i &= \sup\{s \in [0, 1] : V_i + s(V_{i-1} - V_i) \in U_i\}, \\ B_i &= \sup\{s \in [0, 1] : X_i(s) \in U_i\}, \\ C_i &= \sup\{s \in [0, 1] : R_i(s) \in U_i\}. \end{aligned}$$

The V triangle T_i is *supercritical* when $A_i + B_i > 1$, and

$$N_+ := |\{i \in \mathbb{Z}/6\mathbb{Z} : A_i + B_i > 1\}|.$$

Lowercase (a_i, b_i, c_i) always denotes selected lower bounds satisfying

$$0 \leq a_i \leq A_i, \quad 0 \leq b_i \leq B_i, \quad 0 \leq c_i \leq C_i;$$

these quantities are never the maxima defining N_+ .

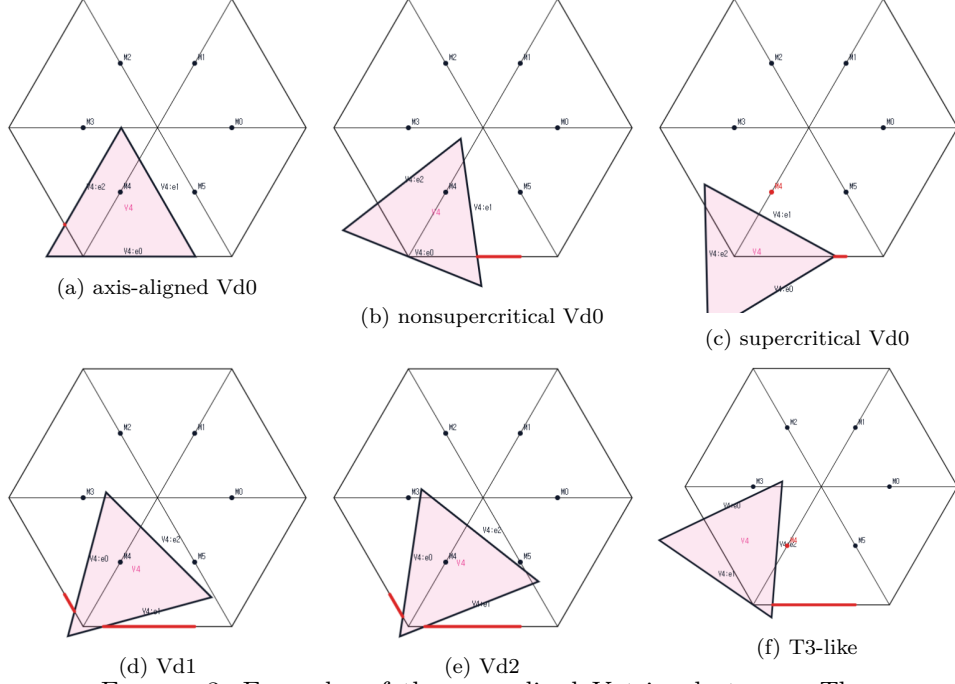


FIGURE 3. Examples of the normalized V triangle types. The top row shows an axis-aligned Vd0, a nonsupercritical Vd0 with $A_i + B_i \leq 1$, and a supercritical Vd0 with $A_i + B_i > 1$. The bottom row shows Vd1, Vd2, and T3-like. The screenshots are illustrative and not to scale; no proof depends on visual inspection.

The two incident open traces on $e_{i,i+1}$ are

$$U_i \cap e_{i,i+1} = X_i([0, B_i]), \quad U_{i+1} \cap e_{i,i+1} = X_i((1 - A_{i+1}, 1]).$$

Define the boundary-gap set and the number of gap edges by

$$G_i := e_{i,i+1} \setminus ((U_i \cap e_{i,i+1}) \cup (U_{i+1} \cap e_{i,i+1})), \quad N_{\text{gap}} := |\{i \in \mathbb{Z}/6\mathbb{Z} : G_i \neq \emptyset\}|.$$

Thus

$$G_i = \begin{cases} X_i([B_i, 1 - A_{i+1}]), & B_i + A_{i+1} \leq 1, \\ \emptyset, & B_i + A_{i+1} > 1. \end{cases}$$

When equality holds in the first branch, G_i is a singleton and remains uncovered by the two incident open V triangles.

For the case tables, put

$$d := |\{i \in \mathbb{Z}/6\mathbb{Z} : (o(T_i), n(T_i)) \in \{(1, 1), (1, 2)\}\}|, \\ t := |\{i \in \mathbb{Z}/6\mathbb{Z} : (o(T_i), n(T_i)) = (2, 1)\}|,$$

and let $N_{\text{sp}} := d + t$. The common trace calculation proves that the number of V triangles that are either supercritical or meet an adjacent radial segment in positive length is

$$N_+ + N_{\text{sp}}.$$

In particular, every CE1/CE2 configuration with $N_+ + N_{\text{sp}} \geq 3$ is impossible by a trace-length bound on S .

1.3. Exhaustive routing. Lemma 2.9 proves the fundamental equivalence

$$N_{\text{gap}} = 0 \iff U_0, \dots, U_5 \text{ cover } \partial H.$$

Thus Table 1 first dispatches every zero-gap case without reference to the center class. The nonzero-gap part is read from top to bottom; the first applicable row is the unique route for a hypothetical cover. The full classification proof is Proposition 2.15.

N_{gap}	N_+	d	t	$N_E(T_C)$	strategy
0	0	$d + t \leq 6$		$N_E(T_C) \in \{0, 1, 2\}$	1
0	1	$d \geq 1$	$t \geq 0$	$N_E(T_C) \in \{0, 1, 2\}$	1
0	1	$d = 0$	$t \geq 1$	$N_E(T_C) \in \{0, 1, 2\}$	2
0	1	$d = 0$	$t = 0$	$N_E(T_C) \in \{0, 1, 2\}$	3
0	≥ 2	$d + t \leq 6$		$N_E(T_C) \in \{0, 1, 2\}$	2
≥ 1	≥ 2	$d + t \leq 6$		$N_E(T_C) \in \{1, 2\}$	1
≥ 1	0	$d + t \geq 3$		$N_E(T_C) \in \{1, 2\}$	1
≥ 1	1	$d + t \geq 2$		$N_E(T_C) \in \{1, 2\}$	1
≥ 1	0	$d = 0$	$t = 0$	$N_E(T_C) \in \{1, 2\}$	3
≥ 1	0	$d \geq 1, d + t \leq 2$		$N_E(T_C) \in \{1, 2\}$	1
≥ 1	0	$d = 0$	$t \in \{1, 2\}$	$N_E(T_C) \in \{1, 2\}$	3
≥ 1	1	$d = 0$	$t = 0$	$N_E(T_C) \in \{1, 2\}$	3
≥ 1	1	$d = 0$	$t = 1$	$N_E(T_C) \in \{1, 2\}$	3
≥ 1	1	$d = 1$	$t = 0$	$N_E(T_C) = 1$	1
≥ 1	1	$d = 1$	$t = 0$	$N_E(T_C) = 2$	1+3

TABLE 1. The exhaustive routing. Strategy 1 is a trace-length argument, Strategy 2 sums area loss, and Strategy 3 forces a finite residual set into the C triangle and proves an equilateral-enclosure obstruction. Nonzero gap count forces center class CE1 or CE2. In that part the high-count rows are applied first; the final two rows separate the CE1 length route from the detailed CE2 finite-enclosure placement assembly.

1.4. Organization of the proof. Sections 2–5 give the exact definitions and compact interfaces for the common reductions, trace-length, area-loss, and finite-enclosure mechanisms. Appendices A–F solve, respectively, the structural, shared-local, and signed-center optimizations; the trace-length optimization; the area-loss optimization; the nonzero-gap finite-enclosure optimization; the zero-gap nine-point optimization; and the exact polynomial positivity certificate. Section 6 applies the zero-gap matrix before the nonzero-gap rows of Table 1 and closes the proof.

2. COMMON DEFINITIONS AND STRUCTURAL REDUCTIONS

Throughout this section indices lie in $\mathbb{Z}/6\mathbb{Z}$, and we use the open triangles $U_C, U_0, \dots, U_5$, their closures $T_C, T_0, \dots, T_5$, and the maps X_i, R_i fixed in section 1. In particular, all classifications and maximal reaches belong to the closed triangles, whereas boundary coverage and endpoint strictness belong to the original open triangles.

2.1. Open covers and the seven distinguished roles.

Proposition 2.1 (Open, closed, and scaled formulations). *The following assertions are equivalent.*

- (1) H is covered by seven open equilateral triangles of side length 1.
- (2) For some $\varepsilon > 0$, H is covered by seven closed equilateral triangles of side length $1 - \varepsilon$.
- (3) For some $L > 1$, $H_L := O + L(H - O)$ is covered by seven closed equilateral triangles of side length 1.

Proof. The uniform inward margin needed for (1) $\Rightarrow$ (2) is supplied by theorem A.1. Enlarging each closed triangle about its incenter proves (2) $\Rightarrow$ (1), and dilation by $(1 - \varepsilon)^{-1}$ identifies (2) and (3). $\square$

Lemma 2.2 (Distinct center and vertex roles). *The covering triangles have a unique labeling for which*

$$O \in U_C, \quad V_i \in U_i \quad (0 \leq i \leq 5),$$

and these seven triangles are distinct. Consequently $O \in \text{int}(T_C)$ and $V_i \in \text{int}(T_i)$.

Proof. The seven displayed points are pairwise at distance at least one, while two points in an open unit equilateral triangle are at distance strictly less than its diameter one. Thus each covering triangle contains at most one of the points, and the asserted labeling is forced. $\square$

2.2. Center and vertex classifications. Recall the exact closed C-triangle count

$$N_E(T_C) = |\{i \in \mathbb{Z}/6\mathbb{Z} : \mathcal{H}^1(T_C \cap e_{i,i+1}) > 0\}|,$$

$$T_C \text{ is CE}k \iff N_E(T_C) = k \quad (k \in \{0, 1, 2\}).$$

A point contact has zero trace length and therefore does not create a second positive trace.

Proposition 2.3 (Exhaustive center classes). *Every C triangle has exactly one of the classes CE0, CE1, CE2; equivalently, $N_E(T_C) \leq 2$. If two edge traces are positive, their edges are adjacent.*

Proof. The two nonadjacent-edge distance calculations in theorem A.2 show that a unit-diameter triangle cannot have positive traces on three hexagon edges. $\square$

For a closed V triangle put

$$o(T_i) := |\text{vert}(T_i) \setminus H|, \quad n(T_i) := |\{j \in \{i-1, i+1\} : \mathcal{H}^1(T_i \cap r_j) > 0\}|.$$

Lemma 2.4 (Local wedge incidence). *Every original V triangle satisfies $1 \leq o(T_i) \leq 3$. If it has positive-length intersection with one of r_{i-1}, r_{i+1} , at least one of its vertices lies in H ; if it has positive-length intersection with both, at least two of its vertices lie in H .*

Proof. This is the wedge-feasibility calculation in theorem A.2. $\square$

Proposition 2.5 (Exact-trace normalization of raw (3,0) roles). *Let U be an original open V_i -triangle, put $T = \overline{U}$, and suppose $(o(T), n(T)) = (3, 0)$. There is a translate $\widehat{U}$ of U , with the same orientation and side length and with closure $\widehat{T}$, such that*

$$V_i \in \widehat{U}, \quad (o(\widehat{T}), n(\widehat{T})) = (2, 0),$$

and

$$T \cap H = \widehat{T} \cap H, \quad U \cap H = \widehat{U} \cap H.$$

Thus all three actual reaches and the inequality $A_i + B_i > 1$ are unchanged. The normalization may be performed simultaneously for every raw $(3, 0)$ role.

Proof. The same-orientation translation and its exact open- and closed-trace verification are supplied by theorem A.3. Rolewise equality on H preserves the open cover, so the translations may be made simultaneously. $\square$

After this normalization, the exact type dictionary is

$(o(T_i), n(T_i))$	type
$(1, 0), (2, 0)$	Vd0
$(1, 1)$	Vd1
$(1, 2)$	Vd2
$(2, 1)$	T3-like.

Proposition 2.6 (Exhaustive normalized vertex classes). *Every normalized V triangle has exactly one of the four types Vd0, Vd1, Vd2, T3-like.*

Proof. We have $n(T_i) \in \{0, 1, 2\}$. By theorem 2.4, the possible raw pairs are

$$(1, 0), (2, 0), (3, 0), (1, 1), (2, 1), (1, 2).$$

The normalization removes exactly $(3, 0)$ and preserves every other pair. $\square$

Proposition 2.7 (T3-like closed-trace majorant). *Let T be the closure of an original T3-like V_i -triangle. There is a translate $\widehat{T}$ of T , with the same orientation and side length, such that*

$$V_i \in \text{relint}(a \text{ side of } \widehat{T}), \quad T \cap H \subsetneq \widehat{T} \cap H,$$

and $(o(\widehat{T}), n(\widehat{T})) = (2, 1)$. Every old open interior point in $H \setminus \{V_i\}$ remains interior to $\widehat{T}$. Since V_i becomes a boundary point, $\widehat{T}$ is only a closed-trace majorant; it is not a replacement open triangle.

Proof. The exhaustive Type-I/Type-II calculation and the strict-interior check are given in theorem A.3. $\square$

2.3. Actual reaches, open traces, and boundary gaps. The maximal reaches are the closed-triangle maxima

$$A_i = \max\{s \in [0, 1] : X_{i-1}(1-s) \in T_i\},$$

$$B_i = \max\{s \in [0, 1] : X_i(s) \in T_i\},$$

$$C_i = \max\{s \in [0, 1] : R_i(s) \in T_i\},$$

and are equally the corresponding suprema with T_i replaced by U_i . Whenever selected demands occur they satisfy, explicitly,

$$0 \leq a_i \leq A_i, \quad 0 \leq b_i \leq B_i, \quad 0 \leq c_i \leq C_i.$$

In particular,

$$N_+ = |\{i \in \mathbb{Z}/6\mathbb{Z} : A_i + B_i > 1\}|$$

is defined only from actual maximal reaches.

Lemma 2.8 (Initial reach segments). *For every normalized V triangle,*

	T_i	U_i
$e_{i-1,i}$	$X_{i-1}([1 - A_i, 1])$	$X_{i-1}((1 - A_i, 1])$
$e_{i,i+1}$	$X_i([0, B_i])$	$X_i([0, B_i))$
r_i	$R_i([0, C_i])$	$R_i([0, C_i))$

Here each table entry denotes the intersection of the triangle heading its column with the segment heading its row.

Proof. The convex initial-segment argument, including the open outer endpoint, is part of theorem A.3. $\square$

The exact gap set is

$$G_i := e_{i,i+1} \setminus ((U_i \cap e_{i,i+1}) \cup (U_{i+1} \cap e_{i,i+1})), \quad N_{\text{gap}} := |\{i \in \mathbb{Z}/6\mathbb{Z} : G_i \neq \emptyset\}|.$$

Lemma 2.9 (Exhaustiveness of the gap split). *For every edge,*

$$G_i = \begin{cases} X_i([B_i, 1 - A_{i+1}]), & B_i + A_{i+1} \leq 1, \\ \emptyset, & B_i + A_{i+1} > 1. \end{cases}$$

Equality in the first branch is a singleton gap. Every nonempty G_i lies in a positive-length trace of U_C . Hence CE0, CE1, CE2 permit, respectively, at most 0, 1, 2 gap edges, and under a hypothetical cover

$$N_{\text{gap}} = 0 \iff \partial H \subseteq \bigcup_{i=0}^5 U_i.$$

Proof. The exact incident trace identities and exclusion of nonincident V triangles are proved in theorem A.3. Thus the displayed set difference is precisely the asserted closed interval, including its singleton endpoint case. A gap point must then lie in the open C triangle; openness supplies a positive relative interval on that edge. Distinct gap edges are therefore counted by $N_E(T_C)$, and the conclusions follow from theorem 2.3. $\square$

Lemma 2.10 (T3-like roles are nonsupercritical). *Every original T3-like V triangle satisfies*

$$(1) \quad A_i + B_i \leq 1.$$

In particular, it is not counted by N_+ .

Proof. This is the exact Type-II reach consequence of theorem A.3. $\square$

2.4. Strict handoffs and midpoint forcing.

Proposition 2.11 (Strict boundary handoffs). *Assume $\partial H \subseteq \bigcup_{i=0}^5 U_i$. There are simultaneous choices*

$$\xi_i \in (1 - A_{i+1}, B_i), \quad a_i := 1 - \xi_{i-1} < A_i, \quad b_i := \xi_i < B_i,$$

for which $0 < a_i$, $0 < b_i$, and

$$X_{i-1}(1 - a_i), X_i(b_i) \in U_i, \quad a_i^2 + a_i b_i + b_i^2 < 1.$$

Moreover, if $N_+ = 1$, every such choice satisfies

$$|\{i : a_i + b_i > 1\}| = 1$$

with the same unique index; if $N_+ \geq 2$, some simultaneous choice satisfies

$$|\{i : a_i + b_i > 1\}| \geq 2.$$

Proof. The interval feasibility, cyclic selector, and preservation assertions are proved in theorem A.8. Thus every selected handoff is strict. $\square$

Lemma 2.12 (Self-midpoint obstruction). *For $0 \leq a, b \leq 1$, if a closed unit equilateral triangle contains*

$$\{V_i, X_{i-1}(1-a), X_i(b), M_i\},$$

then $a + b \leq 1$. Consequently,

$$M_i \in T_i \implies A_i + B_i \leq 1.$$

If the four displayed points lie in an open unit equilateral triangle, the conclusion is strict.

Proof. The four finite calipers and the endpoint-strictness argument are supplied by theorem A.12. $\square$

Proposition 2.13 (CE1/CE2 exactly-one-midpoint property). *If $O \in \text{int}(T_C)$ and $N_E(T_C) \in \{1, 2\}$, then*

$$|T_C \cap \{M_0, \dots, M_5\}| = 1.$$

Proof. After normalizing a positive trace by a hexagon symmetry, the signed-center optimization in theorem A.4 gives the midpoint set $\{M_0\}$. Undoing the symmetry proves the assertion. $\square$

Remark 2.14. The interior hypothesis is essential: $\text{conv}\{O, V_0, V_1\}$ has a boundary edge overlap and contains both M_0 and M_1 .

Proposition 2.15 (Exhaustive structural reduction). *Every hypothetical cover by seven open unit equilateral triangles admits the distinct center and V triangles used in table 1, and its center class, normalized V classes, actual N_+ , and exact gap sets determine exactly one row of that table.*

Proof. The role labeling is theorem 2.2; the two classifications are theorems 2.3 and 2.6. The actual reaches determine N_+ and the open traces determine N_{gap} .

If $N_{\text{gap}} = 0$, theorem 2.9 gives the five zero-gap alternatives in table 1, split by N_+ and then by (d, t) . If $N_{\text{gap}} \geq 1$, the same lemma forces $N_E(T_C) \in \{1, 2\}$, and theorem 2.13 gives the unique radial midpoint. When $N_+ \geq 2$, theorem 3.3 gives $N_{\text{sp}} \geq 1$, so theorem 3.4 applies. For $N_+ \in \{0, 1\}$ the same theorem removes $N_+ + d + t \geq 3$. The remaining formula-only alternatives are exactly the nonzero-gap rows of table 1; they are disjoint and exhaustive. $\square$

3. TRACE-LENGTH MECHANISM

This section gives the exact trace definitions and the compact bounds used by the routing argument. The coordinate optimizations supplying those bounds are solved in Appendix B.

Definition 3.1 (Trace length). For a closed triangle T and a rectifiable set E , define

$$L_E(T) := \mathcal{H}^1(T \cap E).$$

If the seven open triangles cover E , their closures satisfy

$$(2) \quad \mathcal{H}^1(E) \leq L_E(T_C) + \sum_{i=0}^5 L_E(T_i).$$

The two targets are the perimeter ∂H and the full skeleton S . Their lengths are 6 and 12, respectively.



∂H : length 6



S : length 12

the full-skeleton estimate is used only under its stated branch hypotheses

FIGURE 4. The perimeter and the full skeleton. The skeleton is used only under the hypotheses of the corresponding trace bound.

Theorem 3.2 (Boundary trace table). *For the closures of the original open triangles, the following bounds hold:*

(3)	triangle	hypothesis	$L_{\partial H}$
	T_C	$N_E(T_C) = 0$	0
	T_C	$N_E(T_C) = 1$	$\leq \frac{\sqrt{3}}{2} - \frac{3}{4}$
	T_C	$N_E(T_C) = 2$	$< \frac{1}{2}$
	T_i	$(o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\}, A_i + B_i \leq 1$	$\leq \frac{1}{2}$
	T_i	$(o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\}, A_i + B_i > 1$	$\leq \frac{2}{\sqrt{3}}$
	T_i	$(o(T_i), n(T_i)) \in \{(1, 1), (1, 2)\}$	$< \frac{1}{2}$
	T_i	$(o(T_i), n(T_i)) = (2, 1)$	< 1 .

Proof. This is the exact output of the boundary-trace optimization in theorem B.7. $\square$

Lemma 3.3 (A positive-support adjacent exceptional triangle is forced). *Suppose $N_E(T_C) \in \{1, 2\}$ and, after symmetry,*

$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\}.$$

If $N_+ \geq 2$, then there are a supercritical index $s \in \mathbb{Z}/6\mathbb{Z} \setminus \{0\}$ and $j \in \{s-1, s+1\}$ such that

$$M_s \in U_j, \quad \mathcal{H}^1(T_j \cap r_s) > 0.$$

Proof. Choose a supercritical $s \neq 0$. By Lemma 2.12, $M_s \notin T_s$, and the displayed midpoint hypothesis gives $M_s \notin T_C$. Hence the original open cover puts M_s in some U_j . The diameter-one restriction gives $j \in \{s-1, s+1\}$, and openness gives $\mathcal{H}^1(T_j \cap r_s) > 0$. $\square$

Theorem 3.4 (Direct CE1/CE2 skeleton count). *If*

$$N_E(T_C) \in \{1, 2\}, \quad T_C \cap \{M_0, \dots, M_5\} = \{M_0\}, \quad N_+ + N_{\text{sp}} \geq 3,$$

then the seven triangles do not cover S .

Proof. The four optimized skeleton caps are theorem B.1, theorem B.4, theorem B.2, and theorem B.3. The classes counted by N_+ and N_{sp} are disjoint. Therefore

$$L_S(T_C) + \sum_{i=0}^5 L_S(T_i) < \frac{3}{2} + \frac{3}{2}(N_+ + N_{\text{sp}}) + 2(6 - N_+ - N_{\text{sp}}) \leq 12 = \mathcal{H}^1(S),$$

contradicting (2). $\square$

Proposition 3.5 (Vd2 adjacent-midpoint branch). *Assume*

$$\begin{aligned} N_E(T_C) &= 2, & N_+ &= 1, & \{j : (o(T_j), n(T_j)) \in \{(1, 1), (1, 2)\}\} &= \{i\}, \\ (o(T_i), n(T_i)) &= (1, 2), & M_{i-1} &\in T_i \vee M_{i+1} \in T_i, \end{aligned}$$

and

$$\forall j \neq i \quad (o(T_j), n(T_j)) \in \{(1, 0), (2, 0)\}.$$

Then the boundary cannot be covered.

Proof. The exact adjacent-midpoint cap is theorem B.6; its substitution in the perimeter budget is item (5) of theorem B.11. $\square$

Proposition 3.6 (Branches closed by trace length). *The trace-length mechanism excludes exactly the following routing entries and hybrid subcase:*

- (1) $N_{\text{gap}} = 0$ and $N_+ = 0$;
- (2) $N_{\text{gap}} = 0$, $N_+ = 1$, and $\#\{i : T_i \text{ is Vd1 or Vd2}\} \geq 1$;
- (3) $N_E(T_C) \in \{1, 2\}$ and $N_+ + N_{\text{sp}} \geq 3$;
- (4) $N_E(T_C) \in \{1, 2\}$, $N_+ = 0$, and $\#\{i : T_i \text{ is Vd1 or Vd2}\} \geq 1$;
- (5)

$$\begin{aligned} N_E(T_C) &= 1, & N_+ &= 1, \\ \#\{i : T_i \text{ is Vd1 or Vd2}\} &= 1, & \#\{i : T_i \text{ is T3-like}\} &= 0; \end{aligned}$$

- (6) the subcase in Proposition 3.5.

Proof. The complete exact substitutions, including every strict inequality, are given in theorem B.12. Item (3) is also the direct consequence of Theorem 3.4, and item (6) is Proposition 3.5. $\square$

4. AREA-LOSS MECHANISM

This section defines the normalized loss exactly and records the local and cyclic interfaces used by the routing argument. Appendix C solves the associated optimizations.

Put

$$A_\Delta := \frac{\sqrt{3}}{4}.$$

Definition 4.1 (Local feasible family and normalized loss). For $(a, b) \in [0, 1]^2$, define

$$\mathcal{T}(a, b) := \left\{ T : \begin{array}{l} T \text{ is a closed unit equilateral triangle,} \\ \{V_0, V_0 + a(V_5 - V_0), V_0 + b(V_1 - V_0)\} \subseteq T \end{array} \right\}.$$

For every closed unit equilateral triangle T , define

$$\mathcal{L}(T) := \frac{\text{area}(T \setminus H)}{A_\Delta}, \quad \mathcal{L}(a, b) := \inf_{T \in \mathcal{T}(a, b)} \mathcal{L}(T).$$

The corresponding family at V_i is obtained by the hexagon symmetry already fixed in Section 1. In particular, if

$$0 \leq a_i \leq A_i, \quad 0 \leq b_i \leq B_i,$$

then

$$\text{Rot}_{O, -i\pi/3}(T_i) \in \mathcal{T}(a_i, b_i), \quad \mathcal{L}(\text{Rot}_{O, -i\pi/3}(T_i)) = \mathcal{L}(T_i).$$

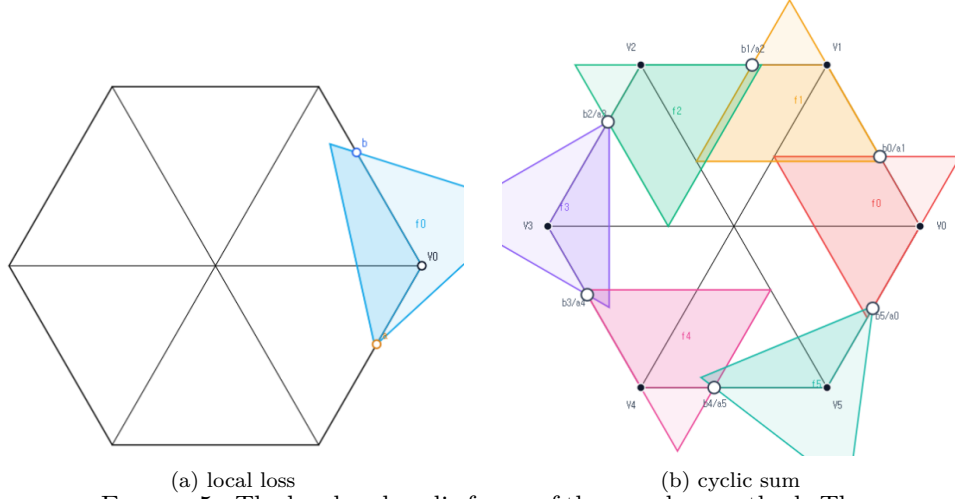


FIGURE 5. The local and cyclic forms of the area-loss method. The proof uses the exact variational definition above, not measurements from the picture.

Theorem 4.2 (Unconditional local square loss). *For every $(a, b) \in [0, 1]^2$ with $\mathcal{T}(a, b) \neq \emptyset$,*

$$(4) \quad \mathcal{L}(a, b) \geq \min\{a, b\}^2.$$

If $a + b > 1$, then

$$(5) \quad \mathcal{L}(a, b) \geq \max\{a, b\}^2.$$

Proof. This is the exact optimum comparison proved in theorem C.2. □

Theorem 4.3 (Direct T3-like loss). *Let T_i be an actual T3-like V triangle. Then*

$$(6) \quad A_i + B_i \leq 1.$$

Moreover, if

$$0 \leq a_i \leq A_i, \quad 0 \leq b_i \leq B_i, \quad 0 \leq \mu \leq \frac{1}{2}, \quad a_i, b_i \geq \mu,$$

then

$$(7) \quad \mathcal{L}(T_i) \geq 2\mu - 4\mu^2.$$

Proof. The nonsupercritical conclusion is Lemma 2.10; the actual-triangle loss optimization is theorem C.3. □

Lemma 4.4 (Cyclic area-loss certificate). *Assume the strict handoff data*

$$\xi_i \in (1 - A_{i+1}, B_i), \quad a_i := 1 - \xi_{i-1} < A_i, \quad b_i := \xi_i < B_i.$$

Then

$$\#\{i : a_i + b_i > 1\} \geq 2 \implies \sum_{i=0}^5 \mathcal{L}(T_i) > 1,$$

and

$$\begin{aligned} \#\{i : a_i + b_i > 1\} &= 1, \\ \forall i [T_i \text{ is Vd0 or T3-like}], \quad \exists j [T_j \text{ is T3-like}] &\implies \sum_{i=0}^5 \mathcal{L}(T_i) > 1. \end{aligned}$$

Proof. Both exact cyclic minima are proved in theorem C.4. $\square$

Proposition 4.5 (Branches closed by area). *The area-loss mechanism excludes:*

- (1) $N_{\text{gap}} = 0$ and $N_+ \geq 2$;
- (2)

$$\begin{aligned} N_{\text{gap}} &= 0, \quad N_+ = 1, \\ \#\{i : T_i \text{ is Vd1 or Vd2}\} &= 0, \quad \#\{i : T_i \text{ is T3-like}\} \geq 1. \end{aligned}$$

Proof. By Lemma 2.9, $N_{\text{gap}} = 0$ is exactly the condition that the six open V triangles cover ∂H . Apply the strict selector of Proposition 2.11. Its at-least-two clause and exact-one clause place the two respective cases under the two implications of Lemma 4.4. Thus the six V triangles have total normalized area inside H strictly below five; the C triangle has normalized area one, whereas H has normalized area six. This contradicts area subadditivity for a cover. $\square$

5. FINITE-ENCLOSURE OBSTRUCTIONS

5.1. Enclosure numbers and local capacities.

Definition 5.1 (Equilateral enclosure number). For $X \in \mathbb{R}^2$ and $r \geq 0$, put

$$\mathbb{B}(X, r) := \{Y \in \mathbb{R}^2 : \|Y - X\| \leq r\}.$$

For every nonempty compact $K \subset \mathbb{R}^2$, put

$$\Lambda(K) := \inf \left\{ \text{side}(Q) : \begin{array}{l} Q \text{ is a closed equilateral triangle,} \\ K \subseteq Q \end{array} \right\}.$$

The support-function evaluation of this infimum is theorem A.9.

Lemma 5.2 (Compact set in an open unit triangle).

$$\left. \begin{array}{l} K \subset \mathbb{R}^2 \text{ is compact,} \quad K \subset U, \\ U \text{ is an open unit equilateral triangle} \end{array} \right\} \implies \Lambda(K) < 1.$$

Proof. The distance from K to each side line of U is positive. Moving the three side lines inward through a common smaller distance leaves a closed equilateral triangle of side strictly below one that still contains K . $\square$

Normalize at V_0 and use the established radial map R_i .

Definition 5.3 (Anchor hull and radial capacity). For $(a, b, c) \in [0, 1]^3$, define

$$K(a, b, c) := \text{conv}\{V_0, X_5(1-a), X_0(b), R_0(c)\},$$

$$\mathcal{A} := \{(a, b, c) \in [0, 1]^3 : \Lambda(K(a, b, c)) \leq 1\}.$$

For

$$0 \leq a, b \leq 1, \quad a^2 + ab + b^2 \leq 1,$$

put

$$c_{\max}(a, b) := \max\{c \in [0, 1] : (a, b, c) \in \mathcal{A}\}.$$

Proposition 5.4 (Projection and symmetry of the anchor capacity). *One has*

$$\{(a, b) \in [0, 1]^2 : \exists c \in [0, 1] ((a, b, c) \in \mathcal{A})\} = \{(a, b) \in [0, 1]^2 : a^2 + ab + b^2 \leq 1\},$$

and the maximum in Definition 5.3 is attained. Moreover,

$$\left. \begin{array}{l} 0 \leq p, q \leq 1, \\ p^2 + pq + q^2 \leq 1 \end{array} \right\} \implies c_{\max}(p, q) = c_{\max}(q, p).$$

Proof. These statements are proved by the solved local enclosure problem theorem A.10. $\square$

Definition 5.5 (Neighboring-radial capacities). For $0 \leq a, b \leq 1$ and $a + b \leq 1$, put

$$C_+(a, b) := \max \left\{ c \in [0, 1] : \begin{array}{l} \exists \Delta \text{ } [\Delta \text{ is a closed unit equilateral triangle,} \\ \{V_0, X_5(1-a), X_0(b), R_1(c)\} \subseteq \Delta] \end{array} \right\},$$

$$C_-(a, b) := C_+(b, a).$$

The maxima are attained and evaluated in theorem D.1.

Definition 5.6 (Type-aware radial witness). Fix $i \in \mathbb{Z}/6\mathbb{Z}$ and selected reaches satisfying

$$\begin{aligned} 0 \leq a_j \leq A_j, \quad 0 \leq b_j \leq B_j \quad (0 \leq j \leq 5), \\ a_j^2 + a_j b_j + b_j^2 \leq 1 \quad (0 \leq j \leq 5), \\ \mathcal{H}^1(T_{i-1} \cap r_i) > 0 \implies a_{i-1} + b_{i-1} \leq 1, \\ \mathcal{H}^1(T_{i+1} \cap r_i) > 0 \implies a_{i+1} + b_{i+1} \leq 1. \end{aligned}$$

Define

$$\begin{aligned} \Gamma_i := \max \bigg(& \{c_{\max}(a_i, b_i)\} \\ & \cup \{C_+(a_{i-1}, b_{i-1}) : \mathcal{H}^1(T_{i-1} \cap r_i) > 0\} \\ & \cup \{C_-(a_{i+1}, b_{i+1}) : \mathcal{H}^1(T_{i+1} \cap r_i) > 0\} \bigg), \quad D_i := (1 - \Gamma_i)V_i. \end{aligned}$$

Lemma 5.7 (Type-aware radial forcing). *For the data in Definition 5.6,*

$$D_i \in r_i \subseteq S, \quad D_i \notin \bigcup_{j=0}^5 U_j.$$

Consequently,

$$S \subseteq U_C \cup \bigcup_{j=0}^5 U_j \implies D_i \in U_C.$$

Proof. This is the endpoint-strict forcing calculation theorem D.3. $\square$

Lemma 5.8 (Common-pair domination). *For $p, q \geq 0$ and $p + q \leq 1$,*

$$C_+(p, q), C_-(p, q) \leq 1 - \min\{p, q\} \leq c_{\max}(p, q).$$

Proof. See theorem D.2. $\square$

5.2. The common nonzero-gap terminals. For $0 \leq p, q \leq 1$ and $p + q \leq 1$, define

$$J(p, q) := X_0([q, 1 - p]), \quad c_* := c_{\max}(p, q),$$

$$\mathcal{D}_{(\sqrt{3}/2)(1-c_*)} := \mathbb{B}\left(O, \frac{\sqrt{3}}{2}(1 - c_*)\right) = \left\{x : \|x - O\| \leq \frac{\sqrt{3}}{2}(1 - c_*)\right\}.$$

Theorem 5.9 (Complementary-gap enclosure).

$$\Lambda\left(\mathcal{D}_{(\sqrt{3}/2)(1-c_*)} \cup J(p, q)\right) \geq 1.$$

Proof. The disk-plus-point optimization and endpoint comparison are theorem D.13. $\square$

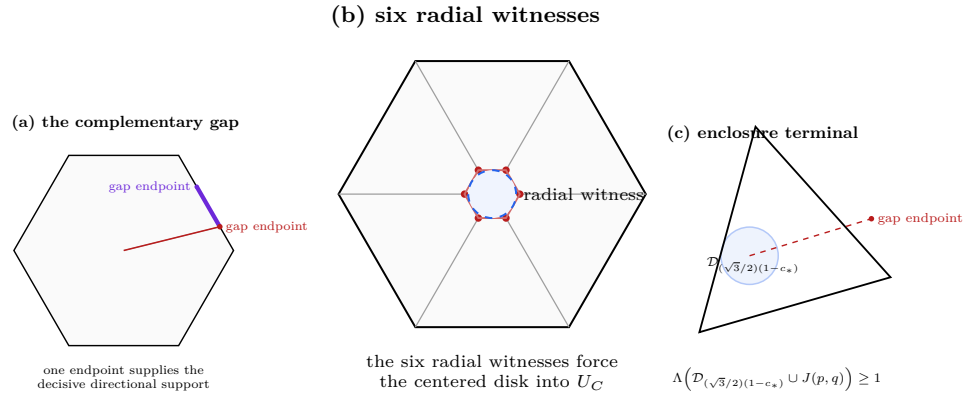


FIGURE 6. The complementary-gap enclosure mechanism. The actual gap $J(p, q)$ and six radial witnesses are center-forced. Their convex hull contains the displayed centered disk, and the enclosure theorem forces side length at least one. The disk is not an additional witness; it is contained in the convex hull of the six finite radial witnesses.

Proposition 5.10 (One-gap common-enclosure terminal). *Assume*

$$S \subseteq U_C \cup \bigcup_{i=0}^5 U_i, \quad N_{\text{gap}} = 1, \quad N_+ = 0, \quad N_E(T_C) \in \{1, 2\}, \quad d = 0, \quad t \in \{0, 1, 2\}.$$

After rotation, let

$$J := G_0 = X_0([\ell, r]), \quad \ell := B_0, \quad r := 1 - A_1,$$

$$p := 1 - r, \quad q := \ell, \quad c_* := c_{\max}(p, q).$$

Then

$$0 \leq p, q \leq 1, \quad p + q \leq 1.$$

If

$$\forall i \in \mathbb{Z}/6\mathbb{Z} \quad [(p \leq A_i \wedge q \leq B_i) \vee (q \leq A_i \wedge p \leq B_i)],$$

then

$$\mathcal{D}_{(\sqrt{3}/2)(1-c_*)} \cup J \subset U_C, \quad \Lambda(\mathcal{D}_{(\sqrt{3}/2)(1-c_*)} \cup J) \geq 1.$$

In particular, these hypotheses are inconsistent.

Proof. Lemmas 5.7 and 5.8 force the six vertices of a regular radial hexagon into U_C ; its convex hull contains the displayed disk. The exact gap definition forces $J \subset U_C$, including when $\ell = r$. Theorem 5.9 and Lemma 5.2 give the contradiction. $\square$

Theorem 5.11 (CE2 short-ray obstruction). *Assume*

$$S \subseteq U_C \cup \bigcup_{i=0}^5 U_i, \quad N_E(T_C) = 2, \quad N_{\text{gap}} = 2, \quad N_+ \in \{0, 1\}, \quad d = 0.$$

Normalize the two positive C-triangle traces to $e_{5,0}$ and $e_{0,1}$, and put

$$p := \min\{s \in [0, 1] : X_5(s) \in T_C\}, \quad q := 1 - \max\{s \in [0, 1] : X_0(s) \in T_C\},$$

$$c_* := c_{\max}(p, q), \quad D_2 := (1 - c_*)V_2, \quad D_4 := (1 - c_*)V_4.$$

Then

$$0 \leq p, q \leq 1, \quad p + q \leq 1.$$

If

$$\forall i \in \{1, 2, 3, 4, 5\} \quad [(p \leq A_i \wedge q \leq B_i) \vee (q \leq A_i \wedge p \leq B_i)],$$

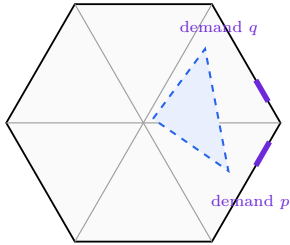
then

$$c_{\max}(q, p) = c_{\max}(p, q) = c_*, \quad D_2, D_4 \in U_C, \quad \{D_2, D_4\} \not\subseteq T_C.$$

In particular, these hypotheses are inconsistent.

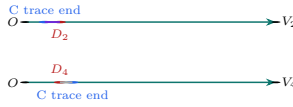
Proof. The exact signed CE2 optimization is theorem D.14; type-aware forcing and Proposition 5.4 give the stated points and symmetry. $\square$

(a) two CE2 gaps



p, q are the common boundary demands

(b) transverse center exits



one radial witness lies beyond the corresponding C trace

(c) the case split

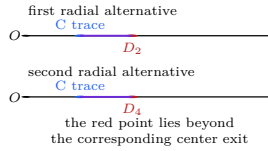


FIGURE 7. The CE2 short-ray terminal. The two boundary gaps determine the common pair (p, q) and the center-forced radial witnesses D_2, D_4 . The theorem proves that at least one of these witnesses lies beyond the corresponding C-triangle trace and is uncovered. The displayed ordering is schematic.

Proposition 5.12 (All nonsupercritical nonzero-gap closures). *Every configuration satisfying the hypotheses of Proposition 5.10 or Theorem 5.11 is impossible.*

Proof. This is the final sentence of the two cited results. $\square$

5.3. The transverse seven-point terminal. Suppose

$$S \subseteq U_C \cup \bigcup_{i=0}^5 U_i, \quad N_{\text{gap}} = 1, \quad N_+ = 1, \quad N_E(T_C) \in \{1, 2\}, \quad (d, t) = (0, 0),$$

$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\}, \quad \{\sigma\} = \{i : A_i + B_i > 1\}, \quad \sigma = 0.$$

$$(o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\} \quad (0 \leq i \leq 5).$$

Put

$$J := G_0 = X_0([\ell, r]), \quad \ell := B_0, \quad r := 1 - A_1,$$

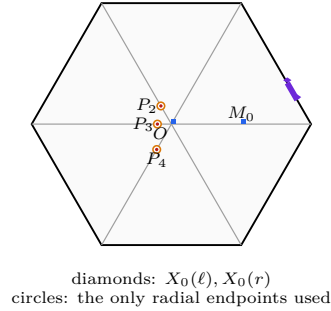
$$P_j := R_j(C_j), \quad K_{\text{tr}} := \{O, M_0, X_0(\ell), X_0(r), P_2, P_3, P_4\}.$$

Lemma 5.13 (The transverse witness is center-forced).

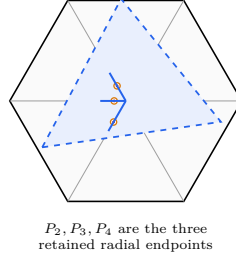
$$K_{\text{tr}} \subset U_C.$$

Proof. The two boundary points belong to the actual gap. Each P_j for $j \in \{2, 3, 4\}$ is the closed endpoint of the own radial trace, adjacent Vd0 roles have no positive trace on r_j , and the nonincident roles are excluded by diameter one. Hence the six open V roles miss all seven displayed points. $\square$

(a) the transverse set K_{tr}



(b) the hypothetical C triangle U_C



(c) the scalar squeeze
the two gap endpoints
impose an upper demand
the three radial endpoints
force an incompatible return
contradiction

FIGURE 8. The simplified one-gap all-Vd0 enclosure certificate. Only the actual gap endpoints and the three transverse radial endpoints are needed. The gap gives the diameter upper squeeze at the supercritical role; the three radial endpoints supply the CE1/CE2 transverse return. No radial point on r_0 is used. Lengths are schematic.

Proposition 5.14 (CE2 transverse threshold terminal). *No CE2 open unit equilateral triangle contains K_{tr} .*

Proof. See theorem D.16. $\square$

Proposition 5.15 (CE1 three-transverse return). *No CE1 open unit equilateral triangle contains K_{tr} .*

Proof. See theorem D.17. □

Proposition 5.16 (Transverse seven-point enclosure).

$$K_{\text{tr}} \subset U_C, \quad \Lambda(K_{\text{tr}}) \geq 1.$$

Consequently the normalized one-gap, one-supercritical all-Vd0 hypotheses above are inconsistent.

Proof. The first conclusion is Lemma 5.13. Every open unit triangle containing O, M_0 and a positive boundary interval is CE1 or CE2, so Propositions 5.15 and 5.14 show that no open unit triangle contains K_{tr} . Lemma 5.2 finishes the contradiction. □

5.4. Supported tails and residual separation. For $0 \leq c < 1/2$, define

$$M_c^{\text{sup}} := \sup\{b \in [0, 1] : \exists a \in [0, 1] ((a, b, c) \in \mathcal{A} \wedge a + b > 1)\},$$

and set

$$M_{1/2}^{\text{sup}} := \frac{1}{2}.$$

The endpoint value is the continuous extension evaluated in theorem D.6, not the supremum of an empty strict feasible set.

Proposition 5.17 (T3-like supported-tail terminal). *Assume*

$$S \subseteq U_C \cup \bigcup_{i=0}^5 U_i, \quad N_{\text{gap}} = 1, \quad N_+ = 1, \quad (d, t) = (0, 1), \quad N_E(T_C) \in \{1, 2\},$$

$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\}, \quad (o(T_0), n(T_0)) = (2, 1), \quad M_1 \in T_0, \quad A_1 + B_1 > 1, \\ (o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\} \quad (1 \leq i \leq 5).$$

If

$$T_0 \cap r_1 = R_1([c, u]), \quad 0 \leq c \leq \frac{1}{2} \leq u < 1, \quad \varepsilon := 1 - u, \quad C_1 \geq c, \quad M := M_c^{\text{sup}},$$

then

$$R_1(u) = \varepsilon V_1 \in U_C, \quad A_0 + \varepsilon \leq 1, \quad A_0 \leq 1 - M, \quad \frac{A_0}{A_0 + \varepsilon} \leq 1 - M,$$

$$B_1 < M, \quad A_i + B_i \leq 1 \quad (2 \leq i \leq 5),$$

and

$$\sum_{i=2}^5 (A_i + B_i) \geq 4 + M - B_1 > 4.$$

Thus the hypotheses are inconsistent.

Proof. The adapter inequalities are theorem D.18, and the common terminal is theorem D.11. □

Proposition 5.18 (Vd1 supported-tail terminal). *Assume*

$$S \subseteq U_C \cup \bigcup_{i=0}^5 U_i, \quad N_{\text{gap}} \in \{1, 2\}, \quad N_+ = 1, \quad (d, t) = (1, 0), \quad N_E(T_C) = 2,$$

$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\}, \quad (o(T_0), n(T_0)) = (1, 1), \quad M_1 \in T_0, \quad A_1 + B_1 > 1, \\ (o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\} \quad (1 \leq i \leq 5).$$

If

$$T_0 \cap r_1 = R_1([c, u]), \quad 0 \leq c \leq \frac{1}{2} \leq u < 1, \quad \varepsilon := 1 - u, \quad C_1 \geq c, \quad M := M_c^{\sup},$$

then

$$R_1(u) = \varepsilon V_1 \in U_C, \quad A_0 + \varepsilon \leq 1, \quad A_0 \leq 1 - M, \quad \frac{A_0}{A_0 + \varepsilon} \leq 1 - M,$$

$$B_1 < M, \quad A_i + B_i \leq 1 \quad (2 \leq i \leq 5),$$

and

$$\sum_{i=2}^5 (A_i + B_i) \geq 4 + M - B_1 > 4.$$

Thus the hypotheses are inconsistent.

Proof. Use theorems D.11 and D.19. $\square$

Lemma 5.19 (Radial separation interface). *For a routed arm r_i , define*

$$d_C := \max\{x \in [0, 1] : xV_i \in T_C\},$$

$$c_{\text{loc}} := \sup\{s \in [0, 1] : R_i(s) \in U_{i-1} \cup U_i \cup U_{i+1}\}.$$

Then

$$T_C \cap r_i = \{xV_i : 0 \leq x \leq d_C\},$$

$$(U_{i-1} \cup U_i \cup U_{i+1}) \cap r_i \subseteq \{xV_i : 1 - c_{\text{loc}} \leq x \leq 1\}.$$

If $c_{\text{loc}} < 1 - d_C$, then

$$\emptyset \neq \{xV_i : d_C < x < 1 - c_{\text{loc}}\} \subseteq S \setminus \left(U_C \cup \bigcup_{j=0}^5 U_j \right).$$

Proof. The first two displays follow from convexity and the extremal definitions. Every nonlocal V role is excluded from the open interval in the last display by diameter one. $\square$

Definition 5.20 (Boundary residual). For a closed interval $I \subseteq [0, 1]$ and $0 \leq p \leq 1$, put

$$\mathcal{R}_I(p) := 1 - \sup\{x \in [0, 1] : [0, x] \subseteq [0, p] \cup I\}.$$

Proposition 5.21 (Adjacent and nonadjacent Vd separation). *Assume*

$$S \subseteq U_C \cup \bigcup_{i=0}^5 U_i, \quad N_E(T_C) = 2, \quad N_{\text{gap}} \in \{1, 2\}, \quad N_+ = 1, \quad (d, t) = (1, 0),$$

$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\},$$

$$\{\sigma\} = \{i : A_i + B_i > 1\}, \quad \{\tau\} = \{i : (o(T_i), n(T_i)) \in \{(1, 1), (1, 2)\}\},$$

$$\sigma = 0, \quad \tau \in \{1, 2, 3, 4, 5\}, \quad (o(T_j), n(T_j)) \in \{(1, 0), (2, 0)\} \quad (j \neq \tau).$$

If $\tau \in \{1, 5\}$, reflect so that $\tau = 1$ and apply Lemma 5.19 with $i = 2$; then

$$c_{\text{loc}} < 1 - d_C.$$

If $\tau \in \{2, 3, 4\}$, define

$$I_R := \{x \in [0, 1] : X_0(x) \in T_C\}, \quad I_L := \{x \in [0, 1] : X_5(1 - x) \in T_C\},$$

$$\rho_R := \mathcal{R}_{I_R}(B_0), \quad \rho_L := \mathcal{R}_{I_L}(A_0), \quad \rho := \min\{\rho_R, \rho_L\}, \quad D_\tau := \rho V_\tau,$$

$$d_\tau^C := \max\{x \in [0, 1] : xV_\tau \in T_C\}.$$

Then

$$d_\tau^C < \rho < 1 - C_\tau, \quad D_\tau \in S \setminus \left(U_C \cup \bigcup_{j=0}^5 U_j \right).$$

Thus both alternatives contradict the assumed cover of S .

Proof. The adjacent inequality is theorem D.20; the two strict nonadjacent estimates are theorem D.21. $\square$

5.5. The two-chart replacement router.

Proposition 5.22 (Two-chart Vd1 replacement and router). *Assume*

$$\begin{aligned} S &\subseteq U_C \cup \bigcup_{i=0}^5 U_i, \quad N_E(T_C) = 2, \quad N_{\text{gap}} \in \{1, 2\}, \quad N_+ = 1, \\ (o(T_0), n(T_0)) &= (1, 1), \quad M_1 \in T_0, \quad A_1 + B_1 > 1, \quad A_i + B_i \leq 1 \quad (i \neq 1), \\ (o(T_i), n(T_i)) &\in \{(1, 0), (2, 0)\} \quad (1 \leq i \leq 5), \quad U_C \cap e_{0,1} = \emptyset, \\ \mathcal{H}^1(T_j \cap r_k) &= 0 \quad (2 \leq j \leq 5, k \in \{0, 1\}). \end{aligned}$$

Then

$$\exists \varepsilon > 0 \exists U'_0, U'_1 \quad [U'_k \text{ is an open unit equilateral triangle and } V_k \in U'_k \quad (k = 0, 1)]$$

such that, on setting

$$\begin{aligned} U'_i &:= U_i \quad (2 \leq i \leq 5), \quad T'_i := \overline{U'_i} \quad (0 \leq i \leq 5), \\ G'_i &:= e_{i,i+1} \setminus ((U'_i \cap e_{i,i+1}) \cup (U'_{i+1} \cap e_{i,i+1})), \quad N'_{\text{gap}} := |\{i : G'_i \neq \emptyset\}|, \end{aligned}$$

one has

$$N'_{\text{gap}} \in \{0, 1, 2\}.$$

Define

$$\begin{aligned} A'_i &:= \max\{s \in [0, 1] : V_i + s(V_{i-1} - V_i) \in T'_i\}, \\ B'_i &:= \max\{s \in [0, 1] : V_i + s(V_{i+1} - V_i) \in T'_i\}, \\ C'_i &:= \max\{s \in [0, 1] : R_i(s) \in T'_i\}. \end{aligned}$$

Then

$$\begin{aligned} S &\subseteq U_C \cup \bigcup_{i=0}^5 U'_i, \quad N_E(T_C) = 2, \\ (o(T'_i), n(T'_i)) &\in \{(1, 0), (2, 0)\} \quad (0 \leq i \leq 5), \quad A'_i + B'_i \leq 1 \quad (0 \leq i \leq 5), \\ A'_0 + B'_0 &= A'_1 + B'_1 = 1 - \varepsilon < 1, \\ (A'_i, B'_i, C'_i) &= (A_i, B_i, C_i) \quad (2 \leq i \leq 5), \quad |\{i : A'_i + B'_i > 1\}| = 0. \end{aligned}$$

For the recomputed gap rank,

$$\begin{aligned} N'_{\text{gap}} = 0 &\implies S \not\subseteq U_C \cup \bigcup_{i=0}^5 U'_i, \\ N'_{\text{gap}} = 1 &\implies S \not\subseteq U_C \cup \bigcup_{i=0}^5 U'_i, \\ N'_{\text{gap}} = 2 &\implies S \not\subseteq U_C \cup \bigcup_{i=0}^5 U'_i. \end{aligned}$$

The three implications use, respectively, Proposition 3.6, Theorem 5.9, and Theorem 5.11.

Proof. The complete two-chart replacement, including every strict margin and the output gap-rank recomputation, is theorem D.25. $\square$

Proposition 5.23 (Direct one-Vd placement assembly). *If*

$$N_E(T_C) = 2, \quad N_{\text{gap}} \in \{1, 2\}, \quad N_+ = 1, \quad d = 1,$$

then

$$S \not\subseteq U_C \cup \bigcup_{i=0}^5 U_i.$$

Proof. The exact placement partition and its hypotheses are proved in theorem D.22. Its positive-support alternative is theorem 3.4; its Vd2 alternatives are theorem 3.5; and its remaining alternatives are theorems 5.18, 5.21 and 5.22. $\square$

5.6. Finite-enclosure register. Rows A–E below share the exact hypothesis

$$S \subseteq U_C \cup \bigcup_{i=0}^5 U_i;$$

row F retains the stronger hypothetical cover of H . All entries in the subcase-identifying columns are formulas. In row F, a, b and $K_{\text{wit}}(a, b)$ are the exact objects defined in equations (9) and (13) in the following zero-gap subsection.

Table 2: Exact finite-enclosure register.

	rank data	normalized predicates	exact conclusion
A		$N_E(T_C) \in \{1, 2\}, \quad G_0 = X_0([\ell, r]),$	
	$N_{\text{gap}} = 1$	$\ell = B_0, \quad r = 1 - A_1, \quad p = 1 - r, \quad q = \ell,$	$\mathcal{D}_{(\sqrt{3}/2)(1-c_*)} \cup G_0$
	$N_+ = 0$	$0 \leq p, q \leq 1, \quad p + q \leq 1, \quad c_* = c_{\max}(p, q),$	$\subset U_C,$
	$d = 0$	$\forall i \in \mathbb{Z}/6\mathbb{Z},$	$\Lambda(\mathcal{D}_{(\sqrt{3}/2)(1-c_*)} \cup G_0) \geq 1$
	$t \in \{0, 1, 2\}$	$(p \leq A_i \wedge q \leq B_i)$ $\vee (q \leq A_i \wedge p \leq B_i)$	
B		$N_E(T_C) = 2, \quad p = \min\{s \in [0, 1] : X_5(s) \in T_C\},$	
		$q = 1 - \max\{s \in [0, 1] : X_0(s) \in T_C\},$	
	$N_{\text{gap}} = 2$	$0 \leq p, q \leq 1, \quad p + q \leq 1,$	$c_{\max}(q, p) = c_*,$
	$N_+ \in \{0, 1\}$	$c_* = c_{\max}(p, q), \quad D_j = (1 - c_*)V_j \quad (j \in \{2, 4\}),$	$D_2, D_4 \in U_C,$
	$d = 0$	$\forall i \in \{1, \dots, 5\},$	$\{D_2, D_4\} \not\subseteq T_C$
C		$(p \leq A_i \wedge q \leq B_i)$ $\vee (q \leq A_i \wedge p \leq B_i)$	
	$N_{\text{gap}} = 1$	$N_E(T_C) \in \{1, 2\}, \quad T_C \cap \{M_0, \dots, M_5\} = \{M_0\},$	
	$N_+ = 1$	$\{i : A_i + B_i > 1\} = \{0\},$	$K_{\text{tr}} \subset U_C \wedge \Lambda(K_{\text{tr}}) \geq 1$
	$(d, t) = (0, 0)$	$\forall i \in \mathbb{Z}/6\mathbb{Z} \quad (o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\}$	
		$N_E(T_C) \in \{1, 2\}, \quad T_C \cap \{M_0, \dots, M_5\} = \{M_0\},$	
D ₁		$(o(T_0), n(T_0)) = (2, 1),$	
	$N_{\text{gap}} = 1$	$M_1 \in T_0, \quad A_1 + B_1 > 1,$	
	$N_+ = 1$	$\forall i \in \{1, \dots, 5\} \quad (o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\},$	$\sum_{i=2}^5 (A_i + B_i) \geq 4 + M -$
	$(d, t) = (0, 1)$	$T_0 \cap r_1 = R_1([c, u]), \quad 0 \leq c \leq \frac{1}{2} \leq u < 1,$	$B_1 > 4$
		$\varepsilon = 1 - u, \quad C_1 \geq c, \quad M = M_c^{\text{sup}}$	

	rank data	normalized predicates	exact conclusion
		$N_E(T_C) = 2, \quad T_C \cap \{M_0, \dots, M_5\} = \{M_0\},$ $(o(T_0), n(T_0)) = (1, 1),$ $M_1 \in T_0, \quad A_1 + B_1 > 1,$ $\forall i \in \{1, \dots, 5\} \quad (o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\}, \quad \sum_{i=2}^5 (A_i + B_i) \geq 4 + M - \frac{B_1}{4}$ $T_0 \cap r_1 = R_1([c, u]), \quad 0 \leq c \leq \frac{1}{2} \leq u < 1,$ $\varepsilon = 1 - u, \quad C_1 \geq c, \quad M = M_c^{\sup}$ $N_E(T_C) = 2, \quad T_C \cap \{M_0, \dots, M_5\} = \{M_0\},$	
D ₂	$N_{\text{gap}} \in \{1, 2\}$ $N_+ = 1$ $(d, t) = (1, 0)$	$\{\sigma\} = \{i : A_i + B_i > 1\},$ $\{\tau\} = \left\{ i : \begin{smallmatrix} (o(T_i), n(T_i)) \in \\ \{(1, 1), (1, 2)\} \end{smallmatrix} \right\},$ $\sigma = 0, \quad \tau \in \{1, 5\},$ $\forall j \neq \tau,$ $(o(T_j), n(T_j)) \in \{(1, 0), (2, 0)\}$ $N_E(T_C) = 2, \quad T_C \cap \{M_0, \dots, M_5\} = \{M_0\},$	$c_{\text{loc}} < 1 - d_C$
E ₁	$N_{\text{gap}} \in \{1, 2\}$ $N_+ = 1$ $(d, t) = (1, 0)$	$\{\sigma\} = \{i : A_i + B_i > 1\},$ $\{\tau\} = \left\{ i : \begin{smallmatrix} (o(T_i), n(T_i)) \in \\ \{(1, 1), (1, 2)\} \end{smallmatrix} \right\},$ $\sigma = 0, \quad \tau \in \{2, 3, 4\},$ $\forall j \neq \tau,$ $(o(T_j), n(T_j)) \in \{(1, 0), (2, 0)\}$	$d_\tau^C < \rho < 1 - C_\tau,$ $D_\tau \in S \setminus (U_C \cup \bigcup_{j=0}^5 U_j)$
E ₂	$N_{\text{gap}} \in \{1, 2\}$ $N_+ = 1$ $(d, t) = (1, 0)$	$\{\sigma\} = \{i : A_i + B_i > 1\},$ $\{\tau\} = \left\{ i : \begin{smallmatrix} (o(T_i), n(T_i)) \in \\ \{(1, 1), (1, 2)\} \end{smallmatrix} \right\},$ $\sigma = 0, \quad \tau \in \{2, 3, 4\},$ $\forall j \neq \tau,$ $(o(T_j), n(T_j)) \in \{(1, 0), (2, 0)\}$	$d_\tau^C < \rho < 1 - C_\tau,$ $D_\tau \in S \setminus (U_C \cup \bigcup_{j=0}^5 U_j)$
F	$N_{\text{gap}} = 0$ $N_+ = 1$ $(d, t) = (0, 0)$	$H \subseteq U_C \cup \bigcup_{i=0}^5 U_i$	$K_{\text{wit}}(a, b) \subset U_C \wedge \Lambda(K_{\text{wit}}(a, b)) \geq 1$

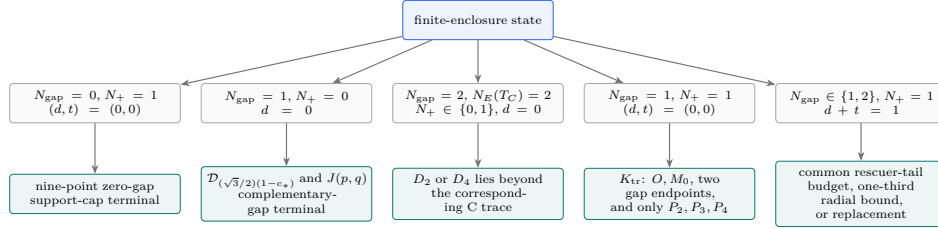


FIGURE 9. Roadmap of the finite-enclosure mechanism. The one-gap all-Vd0 witness uses only three transverse radial endpoints; the two-gap terminal uses one of the two displayed radial witnesses; and the T3-like and Vd1 paths share one rescuer-tail lemma.

Proposition 5.24 (All nonzero-gap finite-enclosure rows). *Every nonzero-gap row of Table 1 assigned to the finite-enclosure strategy is impossible.*

Proof. Rows A and B are Proposition 5.12; row C is Proposition 5.16; row D is Propositions 5.17 and 5.18; and row E is Proposition 5.23. $\square$

5.7. Zero-Gap Nine-Point Obstruction. Under a hypothetical cover, the zero-gap branch considered here is

$$N_{\text{gap}} = 0, \quad N_+ = 1, \quad (d, t) = (0, 0).$$

Rotate the unique supercritical role to index 4. Then

$$(8) \quad \xi_3 < \xi_2 < \xi_1 < \xi_0 < \xi_5 < \xi_4,$$

and, on putting

$$(9) \quad a := 1 - \xi_3, \quad b := \xi_4, \quad p := 1 - b, \quad q := 1 - a,$$

one has

$$(10) \quad \begin{aligned} 0 < a, b < 1, \quad a = a_4 < A_4, \quad b = b_4 < B_4, \\ a + b > 1, \quad a^2 + ab + b^2 < 1, \quad a_i \geq p, \quad b_i \geq q \quad (0 \leq i \leq 5). \end{aligned}$$

These relations are supplied by theorem E.2.

Define the two boundary anchors

$$P_A := V_4 + a(V_3 - V_4), \quad P_B := V_4 + b(V_5 - V_4),$$

and the feasible union

$$\mathcal{R}_{AB}(a, b) := \bigcup \left\{ T \cap H : \begin{array}{l} T \text{ is a closed unit equilateral triangle,} \\ V_4, P_A, P_B \in T \end{array} \right\}.$$

Using the previously defined radial envelope $c_{\max}$, put

$$\begin{aligned} c_* &:= c_{\max}(p, q), \quad h := \frac{\sqrt{3}}{2}, \quad r_* := h(1 - c_*), \\ P_i^{\text{rad}} &:= (1 - c_*)V_i \quad (0 \leq i \leq 5), \quad \mathbb{B}(0, r_*) := \{X : \|X\| \leq r_*\}. \end{aligned}$$

The direct radial forcing lemma gives

$$(11) \quad \mathbb{B}(0, r_*) \subseteq \text{conv}\{P_0^{\text{rad}}, \dots, P_5^{\text{rad}}\} \subset U_C.$$

Indeed, this is precisely theorem E.3.

For the asymmetric witnesses, define the two adjacent-handoff circles by

$$\mathcal{C}_- := \partial\mathbb{B}(X_2(b), 1), \quad \mathcal{C}_+ := \partial\mathbb{B}(X_5(1 - a), 1).$$

The corresponding maximal nondegenerate line pieces of the interior frontier are defined intrinsically by

$$\begin{aligned} \{\ell_-\} &:= \left\{ [X, Y] : \begin{array}{l} X, Y \in \mathbb{R}^2, \quad X \neq Y, \quad [X, Y] \subseteq \partial\mathcal{R}_{AB}(a, b) \cap \text{int}(H), \\ [X, Y] \cap \mathcal{C}_- \neq \emptyset, \\ \nexists X', Y' \in \mathbb{R}^2 \quad [X, Y] \subsetneq [X', Y'] \subseteq \partial\mathcal{R}_{AB}(a, b) \cap \text{int}(H) \end{array} \right\}, \\ \{\ell_+\} &:= \left\{ [X, Y] : \begin{array}{l} X, Y \in \mathbb{R}^2, \quad X \neq Y, \quad [X, Y] \subseteq \partial\mathcal{R}_{AB}(a, b) \cap \text{int}(H), \\ [X, Y] \cap \mathcal{C}_+ \neq \emptyset, \\ \nexists X', Y' \in \mathbb{R}^2 \quad [X, Y] \subsetneq [X', Y'] \subseteq \partial\mathcal{R}_{AB}(a, b) \cap \text{int}(H) \end{array} \right\}. \end{aligned}$$

Finally, define the three frontier points by

$$\{Q_0\} := \ell_- \cap \ell_+, \quad \{Q_-\} := \ell_- \cap \mathcal{C}_-, \quad \{Q_+\} := \ell_+ \cap \mathcal{C}_+.$$

Their existence and uniqueness follow from theorems E.1 and E.4; their direct exclusion from all six V triangles is theorem E.5. Thus

$$(12) \quad Q_-, Q_0, Q_+ \in \text{int}(H) \setminus \bigcup_{i=0}^5 U_i, \quad Q_-, Q_0, Q_+ \in U_C.$$

The public compact witness is

$$(13) \quad \begin{aligned} K_{\text{wit}}(a, b) &:= \mathbb{B}(0, r_*) \cup \{Q_-, Q_0, Q_+\}, \\ K_{\text{wit}}(a, b) &\subset U_C. \end{aligned}$$

For the schematic support picture, for $X \in \mathbb{R}^2$ and $r \geq 0$ write

$$\text{Cap}(X, r) := \{n \in \mathbb{S}^1 : \langle X, n \rangle + r \geq h\}.$$

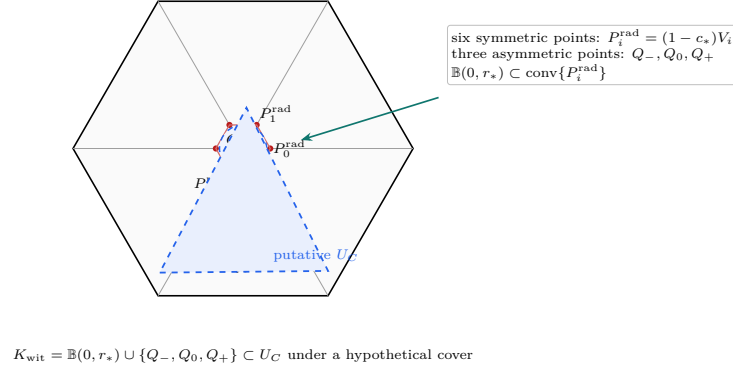


FIGURE 10. The zero-gap nine-point witness. The six symmetric radial points force a centered disk into the convex C triangle, while the three asymmetric frontier points supply the missing directional support. Marker shapes separate the two constructions: circles are radial witnesses and triangles are strict-supercritical AB -frontier witnesses. The displayed placement is illustrative; the point definitions are the exact ones in the preceding lemmas.

appendix E constructs the labeled Newton inner points and proves the four exact cap overlaps depicted below.

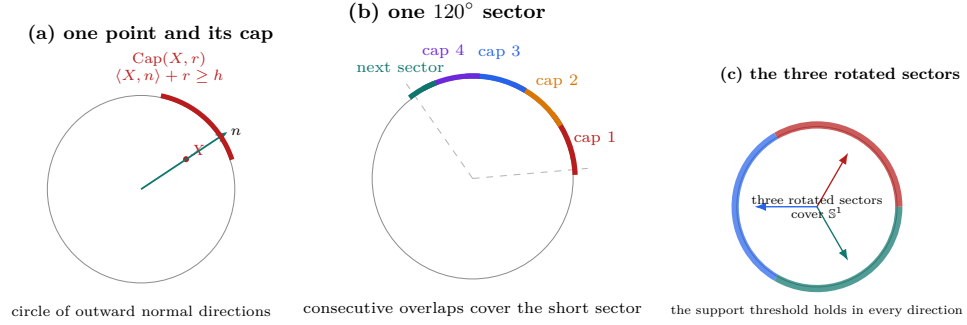


FIGURE 11. The support-cap conclusion for the zero-gap witness. A point and a disk radius determine an arc of normal directions in which their support is at least h . Four consecutive overlap checks cover one sector, and its three 120° rotations cover the full normal circle. The arc widths are schematic; the cyclic order and overlap pattern are the exact hypotheses of the support-arc lemma.

Theorem 5.25 (Witness enclosure). *For every pair (a, b) satisfying (10),*

$$\Lambda(K_{\text{wit}}(a, b)) \geq 1.$$

Proof. This is the zero-gap optimization proved in theorem E.9. $\square$

Theorem 5.26 (Zero-gap obstruction). *There is no cover of H by open unit equilateral triangles $U_C, U_0, \dots, U_5$ for which*

$$N_{\text{gap}} = 0, \quad N_+ = 1, \quad (d, t) = (0, 0).$$

Proof. Equations (11) and (12) give the compact containment $K_{\text{wit}}(a, b) \subset U_C$, whereas theorem 5.25 gives $\Lambda(K_{\text{wit}}(a, b)) \geq 1$. Every compact subset of an open unit equilateral triangle lies in a closed equilateral triangle of side strictly less than one, which would give $\Lambda(K_{\text{wit}}(a, b)) < 1$. $\square$

Proposition 5.27 (Cases excluded by the nine-point obstruction). *Every branch satisfying*

$$N_{\text{gap}} = 0, \quad N_+ = 1, \quad (d, t) = (0, 0)$$

is impossible, independently of the center class.

Proof. This is exactly theorem 5.26. $\square$

6. COMPLETION OF THE PROOF

Proof of Theorem 1.1. Assume that seven open unit equilateral triangles cover H . By Proposition 2.15, the cover lies in exactly one row of Table 1.

First suppose $N_{\text{gap}} = 0$. The $N_+ = 0$ row and the $N_+ = 1, d \geq 1$ row are excluded by Proposition 3.6. The $N_+ \geq 2$ row and the $N_+ = 1, d = 0, t \geq 1$ row are excluded by Proposition 4.5. The remaining $N_+ = 1, (d, t) = (0, 0)$ row is excluded by the explicit finite-enclosure theorem Proposition 5.27.

Now suppose $N_{\text{gap}} \geq 1$. The rows assigned to trace length are excluded by Proposition 3.6. Every former boundary-reach row is excluded by the finite residual-hull theorem Proposition 5.24. In the final CE2 one-Vd1/Vd2 row, the Strategy 1 complements and the finite-enclosure placements are assembled in Proposition 5.23; its replacement subcase recomputes the gap rank before selecting the closing strategy. Thus every zero- and nonzero-gap row is impossible, contradicting the assumed cover. $\square$

Corollary 1.2 follows from Proposition 2.1.

APPENDIX A. STRUCTURAL, SHARED-LOCAL, AND SIGNED-CENTER
OPTIMIZATIONS

For all coordinate evaluations in this appendix, choose the global Euclidean chart

$$O = (0, 0), \quad V_i = \left(\cos \frac{i\pi}{3}, \sin \frac{i\pi}{3} \right).$$

Every conclusion is invariant under the hexagon symmetries already fixed in the body.

A.1. Module A.1: open–closed shrinking feasibility.

Exact interface. The three cover formulations are those of theorem 2.1.

Variables and feasible set. Let $U_1, \dots, U_7$ be open unit equilateral triangles covering the compact hexagon H . For $p \in H$ define

$$m_j(p) := \text{dist}(p, \mathbb{R}^2 \setminus U_j), \quad m(p) := \max_{1 \leq j \leq 7} m_j(p).$$

Objective. Find a single inward displacement that may be applied to all twenty-one side lines while retaining a cover of H .

Exact result.

Lemma A.1 (Uniform open–closed shrinking). *There is $\rho > 0$ such that moving every side of every U_j inward through distance ρ produces seven closed equilateral triangles covering H , each of side length $1 - 2\sqrt{3}\rho > 0$.*

Solution.

Proof. Each m_j is continuous, hence so is m . The open cover makes m positive on H , and compactness gives

$$\eta := \min_{p \in H} m(p) > 0.$$

Choose

$$0 < \rho < \min \left\{ \eta, \frac{\sqrt{3}}{6} \right\}.$$

For every $p \in H$ some j satisfies $m_j(p) \geq \eta > \rho$, so p remains in the closed triangle obtained from U_j by the inward displacement. The inradius of an equilateral triangle of side s is $s\sqrt{3}/6$; moving all three sides inward by ρ therefore decreases the side by $2\sqrt{3}\rho$. The choice of ρ keeps the resulting side positive. $\square$

Body consequence. This is the quantitative implication used in theorem 2.1.

A.2. Module A.2: center and V-triangle incidence feasibility.

Exact interface. The C-triangle count $N_E(T_C)$ and the V-triangle incidence pair $(o(T_i), n(T_i))$ are defined in sections 1 and 2.

Variables and feasible set. For the center calculation take relative-interior parameters $0 < s, t < 1$ on two nonadjacent edges. For the V calculation normalize to $i = 0$ and use the affine wedge chart

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0),$$

with V_0 as the chart origin. Its squared Euclidean norm is

$$(14) \quad \|(x, y)\|^2 = x^2 + y^2 - xy = \left(y - \frac{x}{2}\right)^2 + \frac{3x^2}{4} = (x - y)^2 + xy.$$

Within the closed unit ball about V_0 , the part of H is exactly

$$W := \{(x, y) : x \geq 0, y \geq 0\},$$

and the adjacent radial segments are

$$r_1 = \{(1, y) : 0 \leq y \leq 1\}, \quad r_5 = \{(x, 1) : 0 \leq x \leq 1\}.$$

Objective. Determine the maximum number of positive C-triangle edge incidences and the minimum number of V-triangle vertices forced into the local wedge by one or two positive radial incidences.

Exact result.

Lemma A.2 (Center and V-triangle incidence calculation). *A closed unit equilateral triangle with O in its interior has positive trace on at most two edges of H , and any two such edges are adjacent. An original V triangle has $1 \leq o(T_i) \leq 3$; one positive adjacent radial trace forces at least one of its vertices into H , and two such traces force at least two vertices into H .*

Solution.

Proof. For the C triangle, among any three edges of the six-cycle two are nonadjacent. By symmetry take

$$x = (1 - t)V_0 + tV_1.$$

If $y = (1 - s)V_2 + sV_3$, direct calculation in the global chart gives

$$\|x - y\|^2 - 1 = (1 - t)(2 - t) + s(s + t) > 0.$$

If instead $y = (1 - s)V_3 + sV_4$ and $u = t + s$, then

$$\|x - y\|^2 - 1 = (u - 1)^2 + 2 > 0.$$

Thus a unit-diameter triangle cannot contain relative-interior points of two nonadjacent edges, proving the center assertion.

For the V triangle, (14) proves the asserted description of H inside the unit ball. Suppose $T_0 \cap r_1$ has positive length and let m be the largest x -coordinate of a vertex of T_0 . If $m > 1$, a maximizing vertex (m, y) lies in the unit ball, whence $0 < y < 1$ and that vertex lies in H . If $m = 1$, positive-length contact with the support line $x = 1$ forces a side of T_0 to lie there; both endpoints satisfy $1 + y^2 - y \leq 1$ and hence lie in H . The same argument applies to r_5 .

If both r_1 and r_5 are met but only one triangle vertex lies in H , that vertex must have $x > 1$ and $y > 1$, contradicting $(x - y)^2 + xy > 1$. Finally, if all three vertices belonged to the convex set H , then $T_0 \subseteq H$, which is impossible because the extreme point V_0 lies in the interior of T_0 . This also proves $o(T_0) \geq 1$. $\square$

Body consequence. The first conclusion supplies theorem 2.3; the remaining conclusions supply theorem 2.4 and the finite vertex-type list.

A.3. Module A.3: exact-trace normalization and interval geometry.

Exact interface. The input is an original V triangle U_i , its closure T_i , the incidence pair $(o(T_i), n(T_i))$, and the actual reaches A_i, B_i, C_i defined in section 1. The output must preserve the exact open and closed traces whenever a role is normalized.

Variables and feasible set. Normalize to $i = 0$ in the wedge chart of Module A.2. Put

$$D_t := 1 - t + t^2, \quad z := \sqrt{D_t}, \quad 0 \leq t \leq 1.$$

After relabeling the triangle vertices, every orientation is in one of the two families

	first side vector	second side vector
I	$z^{-1}(1, t)$	$z^{-1}(1 - t, 1)$
II	$z^{-1}(t, -(1 - t))$	$z^{-1}(1, t)$.

For Type I set

$$(15) \quad U = \alpha + y - tx, \quad V = \beta + x - (1 - t)y, \quad T = \{U \geq 0, V \geq 0, U + V \leq z\},$$

and for Type II set

$$(16) \quad U = \alpha + (1 - t)x + ty, \quad V = \beta + tx - y, \quad T = \{U \geq 0, V \geq 0, U + V \leq z\}.$$

In both cases

$$(17) \quad V_0 \in \text{int}(T) \iff \alpha > 0, \quad \beta > 0, \quad \alpha + \beta < z.$$

The Type-II vertex in the wedge is

$$(18) \quad Q = \left(\frac{z - \alpha - t\beta}{D_t}, \frac{t(z - \alpha) + (1 - t)\beta}{D_t} \right),$$

whose coordinates are positive on the feasible set.

Objective. For a raw $(3, 0)$ role, find a same-orientation translation with identical trace in H and type $(2, 0)$. For a T3-like role, find the exact closed-trace majorant and determine its actual boundary-reach sum. Finally, determine all open and closed initial reach intervals and their exact gap set.

Exact result.

Lemma A.3 (Exact-trace translations and initial intervals). *The following hold.*

(1) *Every raw $(3, 0)$ role admits a same-orientation translate $\widehat{U}$ for which*

$$V_i \in \widehat{U}, \quad (o(\widehat{U}), n(\widehat{U})) = (2, 0), \quad U \cap H = \widehat{U} \cap H, \quad \overline{U} \cap H = \widehat{\overline{U}} \cap H.$$

(2) *Every T3-like role has Type II orientation and admits the closed-trace majorant of theorem 2.7. Its exact reaches satisfy $A_i + B_i \leq 1$.*

(3) *For every V triangle,*

	T_i	U_i
$e_{i-1,i}$	$X_{i-1}([1 - A_i, 1])$	$X_{i-1}([1 - A_i, 1])$
$e_{i,i+1}$	$X_i([0, B_i])$	$X_i([0, B_i])$
r_i	$R_i([0, C_i])$	$R_i([0, C_i])$.

Consequently

$$G_i = \begin{cases} X_i([B_i, 1 - A_{i+1}]), & B_i + A_{i+1} \leq 1, \\ \emptyset, & B_i + A_{i+1} > 1, \end{cases}$$

including a singleton in the equality case.

Solution.

Proof. For a raw $(3,0)$ role, the Type-II vertex (18) lies in the wedge; since T contains the origin and has diameter one, theorem A.2 puts it in H . Thus only Type I is feasible. Its vertices are

$$(19) \quad \begin{aligned} Q_0 &= \left(-\frac{(1-t)\alpha + \beta}{D_t}, -\frac{\alpha + t\beta}{D_t} \right), \\ Q_1 &= \left(\frac{z - (1-t)\alpha - \beta}{D_t}, \frac{tz - \alpha - t\beta}{D_t} \right), \\ Q_2 &= \left(\frac{(1-t)z - (1-t)\alpha - \beta}{D_t}, \frac{z - \alpha - t\beta}{D_t} \right). \end{aligned}$$

The vertex Q_0 has two negative coordinates, the first coordinate of Q_1 is positive, and the second coordinate of Q_2 is positive. Hence $o = 3$ is equivalent to

$$(20) \quad \alpha + t\beta > tz, \quad (1-t)\alpha + \beta > (1-t)z,$$

which forces $0 < t < 1$. Put

$$s := \alpha + \beta, \quad L := z - s > 0.$$

Then (20) becomes

$$(21) \quad \alpha > \frac{tL}{1-t}, \quad \beta > \frac{(1-t)L}{t}.$$

On the wedge cut

$$(1-t)x + ty \leq L,$$

the minimum of U occurs at $(L/(1-t), 0)$ and that of V occurs at $(0, L/t)$. Therefore (21) makes both inequalities strictly redundant there and

$$(22) \quad T \cap W = \{(x, y) \in W : (1-t)x + ty \leq L\}.$$

Keep t, s fixed and replace the affine constants by

$$(23) \quad \hat{\alpha} := \frac{tL}{1-t}, \quad \hat{\beta} := s - \hat{\alpha}.$$

Then $0 < \hat{\alpha} < \alpha$, $\hat{\beta} > \beta > 0$, and $\hat{\alpha} + \hat{\beta} = s < z$, so this is a same-orientation translation retaining V_0 in its interior. Its third side still gives the cut in (22); the new $\hat{U}$ inequality has equality only at the already-boundary point $(L/(1-t), 0)$ and the new $\hat{V}$ inequality remains strict. Thus both the closed and open traces on W , hence on H , are unchanged. Formula (23) puts Q_1 on H , leaves Q_0 outside, and keeps Q_2 outside by the second inequality in (21); the new pair is therefore $(2, 0)$. This proves part 1.

For part 2, direct inversion of the Type-I vertices shows that when exactly two vertices are outside W , the sole wedge vertex has both coordinates strictly below one: in the two possible cases the relevant bounds are $(1-t)/z$ and t/z , with strictness from $\alpha > 0$ or $\beta > 0$. The supporting-line argument in Module A.2 then excludes positive contact with $x = 1$ or $y = 1$. A T3-like role must therefore have Type II orientation. Reflect in $x = y$ if needed so that $T_0 \cap r_1$ has positive length. Its outside vertices have respectively negative x - and y -coordinates, its wedge vertex is (18), and $0 < t < 1$. The exact axis reaches are

$$(24) \quad A = \beta, \quad B = z - \alpha - \beta,$$

while the T3-like incidence is equivalent to $Q_x > 1$ and $Q_y \leq 1$.

Replace α by 0 and retain t, β . At the origin the new slacks are $\widehat{U} = 0$ and $0 < \widehat{V} = \beta < z$, placing V_0 in the relative interior of a side. For $(x, y) \in W$,

$$\widehat{U} = (1-t)x + ty \geq 0, \quad \widehat{V} = V, \quad \widehat{U} + \widehat{V} = U + V - \alpha.$$

Hence $T \cap W \subset \widehat{T} \cap W$; the old x -axis reach is B and the new one is $B + \alpha$, so inclusion is strict in H . The two outside vertices remain outside and $\widehat{Q}_x > Q_x > 1$. From $Q_x > 1$ one obtains $t\beta < z - \alpha - D_t$, and therefore

$$tD_t(1 - \widehat{Q}_y) > D_t(1 - z) + (1-t)\alpha > 0.$$

Thus $\widehat{Q}_y < 1$, so the translated role remains of type $(2, 1)$. Every nonzero point of W has $\widehat{U} > 0$, which proves the asserted strict-interior preservation. Finally, (24) gives

$$A + B = z - \alpha \leq 1.$$

For part 3, intersect the convex set T_i with each of the three segments issuing from its interior point V_i . The result is a nonempty compact initial interval, and the defining maximum is its outer endpoint. Each maximum is strictly below one: an interior point of a unit-diameter triangle is at distance strictly below one from every point of its closure. Every point before the outer endpoint is a strict convex combination of V_i and a point of T_i , hence is in $U_i = \text{int}(T_i)$; if the outer endpoint were in U_i , openness would extend the interval past it. This proves all six interval identities and the equality of the closed maxima with the open suprema.

It remains only to exclude nonincident V triangles from the relative interior of $e_{i,i+1}$. Normalize to $e_{0,1}$ and write $x = X_0(s)$ with $0 < s < 1$. For the nearest nonincident vertices one finds

$$\|x - V_2\|^2 = 1 + (1-s)(2-s) > 1, \quad \|x - V_5\|^2 = 1 + s + s^2 > 1;$$

all other cases follow by symmetry or are farther away. Since the distinguished vertex of a V triangle lies in its open interior, its triangle cannot also contain such an x . Subtracting the two incident open initial intervals from the edge now gives the displayed formula for G_i , with the closed singleton retained when their two missing endpoints agree. $\square$

Body consequence. Part 1 proves theorem 2.5. Part 2 proves theorems 2.7 and 2.10; part 3 proves theorems 2.8 and 2.9.

A.4. Module A.4: signed-center optimization.

Exact interface. Let T_C be a closed unit equilateral triangle with $O \in \text{int}(T_C)$ and $N_E(T_C) \in \{1, 2\}$. A hexagon symmetry normalizes a positive trace to $e_{0,1}$. The module must determine its positive boundary traces and radial exits, including the point-contact boundary between CE1 and CE2.

Variables and feasible set. Use the affine chart

$$X = V_0 + b(V_1 - V_0) + a(V_5 - V_0).$$

The orientation parameter and its complements are

$$\begin{aligned} 0 < R < 1, \quad W &:= 1 - R, \quad E := \sqrt{1 - RW}, \\ \eta &:= 1 - E, \quad P := E(1 - E). \end{aligned}$$

The center slacks satisfy $\alpha, \delta \geq 0$, and are strictly positive for the closure of an original open C triangle. Put

$$k := \eta + \alpha + \delta, \quad \Delta_R := P - \alpha - W\delta > 0, \quad \Delta_L := P - R\alpha - \delta.$$

Objective. Determine the exact right and left trace intervals, their sign classification and total length, all six radial exits, the exact midpoint set, and the CE1/CE2 boundary caps.

Exact result.

Proposition A.4 (Signed CE1/CE2 center normal form). *On the feasible set above, define the three side slacks by*

$$(25) \quad \begin{aligned} F_0 &= R + \alpha - a + Wb, \\ F_1 &= Rb + Wa - k, \quad T_C = \{F_0, F_1, F_2 \geq 0\}. \\ F_2 &= W + \delta - b + Ra. \end{aligned}$$

Then

$$(26) \quad T_C \cap e_{0,1} = \left[\frac{k}{R}, W + \delta \right], \quad \mathcal{H}^1(T_C \cap e_{0,1}) = \frac{\Delta_R}{R},$$

and the only possible companion trace is

$$(27) \quad T_C \cap e_{5,0} = \left[\frac{k}{W}, R + \alpha \right] \quad \text{when } \Delta_L > 0, \quad \mathcal{H}^1(T_C \cap e_{5,0}) = \frac{[\Delta_L]_+}{W}.$$

In particular,

$$(28) \quad T_C \text{ is CE1} \iff \Delta_L \leq 0, \quad T_C \text{ is CE2} \iff \Delta_L > 0.$$

The six exits, measured from O toward V_i , are

$$(29) \quad \begin{aligned} d_0^C &= E - \alpha - \delta, \quad d_1^C = \frac{\delta}{R}, \quad d_2^C = \delta, \\ d_3^C &= \min \left\{ \frac{\alpha}{R}, \frac{\delta}{W} \right\}, \quad d_4^C = \alpha, \quad d_5^C = \frac{\alpha}{W}. \end{aligned}$$

Moreover,

$$(30) \quad T_C \cap \{M_0, \dots, M_5\} = \{M_0\}.$$

Solution.

Proof. Write $T_C \cap e_{0,1} = [s, t]$ with $0 \leq s < t \leq 1$. The two sides through these endpoints have a positive slope ratio. Reflect across the perpendicular bisector of $e_{0,1}$ if necessary, call that ratio $R \leq 1$, and set $W = 1 - R$. Direct equations for the three side lines give

$$F_1 = Rb + Wa - Rs, \quad F_2 = -b + Ra + t, \quad F_0 = Wb - a + E + Rs - t,$$

where the unit-side relation forces $E = \sqrt{1 - RW}$. If $R = 1$, then $F_0(O) = s - t < 0$, contrary to $O \in \text{int}(T_C)$, so $0 < R < 1$.

Set $\alpha := F_0(O)$ and $\delta := F_2(O)$. Then

$$t = W + \delta, \quad Rs = \eta + \alpha + \delta = k,$$

and substitution gives (25). The gradients are the three equilateral side-normal directions and

$$F_0 + F_1 + F_2 = E,$$

so these inequalities define the same unit equilateral triangle.

On $e_{0,1}$ one has $a = 0$; the active inequalities are $Rb \geq k$ and $b \leq W + \delta$. Hence

$$W + \delta - \frac{k}{R} = \frac{P - \alpha - W\delta}{R},$$

which proves (26). On $e_{5,0}$ one has $b = 0$; the active inequalities are $Wa \geq k$ and $a \leq R + \alpha$. Their signed interval length is

$$R + \alpha - \frac{k}{W} = \frac{P - R\alpha - \delta}{W},$$

proving (27).

The positive right trace gives

$$\delta < \frac{P}{W} = \frac{ER}{1+E} < \frac{R}{2}.$$

On $e_{1,2}$, parameterized by $b = 1 + q$, $a = q$, the slack F_2 is $\delta - R - Wq < 0$. Thus there is no positive trace there. theorem 2.3 says a C triangle has at most two positive traces and any two are adjacent, so $e_{5,0}$ is the only possible companion. Its signed length proves (28); equality $\Delta_L = 0$ is a point contact and therefore belongs to CE1.

Substituting the six ray parametrizations in (25) gives the exits. On r_1 the decreasing slack is $\delta - Rq$, on r_2 it is $\delta - q$, and on r_3 the two decreasing slacks are $\alpha - Rq$ and $\delta - Wq$. Reflection gives r_4, r_5 , while on r_0 the decreasing slack is $E - \alpha - \delta - q$. This proves (29).

Finally,

$$W(\alpha + \delta) \leq \alpha + W\delta < P,$$

and hence

$$E - \alpha - \delta > E - \frac{P}{W} = \frac{E(E - R)}{W} > \frac{1}{2},$$

where the last inequality is equivalent to $2E(E - R) - W = (E - R)^2 > 0$. Together with

$$\alpha < P < \min \left\{ \frac{R}{2}, \frac{W}{2} \right\}, \quad \delta < \frac{R}{2},$$

this shows $d_0^C > 1/2$ and $d_i^C < 1/2$ for $1 \leq i \leq 5$, proving (30). All three slacks at O are strictly positive for the closure of an original open C triangle, so then $\alpha, \delta > 0$. $\square$

The remaining proof-used consequences belong to the same feasible set.

Lemma A.5 (CE2 total slack). *In the CE2 range, with $T := \alpha + \delta$,*

$$(31) \quad T < \frac{\eta}{E}.$$

Proof. The CE2 inequalities are

$$\alpha + W\delta < P, \quad R\alpha + \delta < P.$$

Multiply them by W and R and add. Since $W + R^2 = W^2 + R = E^2$, this gives $E^2 T < P = E\eta$. $\square$

Corollary A.6 (Common center-boundary formula). *The total C-triangle boundary contribution is*

$$(32) \quad L_{\partial H}(T_C) := \mathcal{H}^1(T_C \cap \partial H) = \frac{\Delta_R}{R} + \frac{[\Delta_L]_+}{W}.$$

In CE1,

$$L_{\partial H}(T_C) \leq P \leq \frac{\sqrt{3}}{2} - \frac{3}{4},$$

whereas in CE2,

$$L_{\partial H}(T_C) = \frac{E}{1+E} - \frac{E^2(\alpha + \delta)}{RW} < \frac{1}{2}.$$

Proof. The first formula is the sum of (26) and (27). In CE1, $R\alpha + \delta \geq P$; multiplying by W and using $\alpha \geq RW\alpha$ yields $\alpha + W\delta \geq WP$. Thus $\Delta_R \leq RP$ and the contribution is at most P . Since $E \geq \sqrt{3}/2$ and $E(1-E)$ decreases on that interval, the stated cap follows.

In CE2, direct addition gives

$$\begin{aligned} L_{\partial H}(T_C) &= \frac{P - \alpha - W\delta}{R} + \frac{P - R\alpha - \delta}{W} \\ &= \frac{P - E^2(\alpha + \delta)}{RW}. \end{aligned}$$

Using $P/(RW) = E/(1+E)$ and positivity of $\alpha + \delta$ gives the strict half-unit cap. $\square$

Lemma A.7 (Adjacent-edge diameter transfer). *For $0 \leq q \leq 1$, define*

$$M_0(q) := \frac{-q + \sqrt{4 - 3q^2}}{2}.$$

If a unit equilateral triangle reaches parameters q, b on two adjacent boundary edges, then $b \leq M_0(q)$. The function $M_0(q)$ is strictly decreasing, and for $q > 0$,

$$M_0(q) < 1 - \frac{q}{2}, \quad 1 - M_0(q) > \frac{q}{2}.$$

Here the argument-free M_0 continues to denote the radial midpoint.

Proof. The squared distance between the two boundary points is $q^2 + qb + b^2 \leq 1$. Solving the quadratic for b gives the envelope. Moreover,

$$M'_0(q) = \frac{-1 - 3q/\sqrt{4 - 3q^2}}{2} < 0.$$

For $q > 0$, $\sqrt{4 - 3q^2} < 2$, which gives both strict linear comparisons. $\square$

Body consequence. The unique-midpoint identity supplies theorem 2.13. The trace formula and its CE1/CE2 caps supply the compact center rows used by the trace-length interface. The exact endpoints, exits, total slack, and diameter transfer supply later appendix calculations without exposing signed parameters in the body.

A.5. Module A.5: strict-handoff feasibility.

Exact interface. Assume the six open V triangles cover ∂H . The body selector is

$$\xi_i \in (1 - A_{i+1}, B_i), \quad a_i := 1 - \xi_{i-1} < A_i, \quad b_i := \xi_i < B_i.$$

Variables and feasible set. Put

$$\ell_i := 1 - A_{i+1}, \quad u_i := B_i.$$

The exact incident open traces are $X_i([0, u_i))$ and $X_i((\ell_i, 1])$. Boundary coverage is therefore equivalent to the six strict overlap conditions $\ell_i < u_i$.

Objective. Choose one $\xi_i \in (\ell_i, u_i)$ on every edge, while preserving the unique actual supercritical index when $N_+ = 1$ and producing at least two selected supercritical indices when $N_+ \geq 2$.

Exact result.

Lemma A.8 (Strict-handoff selector). *The six selector intervals are nonempty. Every simultaneous choice produces $0 < a_i < A_i$, $0 < b_i < B_i$, and*

$$X_{i-1}(1 - a_i), X_i(b_i) \in U_i, \quad a_i^2 + a_i b_i + b_i^2 < 1.$$

If $N_+ = 1$, every choice has exactly one index with $a_i + b_i > 1$, equal to the actual supercritical index. If $N_+ \geq 2$, a simultaneous choice exists with at least two indices satisfying $a_i + b_i > 1$.

Solution.

Proof. Because V_i is interior to a unit-diameter triangle while each other distinguished point is at distance at least one,

$$0 < A_i < 1, \quad 0 < B_i < 1.$$

By theorem A.3, only the two incident V triangles can cover a relative-interior point of $e_{i,i+1}$. Their traces cover the edge and are open at their outer endpoints, so they must overlap: $\ell_i < u_i$. Thus each selector interval is nonempty. Its choices put the two selected anchors in U_i ; their mutual squared distance is $a_i^2 + a_i b_i + b_i^2$ and is strictly less than the squared diameter one.

The actual and selected ascent tests are

$$(33) \quad A_i + B_i > 1 \iff \ell_{i-1} < u_i, \quad a_i + b_i > 1 \iff \xi_{i-1} < \xi_i.$$

If actual role i is nonsupercritical, then

$$\xi_i < u_i \leq \ell_{i-1} < \xi_{i-1},$$

so every selected version is strictly subcritical. Also

$$(34) \quad \sum_{i=0}^5 (a_i + b_i) = \sum_{i=0}^5 (1 - \xi_{i-1} + \xi_i) = 6.$$

When p is the sole actual supercritical index, the other five selected sums are below one, so (34) forces the p th sum above one. This proves the $N_+ = 1$ assertion.

Suppose next that p, q are nonadjacent actual supercritical indices. For each $r \in \{p, q\}$ choose

$$\ell_{r-1} < z_r < u_r,$$

and then

$$\xi_{r-1} \in (\ell_{r-1}, \min\{u_{r-1}, z_r\}), \quad \xi_r \in (\max\{\ell_r, z_r\}, u_r).$$

The two variable pairs are disjoint and give selected ascents at p and q . Any three indices on a six-cycle include a nonadjacent pair, so this settles $N_+ \geq 3$ as well.

It remains that exactly two actual supercritical indices, $p, p+1$, are adjacent. Along the other four nonsupercritical roles the successive inequalities $\ell_i < u_i \leq \ell_{i-1}$ imply $\ell_{p-1} < \ell_{p+1}$. Combining this with supercriticality at $p, p+1$ gives

$$\max\{\ell_{p-1}, \ell_p\} < \min\{u_p, u_{p+1}\}.$$

Choose ξ_p between these bounds and then choose $\xi_{p-1} < \xi_p < \xi_{p+1}$ in their respective overlap intervals. Equation (33) makes both selected roles supercritical. $\square$

Body consequence. This supplies every clause, including endpoint strictness, in theorem 2.11.

A.6. Module A.6: shared enclosure and local admissibility.

Exact interface. The enclosure number Λ , canonical anchor hull $K(a, b, c)$, admissible set $\mathcal{A}$, and radial envelope $c_{\max}$ are defined in theorems 5.1 and 5.3. This module evaluates those objects once for use by all later appendices.

Variables and feasible set. Let $\mathbf{R}$ be counterclockwise rotation through $2\pi/3$. For every nonempty compact $K \subset \mathbb{R}^2$ put

$$h_K(n) := \max_{x \in K} \langle x, n \rangle.$$

Translate the canonical hull's vertex V_0 to the origin and apply a rigid motion so that

$$u = \left(\frac{1}{2}, \frac{\sqrt{3}}{2} \right), \quad v = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2} \right).$$

Thus its generators become

$$O' = 0, \quad A = au, \quad B = bv, \quad C = c(u + v), \quad K = \text{conv}\{O', A, B, C\}.$$

For $a, b, c \in [0, 1]$ set

$$\begin{aligned} s &:= a + b, & d &:= a^2 + ab + b^2, & q &:= s^4 - s^2 + ab, \\ m &:= \min\{a, b\}, & M &:= \max\{a, b\}. \end{aligned}$$

and

$$\begin{aligned} F_L(a, b, c) &:= c^4 - c^2 + mc - m^2, \\ F_T(a, b, c) &:= (s^2 - 1)c^2 + Mc - M^2, \\ F_S(a, b, c) &:= (m^2 - 1)c^2 + (2mM^2 + M)c + M^4 - M^2. \end{aligned}$$

Objective. Evaluate $\Lambda(K)$ by support functions; determine the exact projection of $\mathcal{A}$ to the (a, b) -plane; and, on that projection, calculate the attained maximum $c_{\max}(a, b)$ with the geometric connected-component selectors retained.

Exact result.

Lemma A.9 (Shared equilateral-enclosure gauge). *For every nonempty compact K ,*

$$(35) \quad \Lambda(K) = \frac{2}{\sqrt{3}} \min_{\|n\|=1} \sum_{j=0}^2 h_K(\mathbf{R}^j n).$$

If K is a polygon with at least two distinct points, a minimizing normal may be chosen perpendicular to a hull edge.

Lemma A.10 (Exact local admissible set and radial envelope). *The exact projection is*

$$(36) \quad \{(a, b) \in [0, 1]^2 : \exists c \in [0, 1] (a, b, c) \in \mathcal{A}\} = \{(a, b) \in [0, 1]^2 : a^2 + ab + b^2 \leq 1\}.$$

On $[0, 1]^3$, the admissible set is

$$(37) \quad \mathcal{A} = \mathcal{A}_L \cup \mathcal{A}_T \cup \mathcal{A}_S,$$

where

$$\begin{aligned} \mathcal{A}_L &= \{d \leq 1, s \leq 1, q \leq 0, F_L \leq 0\}, \\ \mathcal{A}_T &= \{d \leq 1, s \leq 1, q \geq 0, F_T \leq 0, c \leq 2M\}, \\ \mathcal{A}_S &= \{d \leq 1, s > 1, F_S \leq 0, c \leq 1/2\}. \end{aligned}$$

It is compact, coordinatewise downward closed, and invariant under exchange of a, b . Every c -fiber therefore has an attained maximum, given by

$$(38) \quad c_{\max}(a, b) = \begin{cases} 1, & a = b = 0, \\ C_L(m), & s \leq 1, \ q \leq 0, \ (a, b) \neq (0, 0), \\ \frac{2M}{1 + \sqrt{4s^2 - 3}}, & s \leq 1, \ q > 0, \\ \frac{M(2mM + 1) - M\sqrt{4d - 3}}{2(1 - m^2)}, & s > 1, \end{cases}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

In particular, $c_{\max}(a, b) = c_{\max}(b, a)$.

Solution.

Proof of theorem A.9. Fix outward unit normals n, Rn, R^2n . Translating the three supporting lines to the support numbers of K gives the least containing triangle with those normals. Its altitude is the sum of the three support numbers, because $n + Rn + R^2n = 0$, and its side is that sum divided by $\sqrt{3}/2$. Minimizing over the unit circle gives (35); continuity of the support function gives the minimum.

For a polygon, partition the unit circle into angular cells on which the three support vertices are fixed. On each open cell the support sum has the form $A \cos \theta + B \sin \theta$ and its second derivative is its negative. An interior critical point is therefore a maximum unless the expression is constant, so a minimum occurs at a cell endpoint, where a support tie makes the normal perpendicular to a hull edge. $\square$

Proof of theorem A.10. Assume first $s > 0$ and $cs \leq ab$. If $a, b > 0$, then

$$\frac{c}{a} + \frac{c}{b} \leq 1,$$

so

$$C = \frac{c}{a}A + \frac{c}{b}B + \left(1 - \frac{c}{a} - \frac{c}{b}\right)O' \in \text{conv}\{O', A, B\}.$$

The zero-coordinate cases follow directly. The distance AB is $\sqrt{d}$. With

$$T := -bu - av,$$

the triangle $\text{conv}\{A, B, T\}$ is equilateral of side $\sqrt{d}$, and

$$O' = \frac{b^2}{d}A + \frac{a^2}{d}B + \frac{ab}{d}T.$$

Thus it contains K , while every enclosing triangle has diameter at least AB . Hence $\Lambda(K) = \sqrt{d}$. When $a = b = 0$, the hull is the segment $[O', C]$ and its enclosure infimum is c .

Now suppose $s > 0$ and $cs > ab$. The hull vertices have cyclic order O', B, C, A . By theorem A.9, it suffices to test the four edge normals. For $O'A$ the three supports give

$$L_{O'A} = b + \max\{a, c\},$$

and reflection gives $L_{BO'} = a + \max\{b, c\}$. For AC , put

$$R_a := \sqrt{a^2 - ac + c^2}, \quad n_0 := \frac{1}{2R_a}(\sqrt{3}a, 2c - a).$$

The supports in the directions n_0, Rn_0, R^2n_0 are

$$\frac{\sqrt{3}ac}{2R_a}, \quad \frac{\sqrt{3}a(a-c)_+}{2R_a}, \quad \frac{\sqrt{3}c \max\{b, (c-a)_+\}}{2R_a}.$$

Consequently

$$L_{AC} = \begin{cases} (a^2 + bc)/R_a, & a \geq c, \\ c(a+b)/R_a, & a < c \leq a+b, \\ c^2/R_a, & c > a+b, \end{cases}$$

and L_{CB} is obtained by interchanging a, b . Therefore

$$(39) \quad \Lambda(K) = \min\{L_{O'A}, L_{BO'}, L_{AC}, L_{CB}\}.$$

Assume by symmetry that $a = m \leq b = M$. Reading the active supports in (39) gives

range	active supports	unit-frontier equation
$s < 1, q < 0$	$\{O'\}, \{C\}, \{A, C\}$	$F_L = 0$
$s < 1, q > 0$	$\{O'\}, \{B, C\}, \{A\}$	$F_T = 0$
$s = 1$	preceding pattern with an O', B tie	$c = M$
$s > 1$	$\{B\}, \{B, C\}, \{A\}$	$F_S = 0.$

The inactive-support inequalities reduce, respectively, to $q \leq 0$, $q \geq 0$, and $s > 1$. Squaring the positive side-length equations gives exactly $F_L = 0, F_T = 0, F_S = 0$.

For the T cell the roots are

$$c_- = \frac{2M}{1 + \sqrt{4s^2 - 3}}, \quad c_+ = \frac{2M}{1 - \sqrt{4s^2 - 3}}.$$

The squared inequality contains $0 \leq c \leq c_-$ and a formal high component $c \geq c_+$. Since

$$c_- \leq 2M \leq c_+,$$

with equality throughout only at root coalescence, $c \leq 2M$ selects exactly the component connected to $c = 0$.

For the S cell,

$$F_S(a, b, 0) = M^2(M^2 - 1) < 0, \quad F_S(a, b, M) = M^2(s^2 - 1) > 0.$$

Moreover, $4F_S(a, b, 1/2)$ is increasing in m , and at $m = 1 - M$ equals $M^2(2M - 1)^2$. Since $s > 1$ gives $m > 1 - M$, one has $F_S(a, b, 1/2) > 0$. The quadratic opens downward, so $1/2$ lies strictly between its roots and $c \leq 1/2$ selects the smaller-root component containing $c = 0$.

For the L cell, the fiber issuing from $c = 0$ reaches one selected root in $[\sqrt{3}/2, 1]$; uniqueness there follows from $4c^3 - 2c + m > 0$. At $q = 0$ the L and T formulas meet and both give $c = s$. This proves the three exact cells and solving their selected frontier equations gives (38).

At $c = 0$, the preceding triangular calculation shows admissibility exactly when $d \leq 1$; if $d > 1$, the two anchors A, B are more than unit distance apart. This proves (36). The support formula makes $\Lambda(K(a, b, c))$ continuous, hence $\mathcal{A}$ is compact. Reflection exchanges a, b . Reducing any coordinate moves its anchor toward the already present point O' , so the new hull lies in the old hull; this proves coordinatewise downward closure. Its nonempty c -fibers are compact initial intervals, proving attainment of $c_{\max}$. $\square$

Body consequence. The support formula is the Appendix A supplier for theorem 5.1. The projection, component selectors, attainment, symmetry, and formula (38) supply theorem 5.3 and every later use of $c_{\max}$.

A.7. Module A.7: midpoint caliper optimization.

Exact interface. The geometric consequence required by theorem 2.12 is the implication

$$\{0, au, bv, (u+v)/2\} \subseteq T,$$

$$T \text{ a closed unit equilateral triangle} \implies a + b \leq 1.$$

Variables and feasible set. Let R and J denote counterclockwise rotations through $2\pi/3$ and $\pi/2$, respectively. For a nonempty finite set $S \subset \mathbb{R}^2$ and a vector N , put

$$(40) \quad W_S(N) := \sum_{k=0}^2 \max_{z \in S} \langle z, R^k N \rangle.$$

For the midpoint instance take

$$u = \left(\frac{1}{2}, \frac{\sqrt{3}}{2} \right), \quad v = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2} \right), \quad S = \left\{ 0, au, bv, \frac{u+v}{2} \right\}, \quad a, b \geq 0.$$

Objective. Reduce the least-equilateral-enclosure test to finitely many hull-edge normals and evaluate all four edge calipers when $a + b > 1$.

Exact result.

Theorem A.11 (Finite caliper certificate). *If a finite set S has at least two distinct points, then S lies in a closed unit equilateral triangle if and only if there are distinct $p, q \in S$ and $\varepsilon \in \{-1, 1\}$ such that*

$$(41) \quad W_S(\varepsilon J(q-p)) \leq \frac{\sqrt{3}}{2} \|p-q\|.$$

For $S_x = \{O, A, B, x\}$, every clause is polynomial of degree at most four in the coordinates of A, B, x , and at most $12 \cdot 4^3$ clauses are required. A singleton lies in equilateral triangles of arbitrarily small positive side.

Lemma A.12 (Four-point midpoint caliper). *If a closed unit equilateral triangle contains $0, au, bv, (u+v)/2$, then $a + b \leq 1$. If all four points have a positive containment margin, then $a + b < 1$.*

Solution.

Proof of theorem A.11. Apply theorem A.9 to the polygonal hull of S . A minimizing unit normal is perpendicular to a hull edge $[p, q]$ and, after cyclically relabeling the three normal directions, equals $\varepsilon J(q-p)/\|p-q\|$. Homogeneity of W_S gives (41). A nondegenerate segment is checked by its two normal directions.

For the polynomial form, fix (p, q, ε) , put $N = \varepsilon J(q-p)$ and $N_k = R^k N$, and choose support labels $z_0, z_1, z_2 \in S_x$. The support cell is

$$(42) \quad \langle z_k, N_k \rangle \geq \langle z, N_k \rangle \quad (z \in S_x, k = 0, 1, 2),$$

with $\|p-q\|^2 > 0$. Since $W_S(N) \geq 0$, the remaining condition is

$$(43) \quad 4 \left(\sum_{k=0}^2 \langle z_k, N_k \rangle \right)^2 \leq 3 \|p-q\|^2.$$

The points are affine-linear in the parameters, so these clauses have degree at most four. There are six unordered pairs, two signs, and 4^3 support triples. $\square$

Proof of theorem A.12. Assume $a + b > 1$ and set

$$K := \text{conv} \left\{ 0, au, \frac{u+v}{2}, bv \right\}.$$

For an outward normal to the edge from 0 to au , direct projection gives

$$S_{0A} = \max \left\{ \frac{\sqrt{3}}{4}, \frac{\sqrt{3}a}{2} \right\} + \frac{\sqrt{3}b}{2} > \frac{\sqrt{3}}{2}.$$

Indeed, for $a \geq 1/2$ this equals $(\sqrt{3}/2)(a + b)$, while for $a < 1/2$ one has $b > 1/2$. The edge through bv is symmetric.

For the edge from au to $(u + v)/2$, put

$$D_a := \sqrt{4a^2 - 2a + 1}.$$

Projection gives

$$S_{AM} = \begin{cases} \frac{\sqrt{3}(a+b)}{2D_a}, & a \leq 1/2, \\ \frac{\sqrt{3}(2a^2+b)}{2D_a}, & a > 1/2. \end{cases}$$

In the first case $D_a \leq 1$ and $a + b > 1$. In the second, $b > 1 - a$ and

$$(2a^2 - a + 1)^2 - D_a^2 = a^2(2a - 1)^2 \geq 0.$$

Thus $S_{AM} > \sqrt{3}/2$ in both cases. The fourth edge is symmetric. By theorem A.11, every enclosing equilateral triangle has side strictly greater than one, a contradiction.

If $a + b = 1$ while all four points have positive margin, both boundary demands may be increased slightly and remain contained. Their new sum is greater than one, contradicting the closed conclusion just proved. $\square$

Body consequence. The second result, with the local V-triangle chart transported by a hexagon symmetry, supplies theorem 2.12. The finite certificate remains available to every later finite-witness module.

APPENDIX B. TRACE-LENGTH OPTIMIZATION

This appendix solves the trace maximization and perimeter-budget problems whose compact outputs are used in Section 3. Each module starts from the exact body trace functional in Definition 3.1.

B.1. Skeleton-trace optimization.

Exact interface. The target is the exact trace functional L_S from Definition 3.1; the compact body consequence is Theorem 3.4.

Variables and feasible set. The C-triangle variables range over the signed CE1/CE2 domain of theorem A.4. A V triangle ranges over the closed unit equilateral triangles containing its distinguished vertex, with the displayed reach and radial-incidence restrictions in the lemmas below.

Objective. For each indicated class, maximize $L_S(T)$.

Exact result. On the four feasible classes below, the exact sufficient caps are, respectively,

$$L_S(T_C) < \frac{3}{2}, \quad L_S(T_i) < \frac{3}{2}, \quad L_S(T_i) \leq 2, \quad L_S(T_i) < \frac{3}{2}.$$

Solution. The following four lemmas give the complete calculations for those caps.

Lemma B.1 (Center skeleton cap). *If T_C is a CE1 or CE2 C triangle and contains exactly one radial midpoint, then*

$$L_S(T_C) < \frac{3}{2}.$$

Proof. Normalize the unique midpoint as M_0 and the positive boundary trace to $e_{0,1}$. Use the signed variables of theorem A.4; in particular

$$W = 1 - R, \quad E = \sqrt{1 - RW}, \quad \eta = 1 - E, \quad P = E(1 - E).$$

The positive trace condition is

$$(44) \quad \alpha + W\delta < P.$$

The two possible boundary contributions are

$$\ell_{01} = W + \delta - \frac{\eta + \alpha + \delta}{R}, \quad \ell_{50} = \frac{[P - (R\alpha + \delta)]_+}{W}.$$

The six radial contributions are bounded by

i	0	1	2	3	4	5
$\mathcal{H}^1(T_C \cap r_i)$	$E - \alpha - \delta$	δ/R	δ	$\min\{\alpha/R, \delta/W\}$	α	α/W

Adding gives

$$L_S(T_C) \leq K_R + \mathcal{E}(\alpha, \delta),$$

where

$$K_R = W + E - \frac{\eta}{R}$$

and

$$\mathcal{E}(\alpha, \delta) = -\frac{\alpha}{R} + \frac{\alpha}{W} + \delta + \min\left\{\frac{\alpha}{R}, \frac{\delta}{W}\right\} + \frac{[P - (R\alpha + \delta)]_+}{W}.$$

Splitting only according to the minimum and the positive part gives

$$(45) \quad \mathcal{E}(\alpha, \delta) \leq \frac{P}{W}.$$

For example, if $\alpha/R \leq \delta/W$, the expression without the positive part is $\alpha/W + \delta$; when the positive part is active, its excess over P/W is $\alpha - R\delta/W \leq 0$. The reflected order gives excess $\delta - W\alpha/R \leq 0$. If the positive part vanishes, (44) gives the same bound strictly.

Put $A = 1 + R - R^2$. Then

$$\frac{3}{2} - K_R - \frac{P}{W} = \frac{1 + A - 2EA}{2RW}.$$

Both sides of $1 + A > 2EA$ are positive, and

$$(1 + A)^2 - 4A^2E^2 = R^2W^2(5 + 4R - 4R^2) > 0.$$

Thus $K_R + P/W < 3/2$. □

Lemma B.2 (Positive-support skeleton cap). *Let T_i be the closure of an original open V triangle, so $V_i \in \text{int}(T_i)$. If*

$$\mathcal{H}^1(T_i \cap r_j) > 0 \quad \text{for some } j \in \{i-1, i+1\},$$

then

$$L_S(T_i) < \frac{3}{2}.$$

Proof. Normalize to $i = 0$ and translate the hexagon by V_0 :

$$\tilde{H} = V_0 + H, \quad \tilde{O} = V_0, \quad W_j = V_0 + V_j.$$

Then $W_3 = O$, $W_2 = V_1$, and $W_4 = V_5$. The five original skeleton components on which a unit triangle containing V_0 can have positive-length trace are

$$[V_0, V_1], [V_0, V_5], [O, V_1], [O, V_0], [O, V_5].$$

All five belong to the skeleton $\tilde{S}$ of $\tilde{H}$. Since $V_0 = \tilde{O}$ is interior to T_0 , the triangle is a legitimate center triangle for $\tilde{H}$. Its positive trace on r_1 or r_5 is a positive boundary-edge trace of $\tilde{H}$, so it is CE1 or CE2 by Proposition 2.3. Proposition 2.13 and Lemma B.1 therefore give

$$L_{\tilde{S}}(T_0) < \frac{3}{2}.$$

A nonlocal component of S is more than unit distance from the interior point V_0 , apart from zero-measure endpoints. Hence $L_S(T_0) \leq L_{\tilde{S}}(T_0)$. $\square$

Lemma B.3 (No-support skeleton cap). *If a V triangle has $A_i + B_i \leq 1$ and*

$$\mathcal{H}^1(T_i \cap r_{i-1}) = \mathcal{H}^1(T_i \cap r_{i+1}) = 0,$$

then

$$L_S(T_i) \leq 2.$$

Proof. Its boundary contribution is $A_i + B_i \leq 1$ by (48). For $P = sV_j \in r_j \setminus \{O\}$,

$$\|V_i - P\|^2 = 1 + s^2 - 2s \cos \frac{(j-i)\pi}{3} > 1$$

when $j \notin \{i-1, i, i+1\}$. The traces on r_{i-1} and r_{i+1} have zero length by hypothesis, so only r_i can contribute positive length. This adds at most one. $\square$

For the remaining cap we use one specialized part of the exact local equilateral enclosure criterion. In local directions u, v meeting at 120° , suppose a closed unit triangle contains

$$0, \quad au, \quad bv, \quad c(u+v),$$

with $0 \leq a \leq b \leq 1$ and $a + b > 1$. The support-line calculation gives the selected supercritical cell

$$(46) \quad \begin{aligned} a^2 + ab + b^2 &\leq 1, & c &\leq \frac{1}{2}, \\ (a^2 - 1)c^2 + (2ab^2 + b)c + (b^4 - b^2) &\leq 0, \end{aligned}$$

where c lies on the connected component containing $c = 0$. To see the selector directly, the quadratic is negative at $c = 0$, positive at $c = 1/2$, and opens downward. For the middle assertion, four times its value at $1/2$ is increasing in a and, at $a = 1 - b$, equals $b^2(2b - 1)^2$; it is therefore positive when $a + b > 1$. Hence $c \leq 1/2$ selects its smaller-root component. Formula (46) follows by fixing in the support

sum the three active contacts $\{B\}$, $\{B, c(u+v)\}$, and $\{A\}$, and squaring the positive unit-side equation.

Lemma B.4 (Supercritical skeleton cap). *If $A_i + B_i > 1$, then*

$$L_S(T_i) < \frac{3}{2}.$$

Proof. By Proposition 2.6, a V triangle is Vd0, Vd1, Vd2, or T3-like. Proposition B.5 makes every Vd1/Vd2 V triangle nonsupercritical, and Lemma 2.10 does the same for T3-like V triangles. Hence a supercritical triangle is Vd0 and, by definition, $\mathcal{H}^1(T_i \cap r_{i-1}) = \mathcal{H}^1(T_i \cap r_{i+1}) = 0$. Thus, after normalizing and writing $a = A_i$, $b = B_i$, the skeleton trace has length $a + b + c$, where c is the reach on r_i . Reflect so that $a \leq b$, and put

$$s = a + b, \quad \delta = b - a, \quad c_0 = \frac{3}{2} - s.$$

The diameter constraint gives

$$1 < s \leq \frac{2}{\sqrt{3}}, \quad 0 \leq \delta \leq \sqrt{4 - 3s^2}.$$

Let $P(c)$ denote the quadratic in (46). Substitution and multiplication by 16 give

$$\begin{aligned} 16P(c_0) = Q(s, \delta) &= \delta^4 + 8\delta^3s - 6\delta^3 + 14\delta^2s^2 - 18\delta^2s + 5\delta^2 \\ &\quad - 8\delta s^3 + 30\delta s^2 - 34\delta s + 12\delta \\ &\quad + s^4 - 6s^3 - 19s^2 + 60s - 36. \end{aligned}$$

Its derivative factors as

$$\frac{\partial Q}{\partial \delta} = 2(2\delta + 4s - 3)(\delta^2 + \delta(4s - 3) + (s - 1)(2 - s)) \geq 0.$$

Therefore

$$Q(s, \delta) \geq Q(s, 0) = (s - 1)(s^3 - 5s^2 - 24s + 36) > 0.$$

Indeed, the cubic is decreasing on $[1, 2/\sqrt{3}]$ and at its right endpoint equals $88/3 - 136\sqrt{3}/9 > 0$. Hence $P(c_0) > 0$. Since (46) selects the smaller-root component, $P(c) \leq 0$ forces $c < c_0$. Thus $a + b + c < 3/2$. $\square$

Body consequence. These four optima supply the four skeleton rows used in Theorem 3.4.

B.2. Vd1/Vd2 corner optimization.

Exact interface. The public outputs are the Vd1/Vd2 boundary row of Theorem 3.2 and the branch in Proposition 3.5.

Variables and feasible set. Let T be the closure of an original Vd1 or Vd2 triangle at V_0 . Use the temporary coordinates

$$X = V_0 + x(V_5 - V_0) + y(V_1 - V_0).$$

Thus $V_0 = (0, 0)$, $V_5 = (1, 0)$, $V_1 = (0, 1)$, $O = (1, 1)$, and $\|(x, y)\|^2 = x^2 + y^2 - xy$. Objective. Determine the exact half-plane family and maximize the boundary sum, first without and then with an adjacent-midpoint containment constraint.

Exact result. The exact half-plane and ray descriptions are (B.5) and (B.5). Their optimized consequences are

$$A + B < \frac{1}{2}, \quad M_{i-1} \in T_i \text{ or } M_{i+1} \in T_i \implies A_i + B_i < \frac{1}{3}.$$

Solution.

Proposition B.5 (Vd1/Vd2 corner normal form). *Let T be the closure of an original Vd1 or Vd2 triangle at V_0 , and let A, B be its exact reaches toward V_5, V_1 . There is a unique $t > 0$ such that, with $d = \sqrt{t^2 + t + 1}$,*

$$T = \left\{ (x, y) : \begin{array}{l} x - (t+1)y \leq A, \\ ty - (t+1)x \leq tB, \\ tx + y \leq d - A - tB \end{array} \right\}.$$

The raw traces, in coordinates measured from the indicated outer vertex, are

$$\begin{array}{l|l} r_0 & 0 \leq s \leq \frac{d - A - tB}{t+1} \\ r_1 & \frac{t(1-B)}{t+1} \leq s \leq \frac{d - A - tB - 1}{t} \\ r_5 & \frac{1-A}{t+1} \leq s \leq d - A - tB - t. \end{array}$$

Intersecting with $[0, 1]$ gives the actual traces. Moreover,

$$(47) \quad A + B < \frac{1}{2}.$$

Proof. A Vd1/Vd2 triangle has one vertex W outside H and two vertices U, Z in H . Since V_0 is an interior convex combination of them, both coordinates of W are negative. The two axis endpoints $(A, 0), (0, B)$ lie on distinct sides incident with W . Put $p = U - W$, $q = Z - W$. Both vectors have positive coordinates and satisfy

$$\|p\| = \|q\| = \|p - q\| = 1.$$

The order in which the two sides cross the positive axes selects the 60° rotation

$$q = (p_x - p_y, p_x), \quad p_x > p_y > 0.$$

Writing $t = (p_x - p_y)/p_y$ gives uniquely

$$p = \frac{(t+1, 1)}{d}, \quad q = \frac{(t, t+1)}{d}.$$

The two side lines through the axis endpoints and the parallel third side are exactly those in (B.5). Substitution of $(s, s), (s, 1), (1, s)$ gives (B.5).

If the r_1 trace has positive length, its lower endpoint is smaller than its upper endpoint. Clearing denominators gives

$$(t+1)A + tB < (t+1)(d-1) - t^2.$$

Since the left side is at least $t(A+B)$,

$$A + B < \frac{(t+1)(d-1) - t^2}{t} < \frac{1}{2}.$$

For the last inequality, both sides are positive and

$$\left(t^2 + \frac{3t}{2} + 1\right)^2 - (t+1)^2(t^2 + t + 1) = \frac{t^2}{4} > 0.$$

Reflection proves the same conclusion when the supported arm is r_5 . $\square$

Lemma B.6 (Vd2 adjacent-midpoint cap). *If an open role U_i has Vd2 closure $T_i = \overline{U_i}$ and T_i contains M_{i-1} or M_{i+1} , then its exact boundary reaches satisfy*

$$A_i + B_i < \frac{1}{3}.$$

Proof. Normalize to $i = 0$ and write $A = A_i$, $B = B_i$. In Proposition B.5, reflection sends (t, A, B, M_1, M_5) to $(1/t, B, A, M_5, M_1)$, so assume $t \geq 1$ and put $U = A + tB$.

If $M_5 = (1, 1/2)$ is contained, the third half-plane in (B.5) gives

$$A + B \leq U \leq d - t - \frac{1}{2} \leq \sqrt{3} - \frac{3}{2} < \frac{1}{3};$$

the middle expression decreases for $t \geq 1$.

If $M_1 = (1/2, 1)$ is contained, the second and third half-planes give

$$B \geq \frac{t-1}{2t}, \quad A + B \leq S(t) := d - t - \frac{1}{2t}.$$

Positive-length contact with r_5 gives, by (B.5),

$$A + B < R(t) := \frac{2(t+1)d - 3t^2 - t - 2}{2t}.$$

Direct differentiation shows that S is increasing and R decreasing on $[1, \infty)$. For example, the sign calculation for S' reduces to

$$(2t+1)^2 t^4 - (t^2 + t + 1)(2t^2 - 1)^2 = (t+1)(t^3 + 3t^2 - 1) > 0,$$

while $R' < 0$ follows from $d > t + 1/2$. Splitting at $t = 4/3$ yields

$$\begin{aligned} 1 \leq t \leq \frac{4}{3} : \quad A + B &\leq S(4/3) = \frac{8\sqrt{37} - 41}{24} < \frac{1}{3}, \\ t \geq \frac{4}{3} : \quad A + B &< R(4/3) = \frac{7\sqrt{37} - 39}{12} < \frac{1}{3}. \end{aligned}$$

The final inequalities follow respectively from $8\sqrt{37} < 49$ and $7\sqrt{37} < 43$. $\square$

Body consequence. The first result supplies the Vd1/Vd2 row of Theorem 3.2; the second supplies Proposition 3.5.

B.3. Boundary-trace optimization.

Exact interface. The objective is the body functional $L_{\partial H}$, and the public result is Theorem 3.2.

Variables and feasible set. The center ranges over the three exact classes determined by $N_E(T_C)$. Each V triangle ranges over its exact normalized V type and its actual maximal reaches (A_i, B_i) .

Objective. Maximize $L_{\partial H}$ separately on every row of the public table.

Exact result. The optimum in each feasible class is the corresponding entry of (3).

Solution. The signed center normal form gives both positive center traces from one formula, and the remaining rows follow from the exact V-triangle constraints.

Lemma B.7 (Boundary-trace table calculation). *For closures of original open triangles, the following bounds hold:*

<i>triangle</i>	<i>hypothesis</i>	$L_{\partial H}$
T_C	CE0	0
T_C	CE1	$\leq \frac{\sqrt{3}}{2} - \frac{3}{4}$
T_C	CE2	$< \frac{1}{2}$
T_i	Vd0, $A_i + B_i \leq 1$	≤ 1
T_i	Vd0, $A_i + B_i > 1$	$\leq \frac{2}{\sqrt{3}}$
T_i	Vd1/Vd2	$< \frac{1}{2}$
T_i	T3-like	< 1 .

Proof. For every V triangle,

$$(48) \quad L_{\partial H}(T_i) = A_i + B_i.$$

Indeed, the two incident traces are initial intervals of these lengths. A relative-interior point of any nonincident edge is more than unit distance from V_i , whereas V_i is interior to the diameter-one triangle T_i . This excludes any other positive-length boundary trace.

The CE0 C-triangle row is the definition. The other two C-triangle rows are the class specializations of the common signed formula in theorem A.6.

For a nonsupercritical Vd0 V triangle, (48) is enough. For a supercritical V triangle, its incident endpoints are at squared distance $A_i^2 + A_i B_i + B_i^2 \leq 1$, and hence

$$\frac{3}{4}(A_i + B_i)^2 \leq 1.$$

This proves the $2/\sqrt{3}$ cap. The Vd1/Vd2 V triangle is (47). Finally, the Type II calculation in Proposition 2.7 gives for an original T3-like triangle

$$A_i + B_i = z - \alpha < z < 1$$

because $0 < t < 1$ and $\alpha > 0$. □

Body consequence. The seven exact maxima are the rows of Theorem 3.2.

B.4. Perimeter-budget optimization.

Exact interface. The center and V-triangle boundary contributions are those of Theorem 3.2. The public routing consequence is Proposition 3.6.

Variables and feasible set. The signed center variables range over the CE1 or CE2 cells of theorem A.4. The remaining variables are the numbers and strict caps of the classified V triangles in the stated routing branch.

Objective. Maximize the total perimeter trace subject to those caps and compare it with $\mathcal{H}^1(\partial H) = 6$.

Exact result. The exact deficit, small-slack, endpoint-slack, and routing conclusions are the statements of the four results below.

Solution. Their proofs perform the signed substitutions and strict endpoint checks.

Lemma B.8 (Master perimeter deficit). *Put*

$$\theta = \frac{2}{\sqrt{3}} - 1.$$

Suppose a branch has a center contribution L_C , n supercritical Vd0 V triangles, $m \geq 1$ distinguished nonsupercritical V triangles with strict boundary caps $q_1, \dots, q_m < 1$, and $6 - n - m$ remaining V triangles whose boundary contributions are at most 1. If

$$(49) \quad L_C + n\theta \leq \sum_{j=1}^m (1 - q_j),$$

then the seven triangles do not cover ∂H .

Proof. The supercritical Vd0 cap is $2/\sqrt{3}$. Hence the total boundary contribution is strictly less than

$$L_C + n\frac{2}{\sqrt{3}} + \sum_{j=1}^m q_j + (6 - n - m) = 6 + L_C + n\theta - \sum_{j=1}^m (1 - q_j) \leq 6.$$

The strictness follows from $m \geq 1$ and the strict distinguished-V triangle caps. A cover of the perimeter would give the opposite weak inequality by subadditivity. $\square$

Lemma B.9 (Small center slacks in the one-Vd1/Vd2 branch). *Assume there is exactly one supercritical Vd0 V triangle, exactly one Vd1/Vd2 V triangle, and four nonsupercritical Vd0 V triangles. Then every candidate surviving the perimeter budget is CE2 and satisfies*

$$(50) \quad \boxed{\alpha + \delta < \frac{1}{24}, \quad \alpha + \delta < \frac{\min\{R, W\}}{6}.}$$

Proof. The Vd1/Vd2 boundary cap is strict and equals $1/2$. If the perimeter were covered, Lemma B.8 would force

$$L_C + \theta > \frac{1}{2}.$$

Put

$$\kappa = \frac{1}{2} - \theta = \frac{3}{2} - \frac{2}{\sqrt{3}}.$$

Thus $L_C > \kappa$. The CE1 cap in theorem A.6 gives

$$L_C + \theta \leq \left(\frac{\sqrt{3}}{2} - \frac{3}{4} \right) + \left(\frac{2}{\sqrt{3}} - 1 \right) < \frac{1}{2},$$

so every survivor is CE2.

Set $T = \alpha + \delta$. The CE2 formula gives

$$T < M(E) := \frac{RW}{E^2} \left(\frac{E}{1+E} - \kappa \right).$$

Since $RW = 1 - E^2$,

$$M(E) = \frac{1-E}{E^2} (E - \kappa(1+E)),$$

and

$$M'(E) = -\frac{E - 2\kappa}{E^3} < 0 \quad \left(\frac{\sqrt{3}}{2} \leq E < 1 \right).$$

Here $2\kappa < \sqrt{3}/2$, because it is equivalent to $6\sqrt{3} < 11$, whose square is $108 < 121$. Therefore

$$T < M\left(\frac{\sqrt{3}}{2}\right) = \frac{8\sqrt{3}}{9} - \frac{3}{2} < \frac{1}{24}.$$

The last inequality is equivalent to $64\sqrt{3} < 111$, and follows after squaring from $12288 < 12321$.

For the second estimate, use $E/(1+E) < 1/2$:

$$T < \frac{RW}{E^2} \left(\frac{1}{2} - \kappa \right) = \frac{RW}{E^2} \theta.$$

Since

$$E^2 = W + R^2 = R + W^2,$$

one has $RW/E^2 \leq R$ and $RW/E^2 \leq W$. Also $\theta < 1/6$, equivalently $12 < 7\sqrt{3}$, whose square is $144 < 147$. This proves (50). $\square$

Lemma B.10 (CE2 initial endpoints dominate the total slack). *Assume $\Delta_R > 0$ and $\Delta_L > 0$. Put*

$$T = \alpha + \delta, \quad x = \frac{\eta + T}{W}, \quad y = \frac{\eta + T}{R}.$$

Then

$$(51) \quad \boxed{T < \frac{1}{2} \min\{x, y\}.}$$

Proof. equation (31) gives $T < \eta/E$. Moreover

$$E^2 - (2R - 1)^2 = 3RW > 0,$$

so $|2R - 1| < E$ and therefore $|2R - 1|T < \eta$. Hence

$$x - 2T = \frac{\eta - (1 - 2R)T}{W} > 0,$$

and

$$y - 2T = \frac{\eta - (2R - 1)T}{R} > 0.$$

This is (51). $\square$

Corollary B.11 (Length branches closed by the common budgets). *The following branches are impossible.*

- (1) $N_{\text{gap}} = 0$, $N_+ = 1$, and at least one Vd1/Vd2 V triangle;
- (2) $N_+ = 0$ and at least one Vd1/Vd2 V triangle, for either CE1 or CE2;
- (3) CE1, $N_+ = 1$, and at least one Vd1/Vd2 V triangle;
- (4) CE2, $N_+ = 1$, and at least two Vd1/Vd2 V triangles;
- (5) CE2, $N_+ = 1$, with a Vd2 triangle T_i containing M_{i-1} or M_{i+1} ;
- (6) CE1 or CE2 with $N_+ + N_{\text{sp}} \geq 3$;
- (7) CE1 or CE2 with $N_+ \geq 2$.

Proof. For item 1, Lemma 2.9 says that the six open V triangles cover ∂H . The unique supercritical triangle is Vd0, so the supercritical cap, one strict Vd1/Vd2 cap, and four unit caps give

$$\frac{2}{\sqrt{3}} + \frac{1}{2} + 4 < 6.$$

For item 2, the center contribution is strictly less than $1/2$ and the chosen Vd1/Vd2 V triangle has strict cap $1/2$; the other five V triangles contribute at most 1 each. This is Lemma B.8 with $n = 0$ and $q_1 = 1/2$.

For item 3, the unique supercritical V triangle is Vd0, while the CE1 center cap, the Vd1/Vd2 cap, and the four remaining unit caps give

$$\left(\frac{\sqrt{3}}{2} - \frac{3}{4}\right) + \frac{1}{2} + \frac{2}{\sqrt{3}} + 4 < 6.$$

For item 4, the corresponding CE2 sum is strictly less than

$$\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{2}{\sqrt{3}} + 3 < 6.$$

For item 5, the M_{i-1}/M_{i+1} Vd2 cap is $1/3$, and the total is strictly less than

$$\frac{1}{2} + \frac{1}{3} + \frac{2}{\sqrt{3}} + 4 < 6.$$

Item 6 is Theorem 3.4.

For item 7, Lemma 3.3 supplies a positive-support triangle distinct from two supercritical triangles. Hence $N_+ + N_{\text{sp}} \geq 3$, and Theorem 3.4 applies. $\square$

Lemma B.12 (Exact trace-routing arithmetic). *Every configuration satisfying one of the six hypotheses in Proposition 3.6 has total available trace strictly smaller than the target trace required in that proposition.*

Proof. For item (1), Lemma 2.9 says that the six open V triangles cover ∂H . Their incident open traces overlap on every edge, so

$$1 < B_i + A_{i+1} \quad (0 \leq i \leq 5).$$

Summing gives $6 < \sum_i (A_i + B_i)$, whereas $N_+ = 0$ gives the opposite weak inequality.

Item (2) is item (1) of Corollary B.11.

Item (3) is Theorem 3.4. Items (4) and (5) are items (2) and (3), respectively, of Corollary B.11. Item (6) is item (5) of that corollary, using Lemma B.6. $\square$

Body consequence. The preceding solved budget problem supplies Proposition 3.6, including its center-independent $N_{\text{gap}} = 0$ rows.

APPENDIX C. AREA-LOSS OPTIMIZATION

This appendix solves the local and cyclic variational problems defined in Section 4. The strict handoff selector is the one fixed in Proposition 2.11; it is not redefined here.

C.1. Local square-loss optimization.

Exact interface. The feasible family $\mathcal{T}(a, b)$ and objective $\mathcal{L}$ are those of Definition 4.1; the public output is Theorem 4.2.

Variables and feasible set. Fix $(a, b) \in [0, 1]^2$ with $\mathcal{T}(a, b) \neq \emptyset$. Normalize the distinguished vertex to V_0 and range over every orientation and translation of a closed unit equilateral triangle in $\mathcal{T}(a, b)$.

Objective. Minimize $\mathcal{L}(T)$ over $T \in \mathcal{T}(a, b)$.

Exact result. The optimized lower bounds are (4) and (5).

Solution.

Lemma C.1 (Local wedge reduction). *After rotating indices to $i = 0$, write*

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0), \quad W = \{(x, y) : x \geq 0, y \geq 0\}.$$

If a closed unit equilateral triangle T contains V_0 , then

$$T \cap H = T \cap W.$$

Normalized Euclidean area is twice the corresponding (x, y) -coordinate area.

Proof. The two basis vectors have length one and angle 120° , so

$$\|(x, y)\|^2 = x^2 + y^2 - xy, \quad |\det(V_1 - V_0, V_5 - V_0)| = \frac{\sqrt{3}}{2}.$$

The hexagon is

$$H = \{0 \leq x, y \leq 2, |x - y| \leq 1\} \subset W.$$

Conversely, if $(x, y) \in T \cap W$, then the diameter-one condition gives $x^2 + y^2 - xy \leq 1$.

Hence

$$|x - y|^2 \leq x^2 + y^2 - xy \leq 1, \quad \frac{3}{4}x^2 \leq x^2 + y^2 - xy,$$

and the same estimate holds for y . Thus $|x - y| \leq 1$ and $0 \leq x, y \leq 2/\sqrt{3} < 2$, proving $(x, y) \in H$. The determinant identity proves the area assertion. $\square$

Lemma C.2 (Solved local square-loss problem). *For every feasible pair $0 \leq a, b \leq 1$ and every realizing closed unit equilateral triangle T ,*

$$(52) \quad \mathcal{L}(T) \geq \min\{a, b\}^2.$$

If $a + b > 1$, then

$$(53) \quad \mathcal{L}(T) \geq \max\{a, b\}^2.$$

Consequently, taking the infimum over $T \in \mathcal{T}(a, b)$ gives the same inequalities for $\mathcal{L}(a, b)$.

Proof. In the coordinates of Lemma C.1, physical rotation through 60° sends (x, y) to $(x - y, x)$. Put

$$0 \leq t \leq 1, \quad D = 1 - t + t^2, \quad z = \sqrt{D}.$$

After choosing one of the six directed sides, every orientation has one of the following two forms:

	first side vector	second side vector
I	$z^{-1}(1, t)$	$z^{-1}(1 - t, 1)$
II	$z^{-1}(t, -(1 - t))$	$z^{-1}(1, t)$.

Indeed these forms cover, respectively, the physical direction sectors $[120^\circ, 180^\circ]$ and $[60^\circ, 120^\circ]$; their endpoints cover the common boundary orientations.

For Type I introduce

$$U = \alpha + y - tx, \quad V = \beta + x - (1 - t)y.$$

Then T is the simplex $U, V \geq 0, U + V \leq z$. Containment of $(0, 0), (0, a), (b, 0)$ gives

$$(55) \quad \alpha \geq tb, \quad \beta \geq (1-t)a, \quad \alpha + \beta + (1-t)b \leq z, \quad \alpha + \beta + ta \leq z.$$

The two inequalities defining W become

$$(1-t)U + V \geq (1-t)\alpha + \beta, \quad U + tV \geq \alpha + t\beta.$$

For fixed t , lowering α or β therefore enlarges $T \cap W$. The outside loss is minimized at $\alpha = tb, \beta = (1-t)a$. At that placement the portion outside W has vertices, in the (U, V) -plane,

$$(0, 0), \quad (a + tb, 0), \quad (tb, (1-t)a), \quad (0, b + (1-t)a).$$

Since the Jacobian of $(x, y) \mapsto (U, V)$ has absolute value D ,

$$(56) \quad \mathcal{L}(T) \geq \mathcal{L}_I(a, b; t) = \frac{(1-t)a^2 + tb^2 + 2t(1-t)ab}{D}.$$

If $a \leq b$, then

$$D(\mathcal{L}_I - a^2) = t\{b^2 - a^2 + (1-t)(a^2 + 2ab)\} \geq 0;$$

the reflected factorization proves $\mathcal{L}_I \geq b^2$ when $b \leq a$. This proves (52) in Type I.

Suppose now that $a + b > 1$ and $a \leq b$. Set

$$r = \frac{a}{b}, \quad t_0 = \frac{1 - r^2}{1 + 2r}.$$

Since $b > 1/(1+r)$, feasibility in (56) and (55) excludes $0 \leq t < t_0$: for $0 < t < t_0$,

$$\left(\frac{1 + r(1-t)}{1+r}\right)^2 - D = \frac{t(1+2r)(t_0 - t)}{(1+r)^2} > 0,$$

so $b + (1-t)a > z$, while $t = 0$ would give $a + b \leq 1$. Hence $t \geq t_0$, and

$$D(\mathcal{L}_I - b^2) = (1-t)b^2\{r^2 - 1 + t(2r + 1)\} \geq 0.$$

If $b \leq a$, put $r = b/a$ and $t_1 = (r^2 + 2r)/(1 + 2r)$. The identical reflected calculation excludes $t > t_1$ and gives

$$D(\mathcal{L}_I - a^2) = ta^2\{r^2 + 2r - t(1 + 2r)\} \geq 0.$$

Thus (53) holds in Type I.

For Type II put

$$U = \alpha + (1-t)x + ty, \quad V = \beta + tx - y.$$

Again T is the simplex $U, V \geq 0, U + V \leq z$. Its exact reaches on the positive y - and x -axes are

$$(57) \quad A = \beta, \quad B = z - \alpha - \beta, \quad A + B \leq z \leq 1.$$

In particular Type II is impossible when $a + b > 1$. Its wedge intersection is the quadrilateral with vertices

$$(0, 0), \quad (B, 0), \quad \left(\frac{B + (1-t)A}{D}, \frac{A + tB}{D}\right), \quad (0, A).$$

Hence

$$(58) \quad \mathcal{L}(T) = \mathcal{L}_{II}(A, B; t) = 1 - \frac{(1-t)A^2 + tB^2 + 2AB}{D}.$$

Let $S = A + B$. If $A \leq B$, write $A = rS$, $B = (1 - r)S$ with $0 \leq r \leq 1/2$. Since $S^2 \leq D$, and with

$$C = (1 - t)r^2 + t(1 - r)^2 + 2r(1 - r),$$

we have $\mathcal{L}_{\text{II}} \geq 1 - C$. The exact factorization

$$1 - C - r^2D = (1 - t)(1 - 2r + tr^2) \geq 0$$

gives $\mathcal{L}_{\text{II}} \geq r^2D \geq A^2$. Reflection gives $\mathcal{L}_{\text{II}} \geq B^2$ when $B \leq A$. Since $A \geq a$ and $B \geq b$, this proves (52). The two orientation types are exhaustive, so the lemma follows. $\square$

Body consequence. The two optimized lower bounds are exactly Theorem 4.2.

C.2. T3-like area-loss optimization.

Exact interface. The input is an actual T3-like triangle and selected reach lower bounds as in Theorem 4.3.

Variables and feasible set. Normalize to V_0 . The triangle ranges only over the T3-like part of the closed unit-triangle family; in particular $(o, n) = (2, 1)$. The translation in Proposition 2.7 is used only as a closed-trace majorant.

Objective. Minimize the normalized loss of the actual T3-like triangle.

Exact result. The exact nonsupercritical and actual-triangle loss conclusions are (6) and (7).

Solution.

Lemma C.3 (Solved T3-like loss problem). *Let T_i be an actual T3-like V triangle, let $0 \leq a_i \leq A_i$ and $0 \leq b_i \leq B_i$, and let $0 \leq \mu \leq 1/2$. Then*

$$(59) \quad A_i + B_i \leq 1.$$

Moreover,

$$(60) \quad a_i, b_i \geq \mu \implies \mathcal{L}(T_i) \geq 2\mu - 4\mu^2.$$

Proof. The first assertion is Lemma 2.10. For the loss bound put $T = T_i$, $a = a_i$, $b = b_i$, and $m = \mu$, and apply the closed-trace domination in Proposition 2.7. It gives a translated Type-II triangle $\hat{T}$ of the same area with

$$T \cap H \subset \hat{T} \cap H, \quad \mathcal{L}(T) \geq \mathcal{L}(\hat{T}).$$

In the notation of that proposition, $0 < t < 1$, $D = 1 - t + t^2$, $z = \sqrt{D}$, and after reflection the translated majorant has exact reaches

$$A = \beta, \quad B = z - \beta$$

and unique wedge vertex

$$Q = \left(\frac{B + (1 - t)A}{D}, \frac{A + tB}{D} \right), \quad Q_x > 1, \quad Q_y \leq 1.$$

Thus $z - tA > D$, so

$$(61) \quad A < L := \frac{z - D}{t} = \frac{z(1 - z)}{t}.$$

Define the exact normalized loss of this majorant by

$$\mathcal{L}_{\text{maj}}(z, A) := \frac{tA^2 + (1 - t)(z - A)^2}{D} = \mathcal{L}(\hat{T}).$$

This decreases for $A < L < (1-t)z$, and therefore

$$(62) \quad \mathcal{L}(T) \geq \mathcal{L}(\hat{T}) > \mathcal{L}_{\text{maj}}(z, L).$$

Set $r = (1-z)/t$. Then $0 < r < 1/2$, and eliminating t, z gives

$$t = \frac{1-2r}{1-r^2}, \quad z = \frac{r^2-r+1}{1-r^2},$$

$$L = \alpha_{\text{lim}}(r) := \frac{r(r^2-r+1)}{1-r^2}, \quad z-L = \beta_{\text{lim}}(r) := \frac{r^2-r+1}{1+r}.$$

The limiting normalized inside area and loss are

$$F_0(r) = \frac{(r^2-r+1)^2}{1-r^2}, \quad \mathcal{L}_0(r) = 1 - F_0(r).$$

Since the exact reach A is at least the required reach lower bound $a \geq m$, (61) gives $\alpha_{\text{lim}}(r) > m$. Direct simplification yields

$$\mathcal{L}_0(r) - \{2\alpha_{\text{lim}}(r) - 4\alpha_{\text{lim}}(r)^2\} = \frac{r^2(5r^4 - 8r^3 + 13r^2 - 8r + 2)}{(1-r^2)^2} \geq 0,$$

because the last polynomial is at least $9r^2 - 8r + 2 \geq 2/9$ on $[0, 1/2]$. Also $\alpha_{\text{lim}}(r) \leq r$, and

$$\mathcal{L}_0(r) - \frac{1}{4} = \frac{(1-2r)(2r^3 - 3r^2 + 6r - 1)}{4(1-r^2)} \geq 0 \quad \text{when } \alpha_{\text{lim}}(r) \geq \frac{1}{4};$$

the cubic is increasing and has value $11/32$ at $r = 1/4$.

Finally $\psi(u) = 2u - 4u^2$ is increasing on $[0, 1/4]$ and never exceeds $1/4$ on $[0, 1/2]$. If $\alpha_{\text{lim}}(r) \leq 1/4$, then $\mathcal{L}_0(r) \geq \psi(\alpha_{\text{lim}}(r)) \geq \psi(m)$; if $\alpha_{\text{lim}}(r) \geq 1/4$, then $\mathcal{L}_0(r) \geq 1/4 \geq \psi(m)$. Together with (62), this proves (60). The reflected crossing is identical. $\square$

Body consequence. This solved restricted optimization supplies the actual-triangle bound in Theorem 4.3; it makes no assertion about unrestricted $\mathcal{L}(a, b)$.

C.3. Cyclic area-loss optimization.

Exact interface. Use the strict handoff selector of Proposition 2.11 and the two hypotheses stated in Lemma 4.4.

Variables and feasible set. The variables are the cyclic selector tuple $(\xi_0, \dots, \xi_5)$ in the product of the six open selector intervals. The local losses range over the six actual V triangles realizing the selected lower bounds.

Objective. Minimize $\sum_{i=0}^5 \mathcal{L}(T_i)$ on each of the two selected-ascent classes.

Exact result. On each of the two feasible selector classes in Lemma 4.4, the exact conclusion is

$$\sum_{i=0}^5 \mathcal{L}(T_i) > 1.$$

Solution.

Lemma C.4 (Solved cyclic area-loss problem). *Under either exact hypothesis in Lemma 4.4,*

$$\sum_{i=0}^5 \mathcal{L}(T_i) > 1.$$

Proof. Put $m = \min_i \xi_i$ and $M = \max_i \xi_i$. The reflection $y_i = 1 - \xi_{-i-1}$ swaps each V triangle's two coordinates and preserves both alternatives and the total loss. Hence assume $m \leq 1 - M$. Every selected V triangle coordinate is then at least m , so (52) gives loss at least m^2 for every V triangle.

In case (1), choose an ascent out of a minimum plateau, $\xi_{p-1} = m < \xi_p$. Its V triangle contains the coordinate $1 - m$ and loses at least $(1 - m)^2$ by (53). A second ascent has one coordinate strictly larger than $1/2$ and hence loses strictly more than $1/4$. The other four V triangles lose at least m^2 each, so the total loss is strictly larger than

$$4m^2 + (1 - m)^2 + \frac{1}{4} = 5 \left(m - \frac{1}{5} \right)^2 + \frac{21}{20} > 1.$$

In case (2), rotate so that the unique ascent leaves the minimum: $\xi_5 = m < \xi_0 = M$. The supercritical V triangle loses at least $(1 - m)^2$. Lemma 2.10 shows that a T3-like V triangle is different from it, and Theorem 4.3 gives that V triangle loss at least $2m - 4m^2$. The remaining four V triangles lose at least m^2 each. Their total loss is at least

$$(1 - m)^2 + (2m - 4m^2) + 4m^2 = 1 + m^2 > 1.$$

□

Body consequence. The two strict minima give Lemma 4.4; together with the structural selector they yield Proposition 4.5 for every center class whenever $N_{\text{gap}} = 0$.

APPENDIX D. NONZERO-GAP FINITE-ENCLOSURE OPTIMIZATION

This appendix supplies the solved calculations used by the nonzero-gap finite-enclosure interfaces. The variational objects Λ , $\mathcal{A}$, c_{max} , and $C_{\pm}$ remain those defined in the body; the shared enclosure gauge and the local admissible set are evaluated in Appendix A. Each calculation below has a separate label so that the body cites only the result it uses.

D.1. Module D.1: neighboring-ray capacity.

Exact interface. The variational capacities C_+ and C_- are the exact closed-triangle maxima of theorem 5.5; the anchor comparison uses c_{max} from theorem 5.3.

Variables and feasible set. Take $0 \leq a, b \leq 1$ with $a + b \leq 1$. The maximizing radial coordinate is $c \in [0, 1]$, with the closed containment and reflection convention fixed in theorem 5.5. For the nonlinear branches below, $0 \leq a \leq 1/2$ and the selected cubic root lies in $[a, 1]$.

For $0 \leq a \leq 1/2$, let $p_*(a)$ be the unique root in $[a, 1]$ of

$$p^3 - (a + 2)p^2 + 2(a + 1)p - 1 = 0,$$

and put

$$\sigma(a) := 1 - p_*(a), \quad \tau(a) := 1 - a - (p_*(a) - a)(1 - p_*(a)).$$

Objective. Determine the attained maximum neighboring-ray reach $C_+(a, b)$, its reflected value $C_-(a, b)$, and the exact comparison of either neighboring capacity with the own-role capacity $c_{\text{max}}(a, b)$.

Exact result.

Proposition D.1 (Exact neighboring-ray formula). *For $0 \leq a, b \leq 1$ and $a + b \leq 1$, the maximum in Definition 5.5 is attained and*

$$C_+(a, b) = \begin{cases} 1 - b, & a > 1/2, \\ 1 - b, & 0 \leq a \leq 1/2, \ 0 \leq b \leq \sigma(a), \\ p_*(a), & 0 \leq a \leq 1/2, \ \sigma(a) \leq b \leq \tau(a), \\ a + \frac{1}{2} - \sqrt{(a+b)^2 - \frac{3}{4}}, & 0 \leq a \leq 1/2, \ \tau(a) < b \leq 1 - a. \end{cases}$$

At $b = \tau(a)$ the selected value is $p_*(a)$, and $C_-(a, b) = C_+(b, a)$.

Lemma D.2 (Common-pair domination). *If $p, q \geq 0$ and $p + q \leq 1$, then*

$$C_+(p, q), C_-(p, q) \leq 1 - \min\{p, q\} \leq c_{\max}(p, q).$$

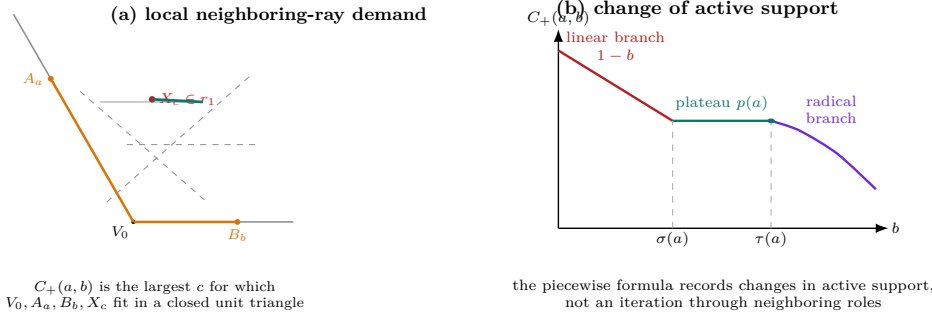


FIGURE 12. Geometry behind the neighboring-ray capacity $C_+(a, b)$. The local problem asks how far a triangle anchored at V_0 and the two boundary points A_a, B_b can reach on the neighboring arm r_1 . As b varies with a fixed, the active support pattern changes from the linear branch to a plateau and then to the radical branch, with transition points $\sigma(a)$ and $\tau(a)$. The graph is schematic; the transition order, endpoint selection, and branch labels are exact.

Solution.

Proof of theorem D.1. Use the cone basis $E_- = V_5 - V_0$, $E_+ = V_1 - V_0$. In these coordinates the four points in the definition of C_+ are

$$V_0 = (0, 0), \quad X_5(1 - a) = (a, 0), \quad X_0(b) = (0, b), \quad R_1(c) = (c, 1),$$

and the metric is $u^2 + v^2 - uv$. Let $R^2 + S^2 - RS = 1$ and define

$$U(u, v) = (R - S)u + Sv, \quad V(u, v) = Su - Rv, \quad W(u, v) = Ru + (S - R)v.$$

For this orientation the least enclosing side is

$$\max_Y W(Y) - \min_Y U(Y) - \min_Y V(Y),$$

where Y ranges over the four displayed points. Compactness of the orientation circle and of $[0, 1]$ proves attainment. At a maximal feasible c , one support side

contains a hull edge. The nondominated contacts are

$$\left. \begin{array}{l} V_0 X_5(1-a) \text{ and } X_0(b) R_1(c) \\ X_5(1-a) R_1(c) \text{ supported, } X_0(b) \text{ inactive} \\ X_5(1-a) R_1(c) \text{ supported, } X_0(b) \text{ active} \end{array} \right| \begin{array}{l} c = 1-b, \\ c = p_*(a), \\ c = a + \frac{1}{2} - \sqrt{(a+b)^2 - \frac{3}{4}}. \end{array}$$

For the first contact take $R = S = 1$. The containing triangle is

$$U \geq 0, \quad V \geq -b, \quad W \leq c;$$

its side is $b + c$, and it contains $(a, 0)$ precisely when $a \leq c$.

For the second contact put $d = c - a$. The supported side gives $R = Sd$ and $S^2(d^2 - d + 1) = 1$. Eliminating S from the unit-side condition gives

$$(c-a)(c^3 - (a+2)c^2 + 2(a+1)c - 1) = 0.$$

For $0 \leq a \leq 1/2$ the nondegenerate root is unique on $[a, 1]$. The remaining containment condition is $b \leq \tau(a)$, and this branch dominates $1 - b$ exactly for $b \geq \sigma(a)$.

For the third contact the active supports give

$$S = -\frac{1}{a+b}, \quad d^2 - d + 1 = (a+b)^2.$$

The selected root $0 \leq d \leq 1/2$ produces the radical in the statement; its inactive-support inequalities are exactly $0 \leq a \leq 1/2$ and $\tau(a) \leq b \leq 1 - a$. This radical is real on that range. Indeed, solving the cubic relation for a in terms of $p = p_*(a)$ gives

$$a + \tau(a) = \frac{p^2 - p + 1}{p(2-p)} \geq \frac{\sqrt{3}}{2} \quad \left(\frac{1}{2} \leq p \leq 1 \right).$$

At the transition, the same cubic relation gives

$$(a + \tau(a))^2 - \frac{3}{4} = \left(a + \frac{1}{2} - p_*(a) \right)^2,$$

so the radical candidate there is at most $p_*(a)$; hence the plateau value is selected at $b = \tau(a)$. For $b > \tau(a)$ the radical candidate is strictly decreasing in b . It dominates the linear candidate because, with $s = a + b \in [\sqrt{3}/2, 1]$,

$$\left(a + \frac{1}{2} - \sqrt{s^2 - \frac{3}{4}} \right) - (1 - b) = s - \frac{1}{2} - \sqrt{s^2 - \frac{3}{4}} \geq 0,$$

the last inequality following from $(s - 1/2)^2 - (s^2 - 3/4) = 1 - s \geq 0$.

The selectors are ordered because

$$\tau(a) - \sigma(a) = (p_*(a) - a)p_*(a) \geq 0, \quad \tau(a) \leq 1 - a,$$

and the same parametrization gives

$$\tau(a) - a = \frac{(2p_*(a) - 1)(p_*(a)^2 - p_*(a) + 1)}{p_*(a)(2 - p_*(a))} \geq 0.$$

These cases exhaust the support contacts. Reflection gives C_- . $\square$

Proof of theorem D.2. Assume first $q \leq p$. The edge-normal unit triangle contains $K(p, q, 1 - q)$, so $c_{\max}(p, q) \geq 1 - q$. The linear branch of theorem D.1 equals $1 - q$. If $p > 1/2$, this is the only branch, so assume $p \leq 1/2$. On the plateau branch the defining cubic, evaluated at $1 - q$, is nonnegative; since the cubic is increasing

on the selected interval, $p_*(p) \leq 1 - q$. The radical branch does not occur here, because $q \leq p \leq \tau(p)$. Hence $C_+(p, q) \leq 1 - q$.

Now assume $p \leq q$. The reflected edge-normal construction gives $c_{\max}(p, q) \geq 1 - p$. Here $p \leq 1/2$. The linear branch satisfies $1 - q \leq 1 - p$, while evaluation of the plateau cubic at $1 - p$ gives

$$p^2(1 - 2p) \geq 0,$$

so $p_*(p) \leq 1 - p$. At its transition the radical value is at most $p_*(p)$ by the calculation above, and it then decreases. Thus $C_+(p, q) \leq 1 - p$. Applying the result just proved to the reflected pair and using $C_-(p, q) = C_+(q, p)$ proves the same bound for C_- . $\square$

Body consequence. The attained piecewise evaluation supplies theorem 5.5; the final comparison supplies the compact body statement theorem 5.8.

D.2. Module D.2: type-aware endpoint forcing.

Exact interface. Use exactly the selected reaches, source-conditioned neighboring terms, Γ_i , and $D_i = (1 - \Gamma_i)V_i$ of theorem 5.6.

Variables and feasible set. The index is $i \in \mathbb{Z}/6\mathbb{Z}$; the selected reaches obey $0 \leq a_j \leq A_j$, $0 \leq b_j \leq B_j$, and $a_j^2 + a_j b_j + b_j^2 \leq 1$. A neighboring capacity is present only under the positive-trace implication displayed in the cited definition, which also places its pair in the domain $a + b \leq 1$.

Objective. Show that the largest admissible local closed radial endpoint still lies outside every open V role, so any seven-role cover must assign it to the center role. Exact result.

Lemma D.3 (Type-aware endpoint forcing). *The point D_i of Definition 5.6 satisfies*

$$D_i \in r_i \subseteq S, \quad D_i \notin \bigcup_{j=0}^5 U_j.$$

Consequently, a cover of S by the seven displayed roles forces $D_i \in U_C$.

Solution.

Proof of theorem D.3. If $\Gamma_i = 1$, then $D_i = O$. Since $\|O - V_j\| = 1$ for every j , no open unit equilateral triangle U_j containing its indexed vertex V_j can also contain O . Thus assume $\Gamma_i < 1$.

If $D_i \in U_i$, openness gives a slightly more inward point of r_i in U_i . Together with the two selected boundary anchors in the closed role, this realizes an own-radial coordinate strictly larger than $c_{\max}(a_i, b_i)$, contrary to the definition of Γ_i .

If $D_i \in U_{i-1}$, openness gives a positive interval of r_i in that role. Thus its C_+ term is present in Γ_i , and the same open neighborhood supplies a neighboring-ray coordinate strictly larger than $C_+(a_{i-1}, b_{i-1})$, a contradiction. The reflected argument excludes U_{i+1} . This also covers a role whose closed intersection with r_i would otherwise be only an isolated point: hypothetical open membership creates positive-length support before the capacity is invoked.

Write $D_i = dV_i$ with $0 < d \leq 1$. For the vertices at cyclic distance two and the opposite vertex,

$$\|dV_i - V_{i\pm 2}\|^2 = 1 + d + d^2 > 1, \quad \|dV_i - V_{i+3}\| = 1 + d > 1.$$

Each nonlocal open V role contains its indexed vertex, and any two points in an open unit equilateral triangle are at distance strictly less than one. Hence the three nonlocal roles also miss D_i . Every V role therefore misses D_i , and the cover forces $D_i \in U_C$. $\square$

Body consequence. This endpoint-strict calculation is the unique Appendix D supplier for theorem 5.7.

D.3. Module D.3: nonsupercritical high-radial output.

Exact interface. Use the canonical admissible set $\mathcal{A}$ and actual reaches of theorem 5.3. The output function below is the attained maximum outgoing reach on its nonsupercritical fiber.

Variables and feasible set. Let $a, b, c \in [0, 1]$ satisfy $(a, b, c) \in \mathcal{A}$ and $a + b \leq 1$, with $c > 1/2$ for the high-radial branches. In the threshold form put $c = 1 - e$ with $0 < e < 1 - \sqrt{3}/2$; closed equality is allowed in the capacity, whereas the hypothesis on the actual incoming reach is strict.

For a nonsupercritical V triangle with incoming reach at least a and own radial reach at least c , define the attained outgoing capacity

$$\mathcal{B}_c(a) := \max\{b \in [0, 1] : (a, b, c) \in \mathcal{A} \wedge a + b \leq 1\}.$$

For $0 < e < 1 - \sqrt{3}/2$, put

$$\epsilon(e) := \frac{1-e}{2} \left(1 - \sqrt{4(1-e)^2 - 3}\right).$$

Set

$$\mu := \frac{2\sqrt{3} - 3}{4}.$$

The selected low root obeys

$$e < \epsilon(e),$$

$$0 < e \leq \mu \implies \epsilon(e) < \frac{2e}{1-2e}, \quad 0 < e \leq \frac{1}{8} \implies \epsilon(e) \leq 2e + 5e^2.$$

For $1/2 < c \leq \sqrt{3}/2$, put

$$h(c) := \frac{2c}{1 + \sqrt{16c^2 - 3}},$$

and, for $\sqrt{3}/2 \leq c < 1$, put

$$\ell(c) := \frac{c}{2} \left(1 - \sqrt{4c^2 - 3}\right), \quad r(c) := c - \ell(c).$$

Finally set

$$Q_-(a, c) := \frac{\sqrt{a^2 - ac + c^2}}{c} - a,$$

$$Q_+(a, c) := \frac{2(1 - a^2)c^2}{2ac^2 + c + \sqrt{(2ac^2 + c)^2 - 4(1 - c^2)(1 - a^2)c^2}}.$$

Objective. Maximize the outgoing reach b on each high-radial nonsupercritical fiber, then determine the exact low-root threshold that forces the outgoing actual reach below the same root.

Exact result.

Proposition D.4 (Four direct high-radial outputs). *For $1/2 < c \leq \sqrt{3}/2$,*

$$\mathcal{B}_c(a) = \begin{cases} 1 - a, & 0 \leq a \leq 1 - c, \\ Q_+(a, c), & 1 - c < a < h(c), \\ h(c), & a = h(c), \\ Q_-(a, c), & h(c) < a < c, \\ 1 - a, & c \leq a \leq 1. \end{cases}$$

For $\sqrt{3}/2 < c < 1$,

$$\mathcal{B}_c(a) = \begin{cases} 1 - a, & 0 \leq a \leq 1 - c, \\ Q_+(a, c), & 1 - c < a \leq \ell(c), \\ \ell(c), & \ell(c) < a < r(c), \\ Q_-(a, c), & r(c) \leq a < c, \\ 1 - a, & c \leq a \leq 1. \end{cases}$$

At $a = \ell(c)$ the selected value is $r(c)$; immediately to its right it is $\ell(c)$.

Lemma D.5 (Direct high-radial threshold). *Let a nonsupercritical V triangle have incoming, outgoing, and own-radial reaches A, B, C . If*

$$0 < e < 1 - \frac{\sqrt{3}}{2}, \quad C \geq 1 - e, \quad A > \epsilon(e),$$

then

$$B \leq \epsilon(e).$$

Solution.

Proof of theorem D.4. Because $c > 1/2$, the radial midpoint is contained and hence $a + b \leq 1$. On the cap $b = 1 - a$, the T -cell inequality from theorem A.10 reduces, with $M = \max\{a, b\}$, to $M(c - M) \leq 0$. Thus the cap is feasible exactly for $a \leq 1 - c$ or $a \geq c$.

Off that cap, a maximal point lies on $F_L = 0$ or $F_T = 0$. When $1/2 < c < \sqrt{3}/2$, the discriminant $c^2(4c^2 - 3)$ of the L contact is negative, so no L -cell root occurs; the two ordered halves of $F_T = 0$ give Q_+ and Q_- and meet at $h(c)$. When $c \geq \sqrt{3}/2$, the selected L cell has $m = \min\{a, b\} \leq \ell(c)$, and its selected radial frontier is only $m = \ell(c)$. Indeed, the algebraic L roots are $\ell(c), r(c)$, but the formal alternative $m \geq r(c)$ is incompatible with the selector $q \leq 0$, except at the coalescence point. The two ordered $q = 0$ contacts are $(\ell(c), r(c))$ and $(r(c), \ell(c))$, producing the exceptional left endpoint and the constant interval.

In the ordered half $a \leq b$, the cap $b \leq 1 - a$ lies strictly between the two formal roots of the F_T quadratic, so its lower root is the selected Q_+ branch and the upper root is excluded. The remaining geometric selector $c \leq 2 \max\{a, b\} = 2b$ holds throughout that retained branch: in the first regime $b \geq h(c) \geq c/2$, and in the second $b \geq r(c) \geq c/2$, with the latter equality only at $c = \sqrt{3}/2$. Reflection gives the selected Q_- branch, for which $c \leq 2a$. These selectors give exactly the displayed pieces, including the stated downward jump. $\square$

Proof of theorem D.5. The value $\epsilon(e) = \ell(1 - e)$ is the smaller root of

$$p_e(z) := z^2 - (1 - e)z + (1 - e)^2 - (1 - e)^4 = 0.$$

Substitution at e gives

$$p_e(e) = e(1 - 3e + 4e^2 - e^3) > 0,$$

and $e < (1 - e)/2$, the midpoint of the two roots. Hence $e < \epsilon(e)$ throughout the stated range. If $0 < e \leq \mu$, then

$$p_e\left(\frac{2e}{1 - 2e}\right) = -\frac{e^2}{(1 - 2e)^2}P(e), \quad P(e) := 4e^4 - 20e^3 + 37e^2 - 28e + 3.$$

On $[0, \mu] \subset [0, 1/8]$,

$$P'(e) = 16e^3 - 60e^2 + 74e - 28 < \frac{16}{512} + \frac{74}{8} - 28 < 0,$$

whereas

$$P(\mu) = \frac{8217}{64} - 74\sqrt{3} > 0, \quad \left(\frac{8217}{64}\right)^2 - 3 \cdot 74^2 = \frac{230001}{4096} > 0.$$

Thus the preceding value of p_e is strictly negative, which proves the rational upper bound on precisely that range. If $e \leq 1/8$, then

$$p_e(2e + 5e^2) = e^2(3e + 4)(8e - 1) \leq 0,$$

proving the third. Proposition D.4, with $c = 1 - e$, shows that after $A > \epsilon(e)$ the selected constant and Q_- pieces have outgoing coordinate at most $\epsilon(e)$. The Q_+ piece has already ended. The upper linear branch $A \geq c$ remains possible, but there $B \leq 1 - A \leq e < \epsilon(e)$. Downward closure handles $C \geq 1 - e$. $\square$

Body consequence. The evaluated output catalogue and threshold supply the high-radial steps in theorems 5.14 and 5.15; later Appendix D modules cite these two exact results rather than re-evaluating the fiber.

D.4. Module D.4: strict-supercritical envelope.

Exact interface. The body defines M_c^{sup} as the supremum of outgoing reaches on the strict-supercritical part of $\mathcal{A}$ in the supported-tail interface preceding theorem 5.17. Variables and feasible set. Take $0 \leq c < 1/2$ and pairs $(a, b) \in [0, 1]^2$ satisfying $(a, b, c) \in \mathcal{A}$ and $a + b > 1$. For the relevant boundary reaches $1/2 \leq b < 1$, put

$$\kappa(b) := \frac{1 - b^2}{2 - b}.$$

The support-slack feasibility condition is the strict inequality $c < \kappa(b)$; it is not replaced by a blanket closure assertion for the whole face $a + b = 1$. At $c = 1/2$ use the body's declared continuous endpoint value.

Objective. Determine the supremal outgoing reach b , decide attainment on the strict fiber, and evaluate its one-sided endpoint limit.

Exact result.

Lemma D.6 (Strict-supercritical envelope). *For $0 \leq c < 1/2$ the variational quantity defined in the body is*

$$M_c^{\sup} = \frac{c + \sqrt{c^2 - 8c + 4}}{2}.$$

It is not attained by a strict-supercritical pair. Moreover

$$\lim_{c \uparrow 1/2} M_c^{\sup} = \frac{1}{2} = M_{1/2}^{\sup}.$$

Thus a strict-supercritical V triangle with own-radial reach at least c has outgoing boundary reach $B < M_c^{\sup}$.

Solution.

Proof of theorem D.6. For $c = 0$, downward closure and the positive- c calculation below give strict-supercritical outgoing reaches arbitrarily close to 1, while always $b \leq 1$. This supremum is not attained: $b = 1$ and $a^2 + ab + b^2 \leq 1$ force $a = 0$, contrary to $a + b > 1$. Fix $0 < c < 1/2$. Only $b \geq 1/2$ matters for the supremum. Put $a = 1 - b$, where $1/2 \leq b < 1$, and use the support gauge of theorem A.9.

Let e_A, e_B be the two unit boundary directions from the corner, at angle $2\pi/3$, so the boundary anchors are ae_A and be_B . We first prove that (a, b, c) has strict support slack, and therefore remains feasible after increasing a slightly into $a + b > 1$, exactly when $c < \kappa(b)$. Reduce a support normal modulo $2\pi/3$ to the two sectors $[-\pi/6, \pi/6]$ and $[\pi/6, \pi/2]$. On the first sector the three-point support sum for $0, ae_A, be_B$ is $\cos \phi \geq \sqrt{3}/2$. On the second, put

$$p = \cos \phi, \quad q = \cos(2\pi/3 - \phi), \quad t = \frac{q}{p}, \quad p^2 + pq + q^2 = \frac{3}{4}.$$

After division by p , the four-point support sum is

$$N(t) = a + bt + \max\{at, b, c(1+t)\}.$$

Set

$$t_* := \frac{2b-1}{1-b^2}.$$

When the maximum in N is b ,

$$(1+bt)^2 - (1+t+t^2) = t((2b-1) - (1-b^2)t).$$

Thus strict slack is possible for $t > t_*$. The two inactive-support conditions $at \leq b$ and $c(1+t) \leq b$ admit such a t exactly when

$$t_* < \frac{b}{c} - 1 \iff c < \kappa(b);$$

the bound $t_* < b/a$ is automatic. Continuity then permits a small strict increase of a .

Conversely, suppose $c \geq \kappa(b)$. For $t \leq t_*$ the term b gives $N(t) \geq \sqrt{1+t+t^2}$. For $t \geq t_*$, monotonicity reduces to $c = \kappa(b)$. With

$$A = a + \kappa(b), \quad B = b + \kappa(b),$$

the quadratic

$$(A + Bt)^2 - (1 + t + t^2)$$

has negative constant term, positive leading coefficient, and positive root $t = t_*$, because $\kappa(b)(1 + t_*) = b$. It is therefore nonnegative for $t \geq t_*$. Increasing a makes the first-sector support sum strict and strictly increases every second-sector expression (the endpoint $p = 0$ is already strict because $b + c > 1$). Hence no strict-supercritical triple with this b is feasible.

The decreasing function κ maps $[1/2, 1)$ onto $(0, 1/2]$. Consequently the supremum is the larger root of

$$b^2 - cb + 2c - 1 = 0.$$

Equality has no strict support slack, so this value is not attained. The limit at $c = 1/2$ follows by direct substitution and equals the body's declared continuous value. $\square$

Body consequence. This formula and its strict nonattainment supply the value M and the bound $B_1 < M$ in theorems 5.17 and 5.18.

D.5. Module D.5: selected-arc chord bounds.

Exact interface. Use the selected Q_+ component of the nonsupercritical output function from Module D.3 and the center-free boundary handoffs in theorem 5.15. Variables and feasible set. Let $0 < e < 1/2$, take $x \in [0, 1]$, and set $q = e + (1 - e)x$. The incoming coordinate p is the smaller-root point on the selected Q_+ component. For $e < 1 - \sqrt{3}/2$ it is the high-radial arc; for $1 - \sqrt{3}/2 \leq e < 1/4$ the CE1 use is the low-radial arc. These ranges keep every occurrence of $\epsilon(e)$ inside its declared domain.

Objective. Parametrize the selected output arc and obtain affine lower bounds for its outgoing reach along the two intervals used by the CE1 reverse path.

Exact result.

Lemma D.7 (Selected-arc chord bounds). *Let $0 < e < 1/2$. On the selected Q_+ component for radial demand $1 - e$, let p be the incoming reach and let q be the next incoming reach forced across a center-free edge. There is $x \in [0, 1]$ with*

$$q = e + (1 - e)x, \quad p = e + (1 - e)x - 1 + \sqrt{1 - x + x^2},$$

and hence

$$q - p = 1 - \sqrt{1 - x + x^2}.$$

This selected arc is increasing and concave. On each interval used by the CE1 reverse path,

$$q \geq p + \lambda(p - e),$$

with the following fully quantified choices. On the high-radial arc $0 < e < 1 - \sqrt{3}/2$, take

$$\lambda_{\text{hi}}(e) := \frac{e}{\epsilon(e) - e};$$

then

$$0 < e \leq \mu \implies \lambda_{\text{hi}}(e) > 1 - 4e, \quad 0 < e < 1 - \frac{\sqrt{3}}{2} \implies \lambda_{\text{hi}}(e) > 1 - 5e.$$

On the low-radial arc $1 - \sqrt{3}/2 \leq e < 1/4$, put

$$t_e := h(1 - e), \quad \lambda_{\text{lo}}(e) := \frac{1 - 2t_e}{t_e - e};$$

then

$$\lambda_{\text{lo}}(e) > \frac{1}{3} > 1 - 5e.$$

Solution.

Proof of theorem D.7. With $x = (q - e)/(1 - e)$ the selected T -cell equation is

$$x(1 - x) = (q - p)(2 - q + p).$$

The smaller root gives the displayed parametrization. If $\psi(x) = 1 - \sqrt{1 - x + x^2}$, then

$$\psi''(x) = -\frac{3}{4(1 - x + x^2)^{3/2}} < 0.$$

The map $x \mapsto p = e + (1 - e)x - \psi(x)$ is strictly increasing and convex; its inverse, and therefore $p \mapsto q$, is strictly increasing and concave. The graph lies above each endpoint chord.

On the high-radial arc the endpoint pairs are

$$(p, q) = (e, e), \quad (p, q) = (\epsilon(e), e + \epsilon(e)),$$

which gives $\lambda_{\text{hi}}(e)$. For $e \leq \mu$, the rational low-root bound proved in Module D.3 gives

$$\lambda_{\text{hi}}(e) > \frac{1 - 2e}{1 + 2e} > 1 - 4e.$$

If $e \leq 1/8$, the bound $\epsilon(e) \leq 2e + 5e^2$ gives

$$\lambda_{\text{hi}}(e) \geq \frac{1}{1 + 5e} > 1 - 5e.$$

If $1/8 < e < 1 - \sqrt{3}/2$, then $\epsilon(e) < (1 - e)/2$ directly from its defining radical, and hence

$$\lambda_{\text{hi}}(e) > \frac{2e}{1 - 3e} > 1 - 5e.$$

On the low-radial arc the other endpoint is $(t_e, 1 - t_e)$, giving $\lambda_{\text{lo}}(e)$. With $c = 1 - e$ and $t_e = 2c/(1 + \sqrt{16c^2 - 3})$, the inequality $\lambda_{\text{lo}}(e) > 1/3$ is equivalent to $4 - c > 7t_e$. After isolating and squaring its positive radical, the difference factors as

$$4(2c - 1)(2c^3 - 15c^2 - 4c + 16) > 0 \quad \left(\frac{1}{2} < c \leq \frac{\sqrt{3}}{2} \right);$$

the cubic decreases on this interval and at $c = \sqrt{3}/2$ equals $(19 - 5\sqrt{3})/4 > 0$. Finally the low-radial range gives $e \geq 1 - \sqrt{3}/2 > 2/15$, and therefore $1/3 > 1 - 5e$ whenever this comparison is needed. $\square$

Body consequence. The two strict chord estimates supply the successive return bounds in the Appendix D certificate for theorem 5.15.

D.6. Module D.6: Vd corner margins.

Exact interface. Use the Vd1/Vd2 corner-form actual reaches and the radial separation objects of theorem 5.21; selected lower bounds remain distinguished from the actual reaches A, B .

Variables and feasible set. Take $0 < A \leq 1$ and $0 \leq B \leq 1$ from the actual corner traces and $\theta > 0$, with $\Delta_\theta = \sqrt{\theta^2 + \theta + 1}$. The endpoint u_+ is used only when $\mathcal{H}^1(T_0 \cap r_1) > 0$. When selected demands are used, write them in lowercase and require $A \geq a_*$ and $B \geq b_*$.

Objective. Evaluate the corner's own-radial reach and far supported endpoint, and prove the strict margins needed to place the residual radial point beyond both.

Exact result.

Lemma D.8 (Vd corner margins). *In the Vd1/Vd2 corner form, let the actual boundary reaches satisfy $0 < A \leq 1$ and $0 \leq B \leq 1$, let $\theta > 0$, and put $\Delta_\theta = \sqrt{\theta^2 + \theta + 1}$. The own-radial reach is*

$$C_0 = \frac{\Delta_\theta - A - \theta B}{\theta + 1}.$$

If $\mathcal{H}^1(T_0 \cap r_1) > 0$, the far endpoint of that positive forward support is

$$u_+ = \frac{\Delta_\theta - A - \theta B - 1}{\theta}.$$

Consequently

$$C_0 < 1 - \min\{A, B\}, \quad u_+ < 1 - B \quad \text{when } \mathcal{H}^1(T_0 \cap r_1) > 0.$$

If $A \geq a_$ and $B \geq b_*$, then*

$$C_0 < 1 - \min\{a_*, b_*\}, \quad u_+ < 1 - b_* \quad \text{when } \mathcal{H}^1(T_0 \cap r_1) > 0.$$

Reflection supplies the backward-support version.

Solution.

Proof of theorem D.8. Since $\Delta_\theta < \theta + 1$,

$$C_0 < 1 - \frac{A + \theta B}{\theta + 1} \leq 1 - \min\{A, B\},$$

while

$$u_+ < 1 - B - \frac{A}{\theta} < 1 - B.$$

Monotonicity gives the selected-demand version. $\square$

Body consequence. The strict corner margins are used by the nonadjacent branch of theorem 5.21.

D.7. Module D.7: disk-finite-set calipers.

Exact interface. The enclosure objective is $\Lambda(\mathcal{D}_\eta \cup P)$ with Λ and the closed centered disk defined in theorem 5.1; here P is the displayed finite set.

Variables and feasible set. Let $\eta \geq 0$, let $\mathcal{D}_\eta = \mathbb{B}(O, \eta)$, and take $P = \{p_1, \dots, p_m\}$. The feasible orientation is a unit normal on the full normal circle; point-disk ties require $\|p_i\| \geq \eta$, and point-point ties require $p_i \neq p_k$.

Objective. Minimize the three-direction support sum for $\mathcal{D}_\eta \cup P$ and reduce every non-disk-only minimizer to a finite list of tie normals.

Exact result.

Proposition D.9 (Disk-finite-set calipers). *Let $\mathcal{D}_\eta$ be centered at O , let $P = \{p_1, \dots, p_m\}$, and let J denote rotation by $\pi/2$. Unless the disk is the active support in all three directions at a minimizing orientation, a minimizing normal belongs to the finite list*

$$\pm \frac{J(p_k - p_i)}{\|p_k - p_i\|} \quad (p_i \neq p_k)$$

or

$$\frac{\eta p_i \pm \sqrt{\|p_i\|^2 - \eta^2} J p_i}{\|p_i\|^2} \quad (\|p_i\| \geq \eta).$$

Thus one side of a least enclosing equilateral triangle either contains two points, or is tangent to the disk and contains one point.

Solution.

Proof of theorem D.9. The tie directions partition the normal circle into cells with fixed support sources. On a cell with q disk supports,

$$\sum_{j=0}^2 h_{\mathcal{D}_\eta \cup P}(\mathbf{R}^j n(\theta)) = q\eta + \langle v, n(\theta) \rangle.$$

If any point is active, the last inner product is positive and the second derivative is its negative. Hence an interior critical point is a maximum. A minimum occurs at a point–point or point–disk tie. On a disk-only cell the support sum is 3η , already the global minimum. $\square$

Body consequence. This finite caliper reduction supplies the disk-plus-point evaluation used by theorem 5.9.

D.8. Module D.8: half-edge one-third radial envelope.

Exact interface. Use the anchor admissible set and attained radial envelope $c_{\max}$ from theorem 5.3.

Variables and feasible set. The ordered boundary demands satisfy $M \geq 1/2$, $0 < m \leq M$, and $M + m < 1$. The candidate radial coordinate is in $[0, 1]$ and is selected from the L or T cell of theorem A.10.

Objective. Give a strict upper bound for the maximum feasible own-radial reach in terms of the smaller boundary demand.

Exact result.

Lemma D.10 (Half-edge one-third radial envelope). *If*

$$M \geq \frac{1}{2}, \quad 0 < m \leq M, \quad M + m < 1,$$

then

$$c_{\max}(M, m) < 1 - \frac{m}{3}.$$

Solution.

Proof of theorem D.10. Put $s = M + m$ and $c_0 = 1 - m/3$. Only the L and T cells of theorem A.10 can occur.

For $m \leq 3/8$, substitution into the L polynomial gives

$$F_L(c_0) = \frac{m}{81}(m^3 - 12m^2 - 63m + 27) > 0;$$

the cubic decreases on $[0, 3/8]$ and equals $891/512$ at $3/8$. Moreover

$$F_L\left(\frac{\sqrt{3}}{2}\right) = -\left(m - \frac{\sqrt{3}}{4}\right)^2 \leq 0,$$

and

$$F'_L(c) = c(4c^2 - 2) + m > 0 \quad \left(c \geq \frac{\sqrt{3}}{2}\right).$$

Thus F_L is strictly increasing from its selected sign change on $[\sqrt{3}/2, 1]$, and the selected L root is below c_0 . At the L/T transition the two roots agree. On the T cell,

$$c_T(s) = \frac{2(s-m)}{1 + \sqrt{4s^2 - 3}},$$

and, writing $r = \sqrt{4s^2 - 3}$,

$$c'_T(s) = \frac{2(r + 4ms - 3)}{r(1+r)^2} < 0$$

because $m \leq s/2$ and $r + 2s^2 - 3 < 0$ for $s < 1$. Hence the T root remains below c_0 .

For $3/8 \leq m < 1/2$, one has

$$s^4 - s^2 + mM \geq \left(\frac{7}{8}\right)^4 - \left(\frac{7}{8}\right)^2 + \frac{3}{16} = \frac{33}{4096} > 0,$$

so only the T cell occurs. It decreases with M ; at $M = 1/2$, $s \geq 7/8$ and its value is at most $4/5 < 5/6 < c_0$. $\square$

Body consequence. The strict one-third bound supplies the local endpoint estimate in the adjacent branch of theorem 5.21.

D.9. Module D.9: rescuer endpoint and boundary-tail budget.

Exact interface. Use the supported interval, actual reaches, strict-supercritical envelope, and four-edge nonsupercritical path in theorems 5.17 and 5.18, together with the exact signed center traces and exits of theorem A.4.

Variables and feasible set. The special role has boundary endpoint a and radial interval $[c, u] \subset r_1$ with $\varepsilon = 1 - u$ and $M = M_c^{\text{sup}}$. The center role is CE1 or CE2 in signed normal form, with $0 < R < 1$, $W = 1 - R$, $E = \sqrt{1 - RW}$, $\eta = 1 - E$, positive slacks α, δ , and $k = \eta + \alpha + \delta$. The four displayed adapter inequalities are assumed, T_1 is uniquely supercritical with radial reach at least c , and $T_2, \dots, T_5$ form the center-free nonsupercritical path.

Objective. Lower-bound the far boundary demand by M in both center-trace alternatives and test it against the four nonsupercritical boundary budgets.

Exact result.

Lemma D.11 (Rescuer endpoint and boundary-tail budget). *Let T_C be a CE1 or CE2 center role in the signed normal form of theorem A.4. Thus*

$$0 < R < 1, \quad W = 1 - R, \quad E = \sqrt{1 - RW}, \quad \eta = 1 - E, \quad \alpha, \delta > 0, \quad k = \eta + \alpha + \delta,$$

when its positive companion boundary trace on $e_{5,0}$ is present, that trace is $[k/W, R + \alpha]$; it may instead be empty or a point contact. Its exit on r_1 , measured from O , is δ/R . Suppose a special role at V_0 has boundary endpoint a on $e_{5,0}$ and supported interval $[c, u] \subset r_1$, measured from V_1 toward O . Put

$$\varepsilon = 1 - u, \quad M = M_c^{\text{sup}}.$$

Assume

$$\begin{aligned} \varepsilon V_1 &\in U_C, & a + \varepsilon &\leq 1, \\ a &\leq 1 - M, & \frac{a}{a + \varepsilon} &\leq 1 - M. \end{aligned}$$

Assume also

$$U_C \cap \bigcup_{i=1}^4 e_{i,i+1} = \emptyset.$$

If T_1 is uniquely supercritical with own-radial reach at least c and T_2, T_3, T_4, T_5 are nonsupercritical along this four-edge path, then the skeleton is not covered.

Solution.

Proof of theorem D.11. Let h_T be the far boundary demand on T_5 . If the companion C trace does not hide a , then $h_T \geq 1 - a \geq M$.

In the hiding configuration, $k/W \leq a$, while $\varepsilon V_1 \in U_C$ gives $\delta/R \geq \varepsilon$. Therefore

$$Wa \geq k = \eta + \alpha + \delta > \alpha + R\varepsilon,$$

so

$$a - R(a + \varepsilon) > \alpha.$$

Also $Wa > \delta \geq R\varepsilon$, so $R < a/(a + \varepsilon)$. Since $a + \varepsilon \leq 1$,

$$R + \alpha < a + R(1 - a - \varepsilon) < \frac{a}{a + \varepsilon} \leq 1 - M.$$

Thus again $h_T \geq M$.

The strict-supercritical envelope gives $B_1 < M$. Adding

$$A_2 \geq 1 - B_1, \quad B_2 + A_3 \geq 1, \quad B_3 + A_4 \geq 1, \quad B_4 + A_5 \geq 1, \quad B_5 \geq h_T$$

gives

$$\sum_{i=2}^5 (A_i + B_i) > 4,$$

contrary to nonsupercriticality. $\square$

Body consequence. Together with the two branch-specific adapters, this common terminal proves theorems 5.17 and 5.18.

D.10. Module D.10: disk-plus-gap enclosure.

Exact interface. Use the enclosure number and centered disk of theorem 5.1, together with the exact gap interval and anchor capacity in theorem 5.9.

Variables and feasible set. For the auxiliary caliper take $\eta \geq 0$ and a point G with $\|G\| = \rho$. For the gap problem take $0 \leq p, q \leq 1$, $p + q \leq 1$, $c_* = c_{\max}(p, q)$, $m = \min\{p, q\}$, and the closed interval $J(p, q) = X_0([q, 1 - p])$; the endpoint case $p + q = 1$ remains feasible.

Objective. Evaluate the minimum enclosing side for a centered disk plus one point, then use that value to lower-bound the enclosure number of the disk-plus-gap set. Exact result.

Lemma D.12 (Disk plus one point). *Let $\mathbb{B}(O, \eta)$ be the centered closed disk and let $\|G\| = \rho$. Then*

$$\Lambda(\mathbb{B}(O, \eta) \cup \{G\}) = \begin{cases} 2\sqrt{3}\eta, & \rho \leq 2\eta, \\ \frac{3\eta + \sqrt{3(\rho^2 - \eta^2)}}{\sqrt{3}}, & \rho > 2\eta. \end{cases}$$

Lemma D.13 (Complementary-gap calculation). *For $0 \leq p, q \leq 1$ with $p + q \leq 1$, set*

$$c_* := c_{\max}(p, q), \quad m := \min\{p, q\}, \quad d_* := 1 - c_*,$$

$$J(p, q) := X_0([q, 1 - p]), \quad \mathcal{D} := \mathbb{B}\left(O, \frac{\sqrt{3}}{2}d_*\right).$$

Then

$$\Lambda(\mathcal{D} \cup J(p, q)) \geq 1.$$

Under the common-pair hypotheses of the body, the exact same set is contained in U_C .

(b) $\rho > 2\eta$

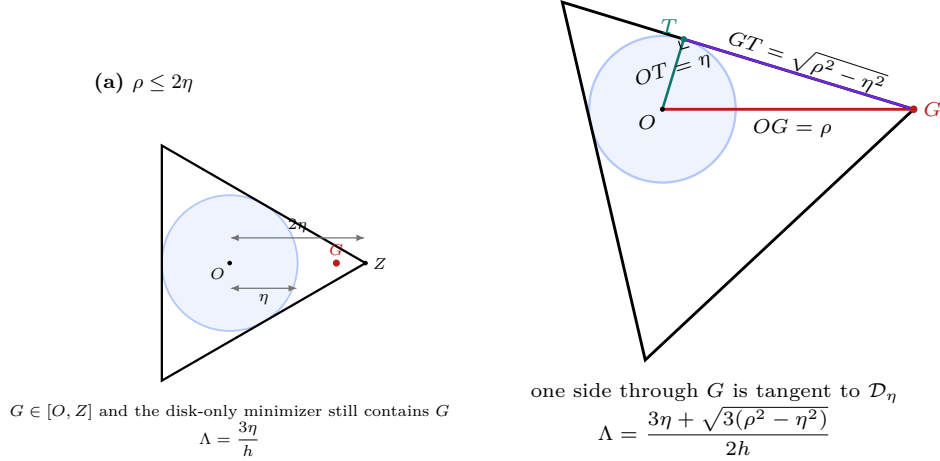


FIGURE 13. The two minimizing configurations in the disk-plus-point formula. For $\rho \leq 2\eta$, the centered equilateral triangle tangent to $\mathcal{D}_\eta$ already contains G . For $\rho > 2\eta$, the point G becomes a vertex of the minimizing triangle and one incident side is tangent to the disk. The tangent length $GT = \sqrt{\rho^2 - \eta^2}$ is the geometric source of the radical term.

Solution.

Proof of theorem D.12. For a unit normal n , put

$$s_j = \langle G, R^j n \rangle \quad (0 \leq j \leq 2).$$

Then $s_0 + s_1 + s_2 = 0$, $\sum_j s_j^2 = 3\rho^2/2$, and the support sum is

$$\sum_{j=0}^2 \max\{\eta, s_j\}.$$

If $\rho \leq 2\eta$, orient the projections as $\rho/2, \rho/2, -\rho$. All three disk supports are active, and the minimum is 3η .

Suppose $\rho > 2\eta$. By theorem D.9, a non-disk minimizer has a point-disk tie. If one projection is η , the sum and sum-of-squares identities determine the other active projection as

$$\frac{-\eta + \sqrt{3(\rho^2 - \eta^2)}}{2}.$$

The minimum support sum is therefore $2\eta + (-\eta + \sqrt{3(\rho^2 - \eta^2)})/2$. Multiplication by the enclosure-gauge factor $2/\sqrt{3}$ gives the two displayed branches and shows that their selector is exactly $\rho \leq 2\eta$ versus $\rho > 2\eta$. $\square$

Proof of theorem D.13. The farther endpoint G of $J(p, q)$ satisfies

$$\|G\|^2 = 1 - m + m^2.$$

Lemma D.12 reduces the nontrivial branch to

$$3d_*(1 - d_*) \geq m(1 - m).$$

By theorem D.2, $c_* \geq 1 - m \geq 1/2$. If $c_* \leq \sqrt{3}/2$, the left side is at least $3(\sqrt{3}/2)(1 - \sqrt{3}/2) > 1/4 \geq m(1 - m)$. Suppose $c_* > \sqrt{3}/2$, and put

$$\zeta_{\pm}(c) := \frac{c}{2} \left(1 \pm \sqrt{4c^2 - 3} \right).$$

We claim that every admissible nonsupercritical pair at radial coordinate c has $m \leq \zeta_-(c)$. On the L cell,

$$f_c(m) := c^4 - c^2 + cm - m^2 \leq 0,$$

whose roots are $\zeta_-(c), \zeta_+(c)$. The high-root alternative $m \geq \zeta_+(c)$ is impossible: $\zeta_+(c) > \sqrt{3}/4$, and for $M \geq m$ the selector $s^4 - s^2 + mM$ is increasing in M and at least $16m^4 - 3m^2 > 0$, whereas the L cell requires it to be nonpositive.

On the T cell,

$$F_T = (s^2 - 1)c^2 + Mc - M^2 \leq 0,$$

and

$$f_c(m) - F_T = (c - s)(Mc^2 - M + c^3 + c^2m + m) \leq 0.$$

Here the selected T value has $c \leq s$, while the second factor is positive (use $c \geq M$ and $c^3 \geq c(1 - c^2) \geq M(1 - c^2)$). Thus $f_c(m) \leq 0$, and the same selector excludes the high root. Consequently

$$m \leq \zeta(c_*) := \zeta_-(c_*) = c_* \left(\frac{1}{2} - \sqrt{c_*^2 - \frac{3}{4}} \right).$$

Here

$$0 \leq m \leq \zeta(c_*) < \frac{1}{2}.$$

Since $z \mapsto z(1 - z)$ is increasing on $[0, 1/2]$, it follows that

$$m(1 - m) \leq \zeta(c_*)(1 - \zeta(c_*)).$$

Direct substitution gives

$$\begin{aligned} 3c_*(1 - c_*) - \zeta(c_*)(1 - \zeta(c_*)) &= \frac{c_*(1 - c_*)}{2} \\ &\cdot \left(5 - 2c_* - 2c_*^2 + 2\sqrt{c_*^2 - \frac{3}{4}} \right) \geq 0. \end{aligned}$$

This proves $3d_*(1 - d_*) \geq m(1 - m)$. In the non-disk branch of theorem D.12, that inequality is exactly $\Lambda(\mathcal{D} \cup \{G\}) \geq 1$ for the farther gap endpoint G . In the disk-only branch, $\rho^2 = 1 - m + m^2 \geq 3/4$ and $\rho \leq 2(\sqrt{3}/2)d_*$ imply $d_* \geq 1/2$, and hence the enclosure value $3d_*$ is at least one. This proves the enclosure inequality in both branches, including $p + q = 1$ and hence singleton gaps. For containment, type-aware forcing puts the six vertices $(1 - c_*)V_i$ in U_C ; convexity contains their incircle $\mathcal{D}$, and the exact set difference defining the gap gives $J(p, q) \subset U_C$. $\square$

Body consequence. The disk-plus-gap bound, including the singleton endpoint, supplies theorem 5.9 and the enclosure conclusion of theorem 5.10.

D.11. Module D.11: CE2 short-ray optimization.

Exact interface. Use the signed CE2 center variables of theorem A.4 and the exact body points D_2, D_4 from theorem 5.11.

Variables and feasible set. The selected boundary pair is $p = W - \alpha$, $q = R - \delta$ with $e = \min\{\alpha, \delta\}$ and $M = \max\{R, W\}$, subject to the strict CE2 inequalities in the signed normal form. The local radial optimization uses the compact fiber defining $c_{\max}(p, q)$.

Objective. Prove the strict capacity estimate $c_{\max}(p, q) < 1 - e$ and identify which of the two opposite body points lies beyond the corresponding C-triangle exit.

Exact result.

Lemma D.14 (CE2 two-gap short-ray calculation). *Use the signed CE2 variables of theorem A.4 and put*

$$p := W - \alpha, \quad q := R - \delta, \quad e := \min\{\alpha, \delta\}, \quad M := \max\{R, W\}.$$

Then

$$0 \leq p, q \leq 1, \quad p + q \leq 1, \quad c_{\max}(p, q) < 1 - e.$$

Consequently the body points D_2, D_4 cannot both lie in T_C .

Solution.

Proof of theorem D.14. The strict CE2 inequalities give $P > (1 + M)e$. Since $E^2 = 1 - M + M^2$,

$$(1 + M)^2 - (3M - 1)^2 E^2 = 3M(1 - M)(3M^2 - 2M + 3) > 0.$$

Thus $\eta = P/E > (3M - 1)e$. The identities $RW = \eta + P$ give

$$Wq > \eta + e > 3Me, \quad Rp > \eta + e > 3Me,$$

and therefore

$$3e < p, q < 1 - 5e.$$

Moreover

$$e < \frac{P}{1 + M} \leq \frac{2\sqrt{3} - 3}{6} < \frac{1}{12}.$$

For $c = 1 - e$ put $f_c(s) = c^4 - c^2 + cs - s^2$. It is concave and

$$f_c(3e) = e(1 - 7e - 4e^2 + e^3) > 0, \quad f_c(1 - 5e) = e(2 - 15e - 4e^2 + e^3) > 0.$$

Hence $f_c(p), f_c(q) > 0$. Also $c > p + q$ and $p + c, q + c > 1$; the four support-contact values in theorem A.10 are

$$p + c, \quad q + c, \quad \frac{c^2}{\sqrt{p^2 - pc + c^2}}, \quad \frac{c^2}{\sqrt{q^2 - qc + c^2}}.$$

The first two exceed one by the preceding inequalities. The last two exceed one exactly when $f_c(p) > 0$ and $f_c(q) > 0$, respectively. Thus every exact support contact exceeds one, so $c_{\max}(p, q) < c = 1 - e$.

If $e = \delta$, then $D_2 = (1 - c_*)V_2$ lies strictly beyond the signed C exit δ ; if $e = \alpha$, then D_4 lies strictly beyond the exit α . This proves the final noncontainment with strictness in both branches. $\square$

Body consequence. The strict estimate and signed-exit alternative supply theorem 5.11.

D.12. Module D.12: transverse seven-point exclusion.

Exact interface. Use the exact seven-point set K_{tr} and the one-gap all-Vd0 normalization preceding theorem 5.13. The CE1 and CE2 center variables are those of theorem A.4; the boundary squeeze uses theorem A.7.

Variables and feasible set. For either signed center type put $X = R - \delta$ and $Q = (\eta + \alpha + \delta)/(2R)$. The five center-free handoffs and the retained radial points are exactly those required by theorems 5.14 and 5.15; no point on r_0 is part of the feasible witness.

Objective. Upper-bound the unique supercritical reach by $1 - Q$ from the actual gap endpoints, then rule out both possible signed center types for the same seven-point witness.

Exact result.

Lemma D.15 (Current transverse endpoint squeeze). *In the one-gap all-Vd0 normalization, suppose an open CE1 or CE2 unit triangle contains K_{tr} . Give that C triangle the signed variables of theorem A.4 and put*

$$X := R - \delta, \quad Q := \frac{\eta + \alpha + \delta}{2R}.$$

Then the actual reach of the unique supercritical role satisfies

$$A_0 < 1 - Q.$$

No radial point on r_0 is used.

Lemma D.16 (CE2 transverse threshold terminal). *No open CE2 unit equilateral triangle contains K_{tr} .*

Lemma D.17 (CE1 transverse return terminal). *No open CE1 unit equilateral triangle contains K_{tr} .*

Solution.

Proof of theorem D.15. The two actual gap endpoints lie in the open C trace. Thus

$$B_0 > \frac{\eta + \alpha + \delta}{R} = 2Q, \quad A_1 > R - \delta = X.$$

The diameter transfer theorem A.7 between the two incident boundary endpoints of T_0 gives

$$A_0 \leq M_0(B_0) < M_0(2Q) = -Q + \sqrt{1 - 3Q^2} < 1 - Q.$$

This is the gap-endpoint squeeze; the three radial points retained in K_{tr} are only P_2, P_3, P_4 . $\square$

Proof of theorem D.16. Suppose one does. The five gap-free handoffs and nonsuper-criticality give

$$A_1 < A_2 < A_3 < A_4 < A_5 < A_0.$$

Lemma D.15 then gives $B_2, B_4 > Q$. Containment of the retained points P_2, P_4 is strict and gives

$$C_2 > 1 - \delta, \quad C_4 > 1 - \alpha.$$

For completeness, put

$$X = R - \delta, \quad Q = \frac{\eta + \alpha + \delta}{2R}.$$

The right-trace inequality gives

$$WX = RW - W\delta = \eta + P - W\delta > \eta + \alpha.$$

Testing the low-root quadratic at this lower bound is equivalent to positivity of the concave function

$$\pi(z) := (\eta + z)(1 - 2z) - 2zW,$$

whose endpoint values are $\pi(0) = \eta > 0$ and $\pi(P) = \eta(\eta + 2ER^2) > 0$. Hence

$$\epsilon(\alpha) < X.$$

Putting $T = \alpha + \delta$, the CE2 inequalities also give

$$E^2T < P = E\eta, \quad T < \frac{\eta}{E}.$$

If $e = \min\{\alpha, \delta\}$, then

$$\min\{\epsilon(\alpha), \epsilon(\delta)\} < \frac{2e}{1 - 2e} \leq \frac{T}{1 - T} < Q.$$

The last comparison is the positivity on $0 \leq T \leq \eta/E$ of

$$\chi(T) := (\eta + T)(1 - T) - 2RT,$$

because $\chi(0) = \eta > 0$ and $\chi(\eta/E) = \eta(E - R)^2/E^2 > 0$.

If $\epsilon(\alpha) < Q$, apply theorem D.5 at T_4 to obtain $B_4 \leq \epsilon(\alpha) < Q$. Otherwise $\epsilon(\delta) < Q$, and its reflected application at T_2 gives $B_2 < Q$. Both conclusions contradict the preceding strict lower bounds. $\square$

Proof of theorem D.17. Suppose one does. The signed CE1 relations satisfy the hypotheses of theorem D.23. Containment of exactly the retained radial points P_4, P_3, P_2 supplies

$$C_4 > 1 - \alpha, \quad C_3 > 1 - \frac{\alpha}{R}, \quad C_2 > 1 - \delta.$$

That certificate gives $A_0 > 1 - Q$, whereas theorem D.15 gives $A_0 < 1 - Q$. This contradiction proves the claim. $\square$

Body consequence. The endpoint squeeze and the two signed exclusions supply theorems 5.14 and 5.15; together with the body's forcing lemma they prove theorem 5.16.

D.13. Module D.13: T3-like supported-tail adapter.

Exact interface. Use the T3-like supported interval, actual boundary reach, and four adapter inequalities displayed in theorem 5.17.

Variables and feasible set. The original T3-like role has actual boundary reach A_0 and radial trace $[c, u]$. Its closed-trace majorant has the same A_0 and u but a possibly smaller near endpoint $\hat{c} \leq c$. The majorant normal form has source parameter $1 \leq \omega \leq 2/\sqrt{3}$, with $\omega = D$ and $\varrho = R_{\text{loc}}$; its rationalized variables x, ϑ obey the exact interval displayed in the solution. The body fixes $0 \leq c \leq 1/2 \leq u < 1$ and $M = M_c^{\text{sup}}$.

Objective. Verify the four endpoint and envelope inequalities needed to invoke the common rescuer-tail budget for the T3-like branch.

Exact result.

Lemma D.18 (T3-like supported-tail adapter). *Under the hypotheses of theorem 5.17, all four adapter inequalities displayed there hold.*

Solution.

Proof of theorem D.18. The O-side endpoint $R_1(u) = (1 - u)V_1$ is missed by the original open T3-like role, the adjacent supercritical role, and all Vd0 or nonlocal roles; hence it belongs to U_C .

Now distinguish the translated closed-trace majorant $\hat{T}_0$ of theorem 2.7 from the original role. In the Type-II offsets of theorem A.3, the translation sets the transverse offset to zero and retains the other offset. Direct intersection with $e_{5,0}$ and r_1 therefore gives

$$\hat{A}_0 = A_0, \quad \hat{u} = u, \quad \hat{c} \leq c.$$

Only the near radial endpoint can move; the two actual quantities used by the body remain unchanged.

For the majorant write $\omega = D$ and $\varrho = R_{\text{loc}}$ as in the Proven 4132 normal form. Then $1 \leq \omega \leq 2/\sqrt{3}$; put

$$\varrho := \frac{\omega + \sqrt{4 - 3\omega^2}}{2}, \quad x := \frac{A_0\omega}{\varrho}, \quad \vartheta := \frac{\omega - 1}{\varrho},$$

so that

$$\hat{c} = x + \vartheta, \quad \frac{1 - 4\vartheta + \vartheta^2}{2(1 - 2\vartheta)} \leq x \leq \frac{1}{2} - \vartheta.$$

The parameter range gives

$$0 \leq \vartheta \leq 2 - \sqrt{3}.$$

Put

$$Q_\vartheta(x) := 2x^2 + (\vartheta - 1)x + \vartheta, \quad x_* := \frac{1 - \vartheta}{4}.$$

If $0 \leq \vartheta \leq 1/5$, the lower feasible endpoint

$$x_L := \frac{1 - 4\vartheta + \vartheta^2}{2(1 - 2\vartheta)}$$

satisfies

$$x_L - x_* = \frac{1 - 5\vartheta}{4(1 - 2\vartheta)} \geq 0.$$

Thus Q_ϑ is increasing on the feasible interval, and

$$Q_\vartheta(x) \geq Q_\vartheta(x_L) = \frac{\vartheta(1 - 5\vartheta + 11\vartheta^2 - \vartheta^3)}{2(1 - 2\vartheta)^2} \geq 0.$$

If $1/5 \leq \vartheta \leq 2 - \sqrt{3}$, then x_* lies in the feasible interval because

$$\left(\frac{1}{2} - \vartheta\right) - x_* = \frac{1 - 3\vartheta}{4} > 0,$$

and hence

$$Q_\vartheta(x) \geq Q_\vartheta(x_*) = \frac{10\vartheta - 1 - \vartheta^2}{8} > 0.$$

Put $\widehat{M} = M_c^{\text{sup}}$ and let $s(z) = z(2 - z)/(1 + z)$. This function is increasing on $[0, 1/2]$ and $s(1 - \widehat{M}) = \widehat{c}$. The inequality $Q_\vartheta(x) \geq 0$ is exactly $s(x) \leq x + \vartheta = \widehat{c}$, so $x \leq 1 - \widehat{M}$. Moreover the explicit formula for M_c^{sup} is decreasing in $c \in [0, 1/2]$, since

$$\frac{d}{dc} M_c^{\text{sup}} = \frac{1}{2} - \frac{4 - c}{2\sqrt{c^2 - 8c + 4}} < 0.$$

Thus $\widehat{c} \leq c$ gives $\widehat{M} \geq M$. Since $\omega \geq \varrho$ and the majorant radial endpoint identity, together with $\widehat{A}_0 = A_0$ and $\widehat{u} = u$, gives $A_0 + 1 - u = \varrho/\omega$, we obtain

$$A_0 \leq x \leq 1 - \widehat{M} \leq 1 - M, \quad \frac{A_0}{A_0 + 1 - u} = x \leq 1 - \widehat{M} \leq 1 - M.$$

Also $A_0 + 1 - u = \varrho/\omega \leq 1$. These are precisely the four displayed adapter inequalities. $\square$

Body consequence. Together with theorem D.11, these four inequalities supply theorem 5.17.

D.14. Module D.14: Vd1 supported-tail adapter.

Exact interface. Use the Vd1 supported interval, actual boundary reaches, and four adapter inequalities displayed in theorem 5.18.

Variables and feasible set. The Vd1 corner parameter satisfies $\theta \geq 1$, with $\Delta_\theta = \sqrt{\theta^2 + \theta + 1}$. The body interval obeys $0 \leq c \leq 1/2 \leq u < 1$, $\varepsilon = 1 - u$, and $M = M_c^{\text{sup}}$; the midpoint inequality selects the feasible corner component used below.

Objective. Bound the actual boundary endpoint and its normalized ratio by $1 - M$, and verify the endpoint containment and unit boundary-sum inequality required by the common rescuer-tail budget.

Exact result.

Lemma D.19 (Vd1 supported-tail adapter). *Under the hypotheses of theorem 5.18, all four adapter inequalities displayed there hold; in fact $A_0 + \varepsilon < 1$.*

Solution.

Proof of theorem D.19. In the Vd1 corner form there is initially $\theta > 0$ with $\Delta_\theta = \sqrt{\theta^2 + \theta + 1}$ such that

$$c = \frac{\theta(1 - B_0)}{\theta + 1}, \quad u = \frac{\Delta_\theta - A_0 - \theta B_0 - 1}{\theta}.$$

The midpoint condition gives

$$A_0 + \theta B_0 \leq \Delta_\theta - 1 - \frac{\theta}{2}.$$

The actual boundary reach A_0 is positive. If $\theta < 1$, the raw r_5 -trace endpoints from the corner form would satisfy

$$\Delta_\theta - A_0 - \theta B_0 - \theta \geq 1 - \frac{\theta}{2} > \frac{1}{\theta + 1} > \frac{1 - A_0}{\theta + 1},$$

giving positive support on both neighboring arms, contrary to the Vd1 type. Thus $\theta \geq 1$.

Eliminating $\theta B_0 = \theta - c(\theta + 1)$ yields

$$A_0 \leq F(\theta, c) := \Delta_\theta - 1 - \frac{3\theta}{2} + c(\theta + 1).$$

For fixed $c \leq 1/2$, F decreases with θ , so

$$A_0 \leq F(1, c) = \sqrt{3} - \frac{5}{2} + 2c =: L(c).$$

The inequality $2L(c) \leq 1 - M_c^{\text{sup}}$ is equivalent to

$$20c^2 + (18\sqrt{3} - 52)c + 47 - 24\sqrt{3} \geq 0,$$

The quadratic is decreasing on $[0, 1/2]$ and its value at $1/2$ is $26 - 15\sqrt{3} > 0$, so the inequality holds throughout that interval. Hence $A_0 \leq 1 - M_c^{\text{sup}}$.

Put $\varepsilon = 1 - u$. Direct substitution gives

$$\varepsilon = \varepsilon_0 + \frac{A_0}{\theta}, \quad \varepsilon_0 = \frac{2\theta + 1 - c(\theta + 1) - \Delta_\theta}{\theta} > 0.$$

Indeed, $c \leq 1/2$ makes its numerator at least $(3\theta + 1)/2 - \Delta_\theta > 0$, since

$$(3\theta + 1)^2 - 4\Delta_\theta^2 = (5\theta - 3)(\theta + 1) > 0 \quad (\theta \geq 1).$$

The map $z \mapsto z/(z + \varepsilon_0 + z/\theta)$ is increasing, while $\varepsilon_0 + F(\theta, c)/\theta = 1/2$. Therefore

$$\frac{A_0}{A_0 + \varepsilon} \leq \frac{F(\theta, c)}{F(\theta, c) + 1/2} \leq 2L(c) \leq 1 - M_c^{\text{sup}}.$$

The strict Vd half-unit cap and $u \geq 1/2$ give $A_0 + \varepsilon < 1$. The O-side endpoint $R_1(u) = \varepsilon V_1$ is missed by all open V roles and hence belongs to U_C . $\square$

Body consequence. Together with theorem D.11, these four inequalities supply theorem 5.18.

D.15. Module D.15: adjacent Vd radial separation.

Exact interface. Use the adjacent alternative $(\sigma, \tau) = (0, 1)$ in theorem 5.21, the exact residual operator theorem 5.20, and the radial gap interface theorem 5.19 at $i = 2$. The signed small-slack and diameter inputs are theorems A.7 and B.9.

Variables and feasible set. Normalize the CE2 traces as in the body. With the signed variables of theorem A.4, put $k = \eta + \alpha + \delta$ and

$$I_L = \left[\frac{k}{W}, R + \alpha \right], \quad I_R = \left[\frac{k}{R}, W + \delta \right],$$

$\rho_R = \mathcal{R}_{I_R}(B_0)$, and $\rho_L = \mathcal{R}_{I_L}(A_0) > 0$ in the reduced candidate. The center-free handoffs give $A_1 \geq \rho_R$, $B_1 \geq \rho_L$, and $B_2 \geq \rho_L$.

Objective. Prove the strict separation margin $c_{\text{loc}} < 1 - d_C$ by comparing the left residual with the C exit and the adjacent Vd0 radial envelope.

Exact result.

Lemma D.20 (Adjacent Vd radial separation). *In the adjacent subcase $(\sigma, \tau) = (0, 1)$ of theorem 5.21, the radial interface at $i = 2$ satisfies*

$$c_{\text{loc}} < 1 - d_C.$$

Solution.

Proof of theorem D.20. With the exact intervals displayed above, set

$$\rho_R = \mathcal{R}_{I_R}(B_0), \quad \rho_L = \mathcal{R}_{I_L}(A_0).$$

The center-free handoffs give

$$A_1 \geq \rho_R, \quad B_1 \geq \rho_L, \quad B_2 \geq \rho_L.$$

Indeed, the definition of $\mathcal{R}_{I_R}(B_0)$ makes the first uncovered tail of $e_{0,1}$ have length ρ_R , so open coverage forces $A_1 \geq \rho_R$. On the reflected boundary chain the corresponding tail first forces $B_5 \geq \rho_L$. Successively, the center-free handoffs $B_4 + A_5 \geq 1$, $B_3 + A_4 \geq 1$, and $B_2 + A_3 \geq 1$, combined with the nonsupercritical caps at T_5, T_4, T_3 , give $B_4 \geq B_5$, $B_3 \geq B_4$, and $B_2 \geq B_3$. Finally $B_1 + A_2 \geq 1$ and the cap at T_2 give $B_1 \geq B_2 \geq \rho_L$. The Vd half-unit cap at T_1 therefore gives

$$\rho_R + \rho_L < \frac{1}{2}.$$

The exact CE2 budget theorem B.9 gives both

$$\alpha + \delta < \frac{1}{24}, \quad \alpha + \delta < \frac{1}{6} \min\{R, W\}.$$

If $\rho_L = W - \alpha$, then

$$4\delta + \alpha \leq 4(\alpha + \delta) < \frac{2W}{3} < W,$$

so $4\delta < \rho_L$. Otherwise

$$\rho_L = 1 - A_0.$$

The other residual cannot be $1 - B_0$, since then $\rho_R + \rho_L < 1/2$ would give $A_0 + B_0 > 3/2$, contrary to $A_0 + B_0 \leq 2/\sqrt{3} < 3/2$ from theorem 3.2. Hence the exact residual alternatives give

$$\rho_R = 1 - (W + \delta) = R - \delta, \quad B_0 \geq \frac{k}{R}.$$

The endpoint diameter bound in theorem A.7 gives

$$\rho_L > \frac{k}{2R} > \frac{\eta}{2R}.$$

Together with $\rho_R + \rho_L < 1/2$, this implies

$$\delta > J(R) := R - \frac{1}{2} + \frac{\eta}{2R}.$$

The function J is strictly increasing: using $E^2 = 1 - RW$,

$$4E(1 + E)^2 J'(R) = 2E(1 + E)(1 + 2E) - W(2R - 1) > 0.$$

For $R \leq 1/2$ the subtracted term is nonpositive; for $R \geq 1/2$ it is at most $1/8$, while the first term is greater than $1/8$. Since $J(3/8) = 1/24$ whereas $\delta < 1/24$, one has $R < 3/8$. Now

$$(2 - 3R)^2 - E^2 = (3 - 8R)(1 - R) > 0,$$

one gets $\eta/R > 1/3$ and again $\rho_L > 1/6 > 4\delta$.

The Vd cap and the handoff $A_1 \geq \rho_R$ give $B_1 < 1/2 - \rho_R$. Coverage of the center-free edge $e_{1,2}$ and the third handoff therefore give

$$A_2 > 1 - B_1 > \frac{1}{2} + \rho_R, \quad B_2 \geq \rho_L.$$

Thus theorem D.10 gives

$$C_2 < 1 - \frac{\rho_L}{3} < 1 - \delta.$$

If T_1 has positive support on r_2 , its supported Vd endpoint is below $1 - \rho_L < 1 - \delta$ by theorem D.8; if it has no positive support there, no endpoint comparison is needed. Thus both local V traces stop strictly before the C interval on r_2 . This is exactly $c_{\text{loc}} < 1 - d_C$. $\square$

Body consequence. This strict local/C separation supplies the adjacent alternative of theorem 5.21.

D.16. Module D.16: nonadjacent Vd radial separation.

Exact interface. Use the nonadjacent alternative $\tau \in \{2, 3, 4\}$ in theorem 5.21, including the intervals I_R, I_L , the residual operator theorem 5.20, and the body point $D_\tau = \rho V_\tau$. The endpoint comparison uses theorem A.7.

Variables and feasible set. Put $x = k/W$, $y = k/R$ and $\rho = \min\{\rho_R, \rho_L\}$ for the two positive residuals of the normalized CE2 trace intervals. The routed C exit is the attained maximum d_τ^C , the actual Vd own-radial reach is C_τ , and all other roles retain the body type restrictions. The residual handoffs to the Vd role are $A_\tau \geq \rho_R$ and $B_\tau \geq \rho_L$.

Objective. Establish strict inequalities on both sides of ρ and exclude every nonlocal unit-diameter role from the selected radial point.

Exact result.

Lemma D.21 (Nonadjacent Vd radial separation). *In the nonadjacent subcase of theorem 5.21,*

$$d_\tau^C < \rho < 1 - C_\tau,$$

and every nonlocal role misses $D_\tau = \rho V_\tau$.

Solution.

Proof of theorem D.21. Let

$$x := \frac{k}{W}, \quad y := \frac{k}{R},$$

and define the farthest already-covered boundary extents by

$$B := \sup\{s \in [0, 1] : [0, s] \subseteq [0, B_0] \cup I_R\} = 1 - \rho_R,$$

$$U := \sup\{s \in [0, 1] : [0, s] \subseteq [0, A_0] \cup I_L\} = 1 - \rho_L.$$

The boundary handoffs through the intervening nonsupercritical roles force

$$A_\tau \geq \rho_R, \quad B_\tau \geq \rho_L.$$

For the first inequality, the residual definition says that the uncovered tail after the component of $[0, B_0] \cup I_R$ has length ρ_R ; coverage forces that demand onto the first role of the right-hand path. On every subsequent center-free edge, $B_j + A_{j+1} \geq 1$ and $A_j + B_j \leq 1$, so induction carries the lower bound unchanged to A_τ . The

reflected path starting from $[0, A_0] \cup I_L$ proves $B_\tau \geq \rho_L$ in the same way. The strict Vd half-unit cap then gives

$$\rho_R + \rho_L < \frac{1}{2}.$$

We claim $B \leq M_0(x)$; the reflected argument gives $U \leq M_0(y)$. By the exact interval-component definition, B is either B_0 or the far endpoint of the C interval I_R . In the latter case the C triangle contains adjacent boundary endpoints with parameters x, B , so theorem A.7 gives $B \leq M_0(x)$. If instead $B = B_0$ and $A_0 < x$, then the component on the other active edge ends at A_0 , so $U = A_0$. The preceding residual cap would imply $A_0 + B_0 > 3/2$, contradicting $A_0 + B_0 \leq 2/\sqrt{3} < 3/2$ from theorem 3.2. Hence $A_0 \geq x$, and the endpoint-distance inequality plus monotonicity of M_0 gives

$$B = B_0 \leq M_0(A_0) \leq M_0(x).$$

Reflection proves $U \leq M_0(y)$.

Because $1 - M_0(z) > z/2$ for $z > 0$, the residuals satisfy

$$\rho_R > \frac{x}{2}, \quad \rho_L > \frac{y}{2}.$$

The signed center-slack estimate in theorem B.10 now yields

$$\alpha + \delta < \min\{\rho_R, \rho_L\} = \rho.$$

Every relevant C exit is at most $\alpha + \delta$, so $d_\tau^C < \rho$. Applying theorem D.8 with the selected lower bounds $a_* = \rho_R$, $b_* = \rho_L$ gives $C_\tau < 1 - \rho$. Adjacent Vd0 roles stop no later, and for a nonlocal vertex V_j the exact distances

$$\|\rho V_\tau - V_{\tau \pm 2}\|^2 = 1 + \rho + \rho^2 > 1, \quad \|\rho V_\tau - V_{\tau+3}\| = 1 + \rho > 1$$

exclude its open unit-diameter role. Hence D_τ is missed by every V role and lies beyond the C exit. $\square$

Body consequence. These two strict radial estimates and the nonlocal distance audit supply the nonadjacent alternative of theorem 5.21.

D.17. Module D.17: CE2 one-Vd placement audit.

Exact interface. The body interface is the implication in theorem 5.23:

$$N_E(T_C) = 2, \quad N_{\text{gap}} \in \{1, 2\}, \quad N_+ = 1, \quad d = 1 \implies S \not\subseteq U_C \cup \bigcup_{i=0}^5 U_i.$$

Variables and feasible set. Use the actual reaches A_i, B_i , the normalized types $(o(T_i), n(T_i))$, and the exact counts

$$d = |\{i : (o(T_i), n(T_i)) \in \{(1, 1), (1, 2)\}\}|, \\ t = |\{i : (o(T_i), n(T_i)) = (2, 1)\}|, \quad N_{\text{sp}} = d + t.$$

All indices lie in $\mathbb{Z}/6\mathbb{Z}$. The only assumed containment in the feasibility test is the hypothetical open cover $S \subseteq U_C \cup \bigcup_i U_i$.

Objective. Exhaust the relative positions of the unique supercritical role, the unique Vd1/Vd2 role, and the center-covered midpoint, while retaining every center-free and radial-locality hypothesis needed by the terminal to which a placement is routed.

Exact result.

Lemma D.22 (CE2 one-Vd placement audit). *Assume*

$$N_E(T_C) = 2, \quad N_{\text{gap}} \in \{1, 2\}, \quad N_+ = 1, \quad d = 1.$$

Under a hypothetical cover of S , if $t \geq 1$, then $N_+ + N_{\text{sp}} \geq 3$. Otherwise $t = 0$, and after a dihedral symmetry

$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\}.$$

There are unique indices

$$\{\sigma\} = \{i : A_i + B_i > 1\}, \quad \{\tau\} = \{i : (o(T_i), n(T_i)) \in \{(1, 1), (1, 2)\}\},$$

with $\sigma \neq \tau$ and

$$(o(T_j), n(T_j)) \in \{(1, 0), (2, 0)\} \quad (j \neq \tau).$$

Exactly one of the following alternatives holds.

- (1) $\sigma = 0$, with $\tau \in \{1, 5\}$ or $\tau \in \{2, 3, 4\}$; these are respectively the adjacent and nonadjacent hypotheses of theorem 5.21.
- (2) $\tau = 0$, $M_\sigma \in T_0$, and $\sigma \in \{1, 5\}$. After reflection take $\sigma = 1$. Type Vd2 satisfies the hypotheses of theorem 3.5; type Vd1 has

$$T_0 \cap r_1 = R_1([c, u]), \quad 0 \leq c \leq \frac{1}{2} \leq u < 1, \quad C_1 \geq c,$$

and satisfies the hypotheses of theorem 5.18.

- (3) $\sigma \neq 0$ and $\tau \neq 0$, with $M_\sigma \in T_\tau$ and $\tau - \sigma \equiv \pm 1 \pmod{6}$. Type Vd2 satisfies the hypotheses of theorem 3.5. In type Vd1, rotate and, if necessary, reflect so that $(\tau, \sigma) = (0, 1)$. Then

$$\begin{aligned} (o(T_0), n(T_0)) &= (1, 1), \quad M_1 \in T_0, \quad A_1 + B_1 > 1, \\ A_i + B_i &\leq 1 \quad (i \neq 1), \quad (o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\} \quad (1 \leq i \leq 5), \\ U_C \cap e_{0,1} &= \emptyset, \quad \mathcal{H}^1(T_j \cap r_k) = 0 \quad (2 \leq j \leq 5, k \in \{0, 1\}), \end{aligned}$$

so every hypothesis of theorem 5.22 holds.

Consequently the hypothetical cover is impossible.

Solution.

Proof of theorem D.22. Assume for contradiction that the seven open roles cover S . By theorem 2.13, a dihedral normalization gives

$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\}.$$

If $t \geq 1$, then $d = 1$ gives

$$N_+ + N_{\text{sp}} = 1 + (d + t) \geq 3,$$

contradicting theorem 3.4. Hence assume $t = 0$.

The equality $N_+ = 1$ defines the unique index σ , and $d = 1$ defines the unique index τ . By the strict half-unit boundary cap in theorem B.5, a Vd1/Vd2 role is nonsupercritical; therefore $\sigma \neq \tau$. The normalized classification theorem 2.6, together with $t = 0$ and $d = 1$, gives

$$(o(T_j), n(T_j)) \in \{(1, 0), (2, 0)\} \quad (j \neq \tau).$$

If $\sigma = 0$, the alternatives $\tau \in \{1, 5\}$ and $\tau \in \{2, 3, 4\}$ exhaust all possible $\tau \neq 0$. They are exactly the adjacent and nonadjacent branches of theorem 5.21, so both contradict the cover.

Suppose next that $\tau = 0$. Then $\sigma \neq 0$. The center misses M_σ by the normalized midpoint identity, while the supercritical role T_σ misses its own midpoint by theorem 2.12. A nonincident V role cannot contain M_σ because its indexed vertex is more than unit distance away. If an incident Vd0 role contained the point in its open interior, it would have positive-length trace on r_σ , contrary to $n = 0$. Thus the cover forces $M_\sigma \in U_0 \subset T_0$, and diameter locality gives $\sigma \in \{1, 5\}$. Reflect so that $\sigma = 1$. For type Vd2, theorem 3.5 contradicts the cover. For type Vd1, its exact corner trace gives

$$T_0 \cap r_1 = R_1([c, u]), \quad 0 \leq c \leq \frac{1}{2} \leq u.$$

In fact $u < 1$: otherwise $O \in T_0$, impossible because $V_0 \in U_0$ and $\|V_0 - O\| = 1$ while an interior point and a point of the closure of an open unit triangle are at distance strictly less than one. The signed exit description gives $d_1^C < 1/2$, so the center trace on r_1 starts, in the coordinate measured from V_1 , strictly after $1/2$. If $c = 0$, the required inequality $C_1 \geq c$ is automatic. If $c > 0$, then immediately before the near endpoint c the Vd1 role is absent; the adjacent Vd0 and nonlocal roles have no positive trace there. The open cover therefore forces the own-radial trace of U_1 up to c , again giving $C_1 \geq c$. All remaining roles are Vd0 and only T_1 is supercritical. These are precisely the hypotheses of theorem 5.18, which again contradicts the cover.

It remains that $\sigma \neq 0$ and $\tau \neq 0$. The same midpoint argument shows that the only possible rescuer is T_τ :

$$M_\sigma \in U_\tau \subset T_\tau, \quad \tau - \sigma \equiv \pm 1 \pmod{6}.$$

The Vd2 alternative is closed by theorem 3.5. In the Vd1 alternative, rotate and reflect so that $(\tau, \sigma) = (0, 1)$. Uniqueness of σ and τ gives the two displayed reach and type lines in the statement. The center's unique midpoint has an index different from 0, 1 in this fresh normalization. Since the two positive CE2 traces are the adjacent traces at that midpoint, the signed trace description in theorem A.4 gives

$$U_C \cap e_{0,1} = \emptyset.$$

For r_0 , the only adjacent role among $T_2, \dots, T_5$ is T_5 ; for r_1 it is T_2 . Both have $n = 0$, so their indicated intersections have zero length. For every remaining pair, write a ray point as dV_k with $0 \leq d \leq 1$. If V_j is at cyclic distance two from V_k , or is its opposite vertex, then respectively

$$\|dV_k - V_j\|^2 = 1 + d + d^2, \quad \|dV_k - V_j\| = 1 + d.$$

These distances exceed one for $d > 0$; at $d = 0$ they permit at most the single endpoint O . Hence the three nonlocal roles have zero-length intersection with the ray. Consequently

$$\mathcal{H}^1(T_j \cap r_k) = 0 \quad (2 \leq j \leq 5, k \in \{0, 1\}).$$

Thus theorem 5.22 applies and contradicts the cover. The positive-support branch and the three positional alternatives are pairwise disjoint and exhaustive, completing the audit. $\square$

Body consequence. This exact placement partition, including the Vd2 length exits, both radial separation positions, the supported Vd1 position, and the full two-chart input hypotheses, supplies theorem 5.23.

D.18. Module D.18: CE1 one-gap reverse-path certificate.

Exact interface. Use the exact seven-point witness and the five center-free handoffs of theorem 5.15. Only the retained radial reaches at T_4, T_3, T_2 enter; no radial point on r_0 is part of this interface.

Variables and feasible set. Take the signed CE1 variables $0 < R < 1$, $w = 1 - R$, $E = \sqrt{1 - Rw}$, $\eta = 1 - E$, and $P = E(1 - E)$. The positive slacks $A = \alpha$ and $D = \delta$ satisfy the two signed CE1 inequalities displayed below. The five V roles have the stated type, handoff, and three closed radial-demand constraints.

Objective. Starting from a hypothetical upper bound on the returning actual reach A_0 , propagate the selected high-radial output twice and force a final contradiction at T_1 .

Exact result.

Proposition D.23 (CE1 three-transverse return certificate). *Use the signed CE1 center variables*

$$\begin{aligned} 0 < R < 1, \quad w = 1 - R, \quad E = \sqrt{1 - Rw}, \\ \eta = 1 - E, \quad P = E(1 - E), \end{aligned}$$

and write

$$A = \alpha, \quad D = \delta.$$

Assume

$$A > 0, \quad D > 0, \quad A + wD < P, \quad D + RA \geq P.$$

Put

$$X = R - D, \quad m = \frac{A}{R}, \quad h_0 = \frac{\eta + A + D}{2R}.$$

Let T_0 be the unique supercritical role and let $T_1, \dots, T_5$ be nonsupercritical Vd0 roles. Assume

$$\begin{aligned} A_1 &\geq X, \\ B_i + A_{i+1} &\geq 1 \quad (1 \leq i \leq 4), \quad B_5 + A_0 \geq 1, \end{aligned}$$

and

$$C_4 \geq 1 - A, \quad C_3 \geq 1 - m, \quad C_2 \geq 1 - D.$$

Then

$$\boxed{A_0 > 1 - h_0.}$$

Solution.

Proof of theorem D.23. The sign difference in (D.23) gives

$$RD > wA,$$

so the center exit on r_3 is $m = A/R$. Moreover

$$0 < A < P < \frac{1}{8}, \quad 0 < D < \frac{R}{2}, \quad 0 < m < \frac{w}{2} < \frac{1}{2}.$$

We first record

$$\epsilon(A) < X, \quad h_0 > A.$$

Indeed, $A + wD < P$ and $D = R - X$ give $A < wX - \eta$. Convexity of

$$wX - \frac{X}{2(1+X)}$$

CE1 reverse path: from a deficient terminal to the final contradiction

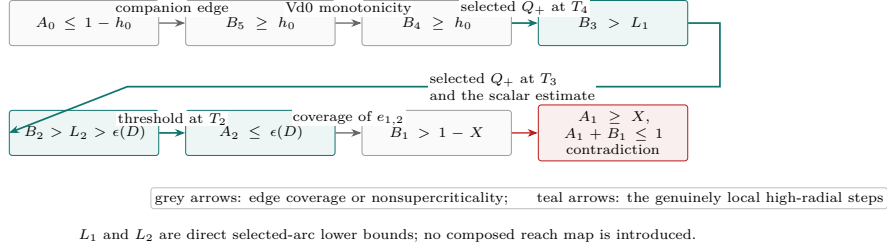


FIGURE 14. Logical structure of the CE1 direct reverse-path certificate. The proof starts from the negation of the desired terminal bound at T_0 , moves through the actual boundary reaches, uses two selected-arc estimates and one high-radial threshold, and returns to T_1 with incompatible incoming and outgoing demands.

on $R/2 < X < R$ gives

$$A < \frac{X}{2(1+X)}.$$

The low-root estimates preceding theorem D.5 yield $\epsilon(A) < X$. Also

$$2R(h_0 - A) = \eta + D + (1 - 2R)A > 0;$$

when $R > 1/2$, use $A < P = \eta E$ for the last sign.

Suppose, contrary to (D.23), that

$$A_0 \leq 1 - h_0.$$

The assumed final handoff gives $B_5 \geq h_0$. Nonsupercriticality of T_5 and the handoff $B_4 + A_5 \geq 1$ imply

$$B_4 \geq B_5 \geq h_0.$$

The assumed transverse radial reach at T_4 is at least $1 - A$. Apply the exact local catalogue to the reflected boundary pair (B_4, A_4) . If the local point is not on the selected Q_+ component, the constant, Q_- , and linear components together with (D.18) give

$$1 - A_4 \geq 1 - \epsilon(A) > 1 - X.$$

The center-free edges then carry this lower bound backward to $B_1 > 1 - X$, contradicting $A_1 \geq X$ and $A_1 + B_1 \leq 1$.

Hence the T_4 state is selected Q_+ . Lemma D.7 and (D.18) give

$$B_3 > L_1, \quad L_1 = (2 - 4A)h_0 - (1 - 4A)A.$$

The selected-state condition also forces

$$X > \frac{1}{2}.$$

To see this, selectedness gives

$$h_0 \leq \epsilon(A) < \frac{2A}{1 - 2A}.$$

If $X \leq 1/2$, the lower CE1 inequality gives

$$2Rh_0 = \eta + A + D \geq w(R + A).$$

Consequently

$$f_R(A) < 0, \quad f_R(z) = w(R+z)(1-2z) - 4Rz.$$

The quadratic is strictly concave and $f_R(0) > 0$. If $R \leq 1/2$, then

$$f_R(P) \geq \frac{1-E}{2}(6E^3 - 2E^2 - 5E + 2) > 0.$$

If $R > 1/2$, then $A < A_* = w/2 - \eta$ and

$$f_R(A_*) = \frac{1-E}{2}(E + 11R - 3ER - 5) > 0.$$

Concavity contradicts $f_R(A) < 0$ in both cases, proving (D.18). Thus

$$m < \frac{w}{2} < \frac{1}{4}.$$

If $B_3 \geq 1/2$, then $B_3 > 1 - X$ and the preceding immediate contradiction applies. Assume $B_3 < 1/2$. At T_3 the reflected incoming reach is $B_3 > h_0 > m$, while its assumed transverse radial reach is at least $1 - m$. If this state is not selected Q_+ , the exact local catalogue gives

$$B_2 \geq 1 - B_3 > \frac{1}{2} > 1 - X,$$

and the remaining center-free edge again contradicts T_1 .

Thus both T_4 and T_3 are selected Q_+ . A second use of Lemma D.7 gives

$$B_2 > (2 - 5m)B_3 - (1 - 5m)m.$$

Since $2 - 5m > 0$, (D.18) yields

$$B_2 > L_2, \quad L_2 = (2 - 5m)L_1 - (1 - 5m)m.$$

We now prove the scalar estimate

$$D < \frac{1}{10}, \quad L_2 > 2D + 3D^2 > \epsilon(D), \quad \epsilon(D) < X.$$

The T_4 selected-state inequality and the bound $\epsilon(A) \leq 2A + 5A^2$ give

$$D \leq D_h := A(4R - 1) + 10RA^2 - \eta.$$

If $D \geq 1/10$, then $A + wD < P$ gives

$$A < \bar{A} = \frac{w(5R - 1)}{10}.$$

The right side of (D.18) is increasing in A , while $\eta > Rw/2$, so

$$D < U(R) := \bar{A}(4R - 1) + 10R\bar{A}^2 - \frac{Rw}{2}.$$

Direct expansion gives

$$\frac{1}{10} - U(R) = -\frac{R}{10}P_4(R),$$

where

$$P_4(R) = 25R^4 - 60R^3 + 26R^2 + 22R - 14.$$

For $y = 2R - 1 \in (0, 1)$,

$$16P_4(R) = 25y^4 - 20y^3 - 106y^2 + 124y - 39 \leq -101y^2 + 124y - 39 < 0.$$

Hence $U(R) < 1/10$, a contradiction. This proves $D < 1/10$.

Put

$$\Psi(D) = L_2 - 2D - 3D^2, \quad J(D) = L_2 - \frac{23}{10}D.$$

Since $D < 1/10$,

$$\Psi(D) - J(D) = 3D \left(\frac{1}{10} - D \right) > 0.$$

For fixed R, A , differentiation of (D.18) gives

$$J'(D) = \frac{S(R, A)}{10R^2},$$

where

$$S(R, A) = 100A^2 - (40R + 50)A + R(20 - 23R).$$

The nonempty interval $P - RA \leq D \leq D_h$ implies

$$A > \frac{4Rw}{25R - 4} =: A_*.$$

On the relevant interval $\partial S / \partial A < -45$, and

$$S(R, A) < S(R, A_*) = -\frac{Rq_3(R)}{(25R - 4)^2},$$

where

$$q_3(R) = 8775R^3 - 14260R^2 + 7928R - 1120.$$

For $y = 2R - 1$,

$$8q_3(R) = 8775y^3 - 2195y^2 + 997y + 3007 > 0.$$

Thus J decreases with D , so $J(D) \geq J(D_h)$.

At $D = D_h$, one has $h_0 = 2A + 5A^2$. If $L_{2,h}$ is the corresponding value, then

$$L_{2,h} - A(1 + 4R) = \frac{A}{R^2}Q(R, A),$$

where

$$Q(R, A) = 100A^3R - 40A^2R^2 - 30A^2R + 12AR^2 - 15AR + 5A - 4R^3 + 5R^2 - R.$$

This polynomial is concave in A on $0 \leq A \leq Rw/2$. Its endpoint values are

$$Q(R, 0) = Rw(4R - 1) > 0$$

and

$$Q\left(R, \frac{Rw}{2}\right) = \frac{Rw}{2}(25R^5 - 30R^4 + 20R^3 - 3R^2 - 7R + 3) > 0.$$

The last positivity follows after putting $y = 2R - 1$:

$$32p(R) = 25y^5 + 65y^4 + 90y^3 + 106y^2 - 35y + 5 > 0.$$

Hence $L_{2,h} > A(1 + 4R)$.

Finally $A < P = \eta E$, $A < Rw/2$, and $1/E > 1 + Rw/2$ give

$$\frac{D_h}{A} < C_* := 4R - 2 + 5R^2w - \frac{Rw}{2}.$$

Moreover

$$1 + 4R - \frac{23}{10}C_* = \frac{h_3(R)}{20}, \quad h_3(R) := 230R^3 - 253R^2 - 81R + 112.$$

Here (D.18) gives $R > 1/2$, and therefore $h_3''(R) = 1380R - 506 \geq 184$. Strong convexity about $R_0 = 20/23$, where

$$h_3(R_0) = \frac{788}{529}, \quad h_3'(R_0) = \frac{17}{23},$$

gives the exact lower bound

$$h_3(R) \geq h_3(R_0) + h_3'(R_0)(R - R_0) + 92(R - R_0)^2 \geq \frac{289695}{194672} > 0.$$

Thus $J(D_h) > 0$, so

$$L_2 > 2D + 3D^2.$$

To compare this value with the selected low root, substitute $z = 2D + 3D^2$ directly into

$$p_D(z) := z^2 - (1 - D)z + (1 - D)^2 - (1 - D)^4.$$

The result is

$$p_D(2D + 3D^2) = D^2(8D^2 + 19D - 2) < 0 \quad (0 < D < 1/10).$$

Thus $2D + 3D^2$ lies strictly between the two roots of p_D , and in particular $\epsilon(D) < 2D + 3D^2$. The argument proving $\epsilon(A) < X$ also gives $\epsilon(D) < X$. This proves (D.18).

The assumed transverse radial reach at T_2 is at least $1 - D$. Since $B_2 > \epsilon(D)$, Lemma D.5, applied after reflection, gives

$$A_2 \leq \epsilon(D).$$

Coverage of $e_{1,2}$ therefore forces

$$B_1 \geq 1 - A_2 \geq 1 - \epsilon(D) > 1 - X,$$

contradicting $A_1 \geq X$ and $A_1 + B_1 \leq 1$. Hence (D.18) is impossible. $\square$

Body consequence. This strict return estimate is the long CE1 supplier used by theorem D.17, and hence proves the compact body statement theorem 5.15.

D.19. Module D.19: two-chart Vd1 replacement.

Exact interface. Use exactly the normalized input roles, output reaches, output open traces, and recomputed gap rank of theorem 5.22. The input and output gap ranks are distinct objects.

Variables and feasible set. The auxiliary admissible triple $(x, y, z) \in \mathcal{A}$ has $x \geq 1/2$ and $0 \leq z \leq 1/2$. For the replacement, the input is CE2 with $N_{\text{gap}} \in \{1, 2\}$, one supercritical role, the displayed Vd1 role, and the exact center-free trace hypotheses below. The chart parameters p_1, p_2, ε are selected by strict inequalities inside their nonempty open feasible intervals.

Objective. Construct distinct V_0 and V_1 replacement charts that preserve the five affected skeleton pieces with strict margins, make all six V roles nonsupercritical Vd0, and route from the recomputed output gap rank.

Exact result.

Lemma D.24 (Half-square admissibility). *If $(x, y, z) \in \mathcal{A}$ satisfies*

$$x \geq \frac{1}{2}, \quad 0 \leq z \leq \frac{1}{2},$$

then

$$y \leq 1 - z.$$

Theorem D.25 (Two-chart replacement and recomputed-gap router). *Assume*

$$\begin{aligned} S &\subseteq U_C \cup \bigcup_{i=0}^5 U_i, \quad N_E(T_C) = 2, \quad N_{\text{gap}} \in \{1, 2\}, \quad N_+ = 1, \\ (o(T_0), n(T_0)) &= (1, 1), \quad M_1 \in T_0, \quad A_1 + B_1 > 1, \quad A_i + B_i \leq 1 \quad (i \neq 1), \\ (o(T_i), n(T_i)) &\in \{(1, 0), (2, 0)\} \quad (1 \leq i \leq 5), \quad U_C \cap e_{0,1} = \emptyset, \\ \mathcal{H}^1(T_j \cap r_k) &= 0 \quad (2 \leq j \leq 5, \quad k \in \{0, 1\}). \end{aligned}$$

Then there are two explicitly constructed open unit equilateral triangles U'_0, U'_1 , each containing its indexed vertex, for which all conclusions of theorem 5.22 hold. In particular, the output gap rank is recomputed from the output open traces and is not identified with the input gap rank.

Solution.

Proof of theorem D.24. Suppose $y > 1 - z$. Then $x + y > 1$, so the selected supercritical cell of theorem A.10 applies. In the ordered half $x \leq y$ its necessary polynomial is

$$F(x, y, z) := (x^2 - 1)z^2 + (2xy^2 + y)z + y^4 - y^2 \leq 0.$$

It is nondecreasing in x ; its derivative in y , starting at $x = 1/2$, is positive for $y \geq 1 - z$; and

$$F\left(\frac{1}{2}, 1 - z, z\right) = \frac{z^2(2z - 5)(2z - 1)}{4} \geq 0.$$

This contradicts the selected-cell inequality, with equality lying only on its boundary. In the reflected half $y < x$, apply the same argument to $F(y, x, z)$. At the corner one has

$$F(1 - z, 1 - z, z) = z(1 - 2z) \geq 0,$$

which gives the same contradiction. $\square$

Proof of theorem D.25. The normalized Vd1 role is T_0 , the unique supercritical role is T_1 , and only these two roles have positive trace on r_0 or r_1 . The exact Vd1 corner form supplies $\theta \geq 1$ and

$$\Delta_\theta := \sqrt{\theta^2 + \theta + 1}$$

such that the actual own-radial reach of T_0 , the near endpoint of its supported interval on r_1 , and its far endpoint are respectively

$$C_0 = \frac{\Delta_\theta - A_0 - \theta B_0}{\theta + 1}, \quad \lambda_0 = \frac{\theta(1 - B_0)}{\theta + 1}, \quad u_{\text{adj}} = \frac{\Delta_\theta - A_0 - \theta B_0 - 1}{\theta}.$$

The strict midpoint and one-sided-support conditions give

$$A_0 + C_0 < 1, \quad A_0 < \lambda_0 \leq \frac{1}{2}, \quad B_0 < \frac{1}{2}, \quad u_{\text{adj}} < 1 - A_0.$$

For the first inequality, the closed midpoint-face comparison reduces to

$$(2\theta^2 + 4\theta + 1)^2 - 4(\theta + 1)^2(\theta^2 + \theta + 1) = 4\theta^3 + 4\theta^2 - 4\theta - 3 > 0$$

for $\theta \geq 1$; strictness comes from the original open midpoint condition. The other three follow directly from the endpoint formulas and the strict midpoint inequalities.

The center-free shared edge and open coverage imply

$$A_1 \geq 1 - B_0 > \frac{1}{2}, \quad C_1 \geq \lambda_0.$$

The selected supercritical component has $C_1 \leq 1/2$, so theorem D.24 gives

$$B_1 \leq 1 - C_1 \leq 1 - \lambda_0.$$

Since $A_0 < \lambda_0$,

$$A_0 + B_1 < 1.$$

Let the C-triangle reach on r_1 , measured from O , be

$$d_1^C := \max\{x \in [0, 1] : xV_1 \in T_C\}, \quad c_1^{\text{req}} := 1 - d_1^C.$$

Only U_C, U_1 , and the supported trace of U_0 contribute a positive interval on r_1 . Openness of the skeleton cover gives

$$c_1^{\text{req}} < \max\{C_1, u_{\text{adj}}\} \leq \max\{1 - B_1, 1 - A_0\}.$$

Put $L := 1 - B_1$. The strict inequalities just proved permit choices

$$A_0 < p_1 < p_2 < 1 - B_1$$

such that

$$p_1 < \frac{1}{2}, \quad 1 - p_1 > C_0, \quad \max\{p_2, 1 - p_2\} > c_1^{\text{req}}.$$

Indeed, the supremum of $p \mapsto \max\{p, 1 - p\}$ on (A_0, L) is $\max\{L, 1 - A_0\}$, and all preceding comparisons are strict. Choose

$$0 < \varepsilon < \min\{p_1 - A_0, p_2 - p_1, 1 - B_1 - p_2, 1 - p_1 - C_0, \max\{p_2, 1 - p_2\} - c_1^{\text{req}}\}.$$

Every term in this minimum is positive.

Use distinct affine vertex charts

$$\Xi_0(x, y) := V_0 + x(V_5 - V_0) + y(V_1 - V_0),$$

$$\Xi_1(x, y) := V_1 + x(V_0 - V_1) + y(V_2 - V_1).$$

Both carry the metric $x^2 + y^2 - xy$. For $0 \leq p \leq 1/2$ put

$$\Delta_p^- := \text{conv}\{(0, 1 - p), (1, 1 - p), (0, -p)\}, \quad \Omega_{p, \varepsilon}^- := \text{int}(\Delta_p^-) + (-\varepsilon, 0),$$

and, for $1/2 < p \leq 1$, put

$$\Delta_p^+ := \text{conv}\{(p, 0), (p, 1), (p - 1, 0)\}, \quad \Omega_{p, \varepsilon}^+ := \text{int}(\Delta_p^+) + (0, -\varepsilon).$$

Direct substitution in the three half-planes shows that the minus template contains $(0, 0)$ and has closed reaches

$$(p - \varepsilon, 1 - p, 1 - p),$$

while the plus template contains $(0, 0)$ and has closed reaches

$$(p, 1 - p - \varepsilon, p).$$

Neither template has positive trace on a neighboring support line. The epsilon bounds imply the needed conditions $\varepsilon < p_1$ and, in the plus branch, $\varepsilon < 1 - p_2$.

Define

$$U'_0 := \Xi_0(\Omega_{p_1, \varepsilon}^-),$$

and

$$U'_1 := \begin{cases} \Xi_1(\Omega_{p_2, \varepsilon}^-), & p_2 \leq 1/2, \\ \Xi_1(\Omega_{p_2, \varepsilon}^+), & p_2 > 1/2. \end{cases}$$

These are open unit equilateral triangles and contain V_0, V_1 . Their exact closed reach triples are

$$(A'_0, B'_0, C'_0) = (p_1 - \varepsilon, 1 - p_1, 1 - p_1),$$

and

$$(A'_1, B'_1, C'_1) = \begin{cases} (p_2 - \varepsilon, 1 - p_2, 1 - p_2), & p_2 \leq 1/2, \\ (p_2, 1 - p_2 - \varepsilon, p_2), & p_2 > 1/2. \end{cases}$$

Thus

$$A'_0 + B'_0 = A'_1 + B'_1 = 1 - \varepsilon < 1,$$

and both new roles are nonsupercritical Vd0.

Only the following five skeleton pieces can change:

$$e_{5,0}, \quad e_{0,1}, \quad e_{1,2}, \quad r_0, \quad r_1.$$

They are all preserved, with strict margins:

$$\begin{aligned} A'_0 &= p_1 - \varepsilon > A_0 && (e_{5,0}), \\ B'_0 + A'_1 &\geq (1 - p_1) + (p_2 - \varepsilon) = 1 + (p_2 - p_1 - \varepsilon) > 1 && (e_{0,1}), \\ B'_1 &\geq 1 - p_2 - \varepsilon > B_1 && (e_{1,2}), \\ C'_0 &= 1 - p_1 > C_0 && (r_0), \end{aligned}$$

and

$$C'_1 = \max\{p_2, 1 - p_2\} > c_1^{\text{req}} \quad (r_1).$$

The last inequality makes the new open trace overlap the open C trace. Every other skeleton component is unchanged. On setting $U'_i = U_i$ for $2 \leq i \leq 5$, the seven open roles still cover S and T_C remains CE2. The unchanged roles were nonsupercritical Vd0, so all six output V roles are nonsupercritical Vd0 and

$$(A'_i, B'_i, C'_i) = (A_i, B_i, C_i) \quad (2 \leq i \leq 5), \quad |\{i : A'_i + B'_i > 1\}| = 0.$$

Finally define from the output open traces

$$\begin{aligned} G'_i &:= e_{i,i+1} \setminus ((U'_i \cap e_{i,i+1}) \cup (U'_{i+1} \cap e_{i,i+1})), \\ N'_{\text{gap}} &:= |\{i : G'_i \neq \emptyset\}|. \end{aligned}$$

Since the CE2 role is unchanged and every output gap is center-covered,

$$N'_{\text{gap}} \in \{0, 1, 2\}.$$

If it is zero, row Z0 of theorem 3.6 contradicts the output cover.

Suppose $N'_{\text{gap}} = 1$, rotate its unique gap to

$$G'_0 = X_0([\ell', r']), \quad \ell' = B'_0, \quad r' = 1 - A'_1,$$

and put $p' = 1 - r' = A'_1$, $q' = \ell' = B'_0$. On each of the other five gap-free edges choose a strict handoff

$$x'_i \in (1 - A'_{i+1}, B'_i) \quad (1 \leq i \leq 5).$$

The nonsupercritical caps give

$$q' < x'_5 < x'_4 < x'_3 < x'_2 < x'_1 < r'.$$

Consequently every output role realizes the same selected pair:

$$p' \leq A'_i, \quad q' \leq B'_i \quad (0 \leq i \leq 5).$$

(At $i = 1$ and $i = 0$ one inequality is equality; for the other indices use $A'_i > 1 - x'_{i-1} > p'$ and $B'_i > x'_i > q'$.) Thus the exact hypotheses of theorem 5.10 hold, and that proposition contradicts the output cover.

Finally suppose $N'_{\text{gap}} = 2$. Because every output V-gap is covered by the unchanged CE2 role, these are its two active adjacent trace edges. In the signed normalization put

$$p' = W - \alpha, \quad q' = R - \delta.$$

Gap containment gives $A'_1 \geq q'$ and $B'_5 \geq p'$. Choose handoffs on the four center-free edges,

$$x'_i \in (1 - A'_{i+1}, B'_i) \quad (1 \leq i \leq 4).$$

The signed endpoints, gap containment, and the nonsupercritical caps give

$$p' < x'_4 < x'_3 < x'_2 < x'_1 < 1 - q'.$$

It follows that

$$A'_i \geq q', \quad B'_i \geq p' \quad (1 \leq i \leq 5),$$

with strict inequalities for the internal roles. Hence all five roles realize the reflected common pair (q', p') , exactly as required by theorem 5.11. That theorem contradicts the output cover. Thus in every case

$$S \not\subseteq U_C \cup \bigcup_{i=0}^5 U'_i.$$

This contradicts the preservation already proved. The input gap rank was never used as the output rank, and no equality between them has been asserted. $\square$

Body consequence. The half-square bound, both affine chart constructions, every strict preservation margin, and the three-way recomputed-gap routing supply theorem 5.22.

APPENDIX E. ZERO-GAP NINE-POINT OPTIMIZATION

This appendix supplies the exact coordinate evaluation of the public AB -union, the line-circle frontier points, their exclusions from the six V triangles, the Newton inner reduction, and the support-arc optimization used in the zero-gap body argument. Each solved module below states its public interface, feasible variables, objective, exact result, complete solution, and body consequence explicitly.

E.1. Strict AB -union and frontier module.

Exact interface. The body fixes

$$P_A = V_4 + a(V_3 - V_4), \quad P_B = V_4 + b(V_5 - V_4),$$

and the exact geometric family

$$\mathcal{R}_{AB}(a, b) = \bigcup \left\{ T \cap H : \begin{array}{l} T \text{ is a closed unit equilateral triangle,} \\ V_4, P_A, P_B \in T \end{array} \right\}.$$

Variables and feasible set. Let e_A, e_B be unit vectors with angle 120° , put

$$O = 0, \quad A = ae_A, \quad B = be_B, \quad C = \{ue_A + ve_B : u, v \geq 0\},$$

identifying O, A, B with the images of V_4, P_A, P_B , respectively. In this chart the public feasible union is

$$(63) \quad \mathcal{R}_{AB}(a, b) = H \cap C \cap \bigcup \left\{ T : \begin{array}{l} T \text{ is a closed unit equilateral triangle,} \\ O, A, B \in T \end{array} \right\}.$$

The finite description in theorem A.11 applies to the four-point set $\{O, A, B, x\}$ and makes membership in (63) an exact finite semialgebraic condition of degree at most four.

The strict supercritical branch admits a simpler closed form. In the next statement write $x = ue_A + ve_B$ and $\|x\|^2 = u^2 + v^2 - uv$.

Objective. Determine the feasible union $\mathcal{R}_{AB}(a, b)$ exactly and identify the cyclic order of its two circular and two linear non-axis frontier pieces.

Exact result.

Theorem E.1 (Strict supercritical AB -union). *Assume*

$$a + b > 1, \quad \rho = a^2 + ab + b^2 < 1,$$

put $h = \sqrt{3}/2$, $D = \sqrt{4\rho - 3}$, and define

$$\begin{aligned} \alpha &= \frac{h(a + 2b - aD)}{2\rho}, & \beta &= \frac{h(a - b + (a + b)D)}{2\rho}, \\ \gamma &= \frac{h(-a + b + (a + b)D)}{2\rho}, & \delta &= \frac{h(2a + b - bD)}{2\rho}. \end{aligned}$$

Then all four coefficients are positive and

$$(64) \quad \mathcal{R}_{AB}(a, b) = H \cap ((C \cap \mathcal{D}_A \cap \mathcal{H}_A) \cup (C \cap \mathcal{D}_B \cap \mathcal{H}_B)),$$

where

$$\begin{aligned} \mathcal{D}_A &= \{(u, v) : (u - a)^2 + v^2 - (u - a)v \leq 1\}, \\ \mathcal{D}_B &= \{(u, v) : u^2 + (v - b)^2 - u(v - b) \leq 1\}, \\ \mathcal{H}_A &= \{(u, v) : \alpha(u - a) + \beta v \leq 0\}, \\ \mathcal{H}_B &= \{(u, v) : \gamma u + \delta(v - b) \leq 0\}. \end{aligned}$$

Its non-axis frontier consists, in counterclockwise order, of the A -circle, the A -line, the B -line, and the B -circle.

Solution.

Proof. The identities

$$4\rho - 3(a + b)^2 = (a - b)^2, \quad (a + b)^2 D^2 - (a - b)^2 = 4\rho((a + b)^2 - 1)$$

give $0 < D < 1$ and positivity of the four coefficients.

We apply the caliper criterion of theorem A.11. For a point $x = (u, v)$ in the relative interior of C but outside $\triangle OAB$, the four hull vertices occur in the order O, B, x, A . The two cone-axis hull edges cannot certify a unit triangle: their support sums, divided by h , are respectively

$$\max\{a, u\} + \max\{b, v - u\}, \quad \max\{b, v\} + \max\{a, u - v\},$$

and are at least $a + b > 1$. It remains to test the Ax and Bx calipers.

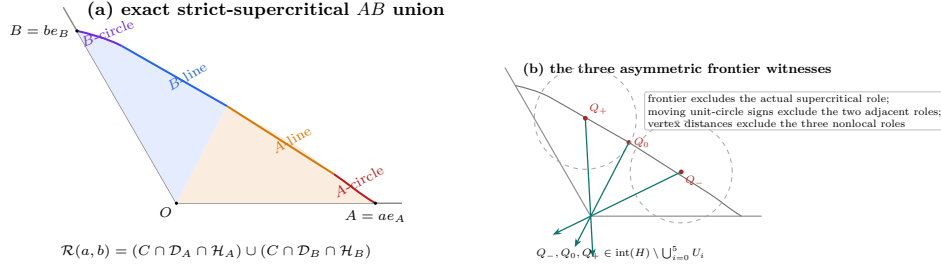


FIGURE 15. The strict-supercritical AB frontier and the three asymmetric witnesses. The source-conditioned union has, in counterclockwise order, an A -circle arc, an A -line segment, a B -line segment, and a B -circle arc. The points Q_-, Q_0, Q_+ are selected on this exact frontier and are then excluded from the six actual V roles by frontier, moving-circle, and vertex-distance arguments. The drawing is schematic; the labelled frontier order and witness roles are exact.

For the Ax caliper put $p = u - a$ and $r^2 = p^2 + v^2 - pv$. Exact evaluation of the three supports gives

$$(65) \quad \frac{G_A}{h} = \frac{\max\{r^2, -ap + (a+b)v\} + N_A}{r}, \quad N_A = \begin{cases} 0, & p \leq 0, \\ ap, & 0 \leq p \leq v, \\ (a+b)p - bv, & p \geq v. \end{cases}$$

No support label is omitted in these three sectors. If $p \leq 0$, $G_A \leq h$ is equivalent to

$$r \leq 1, \quad -ap + (a+b)v \leq r.$$

The second inequality is the half-plane condition in (64), because

$$r^2 - (-ap + (a+b)v)^2 = \frac{(cp + dv)(c'p + d'v)}{4\rho},$$

where

$$c = a + 2b - aD, \quad d = a - b + (a+b)D, \quad c' = a + 2b + aD, \quad d' = a - b - (a+b)D,$$

and $c, c', d > 0 > d'$. Thus the low sector is exactly $\mathcal{D}_A \cap \mathcal{H}_A$. In the middle sector $0 < p \leq v$ one has $r \leq v$, whereas the numerator in (65) is at least $(a+b)v > r$, so no fit is possible.

In the high sector $p \geq v$, the inequalities $G_A \leq h$ imply

$$(66) \quad r^2 + (a+b)p - bv \leq r, \quad bp + av \leq r.$$

Write $x - A = ry(\theta)$, $0 \leq \theta \leq 60^\circ$, with cone coordinates

$$y(\theta) = te_A + we_B, \quad t = \frac{2}{\sqrt{3}} \cos(\theta - 30^\circ), \quad w = \frac{2}{\sqrt{3}} \sin \theta.$$

Then (66) gives

$$r \leq f(\theta) = 1 - (a+b)t + bw, \quad S(\theta) = bt + aw \leq 1.$$

Let n_B be the unit vector with dual coordinates (γ, δ) . The identities

$$b\gamma + (a+b)\delta = h, \quad (a+b)\gamma + a\delta = \frac{h}{2}(1+D), \quad b\delta - a\gamma = \frac{h}{2}(1-D)$$

show that the unique angle θ_B of this normal satisfies $S(\theta_B) = 1$ and $f(\theta_B) = (1-D)/2$. The function S has at most one critical point, a maximum, so $S(\theta) \leq 1$ implies $\theta \leq \theta_B$. On the feasible part $f, f' \geq 0$; hence $f(\theta) \cos(\theta_B - \theta)$ is increasing. Consequently

$$\langle n_B, x - B \rangle \leq a\gamma - b\delta + \frac{h}{2}(1-D) = 0.$$

The diameter bound also gives $\|x - B\| \leq 1$. Thus every high sector fit belongs to $\mathcal{D}_B \cap \mathcal{H}_B$. Reflection exchanges a, u with b, v and supplies the corresponding result for the Bx caliper. This proves (64) in the cone interior; the axes follow by taking limits, since both sides are closed.

For completeness, the line-circle junctions are obtained by solving the two displayed line and circle equations. The line normals satisfy

$$\alpha^2 + \alpha\beta + \beta^2 = h^2, \quad \gamma^2 + \gamma\delta + \delta^2 = h^2,$$

and the line determinant is

$$\alpha\delta - \gamma\beta = \frac{3(1-D)(D+3)}{16\rho} > 0.$$

The successive cone slopes are

$$+\infty, \quad \frac{2}{1-D}, \quad \frac{a(1-D) + 2b}{2a + b(1-D)}, \quad \frac{1-D}{2}, \quad 0;$$

the two middle comparisons reduce to $a(4 - (1-D)^2) > 0$ and $b(4 - (1-D)^2) > 0$. This proves the asserted four-piece order. $\square$

Body consequence. This supplies the exact feasible-union and maximal-line-piece evaluation used to define ℓ_-, ℓ_+ and Q_0 in the body, and is cited there as theorem E.1.

E.2. Handoff and nine-point frontier-forcing module.

Exact interface. The body interface is the exact chain and parameter domain in equations (8) to (10), the forced disk in equation (11), the three frontier points in equation (12), and the witness set in equation (13).

Variables and feasible set. Assume $N_{\text{gap}} = 0$, $N_+ = 1$, and $(d, t) = (0, 0)$; rotate the unique supercritical role to index 4 and use the strict handoffs ξ_i selected in the body. The feasible parameter pair is the strict component $0 < a, b < 1$, $a + b > 1$, $a^2 + ab + b^2 < 1$ determined below.

Objective. Determine the common radial envelope and the two line-circle roots, then prove that the resulting six radial points and three asymmetric frontier points are excluded from every V triangle.

Exact result. The exact outputs are supplied by theorems E.2 to E.5: they give the strict endpoint inequalities, the forced centered disk, the unique frontier roots, and $K_{\text{wit}}(a, b) \subset U_C$.

Solution.

Handoff stage.

Lemma E.2 (Exact-one handoff chain). *Assume the six V triangles cover ∂H and exactly one actual V triangle is supercritical. Rotate its index to 4. The strict handoffs supplied by theorem 2.11 satisfy*

$$\xi_3 < \xi_2 < \xi_1 < \xi_0 < \xi_5 < \xi_4.$$

With

$$a = 1 - \xi_3, \quad b = \xi_4, \quad p = 1 - b, \quad q = 1 - a,$$

one has

$$(67) \quad \begin{aligned} 0 < a, b < 1, \quad a &= a_4 < A_4, \quad b = b_4 < B_4, \\ a + b &> 1, \quad a^2 + ab + b^2 < 1, \end{aligned}$$

and every selected V triangle obeys

$$a_i \geq p, \quad b_i \geq q.$$

Proof. Every selected V triangle other than 4 is strictly nonsupercritical, so $\xi_i < \xi_{i-1}$ for $i \neq 4$, which is precisely (E.2). In particular ξ_3 is the global minimum and ξ_4 the global maximum. Strictness of the selected handoffs gives $a = 1 - \xi_3 = a_4 < A_4$ and $b = \xi_4 = b_4 < B_4$. The two anchors of V triangle 4 lie in its open triangle; their mutual squared distance is $a^2 + ab + b^2$, strictly below the diameter squared of the closed unit triangle. This proves (67). Finally every $\xi_j \in [\xi_3, \xi_4]$, so $1 - \xi_{i-1} \geq 1 - \xi_4 = p$ and $\xi_i \geq \xi_3 = q$. $\square$

Radial stage. Put $h = \sqrt{3}/2$ and

$$s = p + q, \quad m = \min(p, q), \quad M = \max(p, q), \quad \chi = s^4 - s^2 + pq.$$

The strict domain (67) gives

$$(68) \quad pq > (1 - s)(2 - s), \quad m > 1 - s, \quad s > 1 - m.$$

Let $c_* = c_{\max}(p, q)$ be the exact radial envelope of theorem A.10. On the present subcritical domain, equation (38) becomes

$$(69) \quad c_* = \begin{cases} C_L(m), & \chi \leq 0, \\ \frac{2M}{1 + \sqrt{4s^2 - 3}}, & \chi > 0, \end{cases}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

At $\chi = 0$ the two expressions agree and equal s .

Lemma E.3 (Direct radial forcing). *One has*

$$(70) \quad 0 < c_* < 1, \quad c_* > 1 - m.$$

Assume that every U_i is Vd0 and that $U_C, U_0, \dots, U_5$ cover H . Then, for every i ,

$$P_i^{\text{rad}} = (1 - c_*)V_i \in U_C.$$

Consequently

$$(71) \quad \mathbb{B}(0, r_*) = \{x : \|x\| \leq r_*\} \subseteq \text{conv}\{P_0^{\text{rad}}, \dots, P_5^{\text{rad}}\} \subset U_C, \quad r_* = h(1 - c_*).$$

Proof. On the C_L branch, $C_L(m) < 1$ because the defining polynomial is positive at 1; if $s < \sqrt{3}/2$, then $c_* \geq \sqrt{3}/2 > s > 1 - m$, while if $s \geq \sqrt{3}/2$ its value at s is $\chi \leq 0$, so $c_* \geq s > 1 - m$. On the other branch put $E = \sqrt{4s^2 - 3}$. The selector gives $sE > M - m$, whence $c_* < s < 1$. Moreover, with

$$A_* = \frac{2M}{1 - m} - 1,$$

one has

$$A_*^2 - E^2 = \frac{4(1 - s)\{(1 - m)(1 - 2m) + m(2 - m)(1 - s)\}}{(1 - m)^2} > 0.$$

Thus $A_* > E$ and $c_* > 1 - m$. This proves (70).

Fix i . If $P_i^{\text{rad}} \in U_i$, openness supplies $\varepsilon > 0$ with $c_* + \varepsilon < 1$ such that $(1 - c_* - \varepsilon)V_i \in U_i$. The same triangle contains its two selected handoff anchors. Since their incoming and forward-reach lower bounds dominate p and q , convexity and passage to the closure give a closed unit triangle realizing $(p, q, c_* + \varepsilon)$. This contradicts the definition of $c_* = c_{\max}(p, q)$. Hence $P_i^{\text{rad}} \notin U_i$.

If P_i^{rad} belonged to U_{i-1} or U_{i+1} , a small plane ball inside that open triangle would meet r_i in a positive-length interval. This contradicts the Vd0 definition. For the remaining triangles put $r = 1 - c_* > 0$. Directly,

$$\|rV_i - V_{i\pm 2}\|^2 = 1 + r + r^2 > 1, \quad \|rV_i - V_{i+3}\| = 1 + r > 1.$$

Since $V_j \in U_j$ and two points of an open unit triangle are at distance strictly below 1, these inequalities exclude P_i^{rad} from the other three V triangles. Thus no V triangle contains P_i^{rad} . The point lies in $\text{int}(H)$ by (70), so the assumed cover forces $P_i^{\text{rad}} \in U_C$.

The six points P_i^{rad} form a regular hexagon of circumradius $1 - c_*$. Their convex hull contains its centered incircle of radius $h(1 - c_*)$, and convexity of U_C gives (71). $\square$

Asymmetric frontier stage. We next define the three asymmetric witnesses. At V_4 use local coordinates

$$\Phi(u, v) := V_4 + u(V_5 - V_4) + v(V_3 - V_4), \quad \mathbf{q}(u, v) = u^2 + v^2 - uv.$$

In this chart the two public adjacent-handoff circles have centers

$$\begin{aligned} X_2(b) &= \Phi(E_-), \quad E_- = (1 - b, 2 - b), \\ X_5(1 - a) &= \Phi(E_+), \quad E_+ = (2 - a, 1 - a), \end{aligned}$$

and hence

$$C_- = \partial\mathbb{B}(\Phi(E_-), 1), \quad C_+ = \partial\mathbb{B}(\Phi(E_+), 1).$$

The forward-reach lower bound is b and the backward-reach lower bound is a . Accordingly, set

$$\begin{aligned} \alpha &= \frac{h(b + 2a - bD)}{2\rho}, & \beta &= \frac{h(b - a + (a + b)D)}{2\rho}, \\ \gamma &= \frac{h(-b + a + (a + b)D)}{2\rho}, & \delta &= \frac{h(2b + a - aD)}{2\rho}, \end{aligned}$$

where now $\rho = a^2 + ab + b^2$ and $D = \sqrt{4\rho - 3}$. The two line pieces of theorem E.1 are

$$L_-(\lambda) = (b - \beta\lambda, \alpha\lambda), \quad L_+(\mu) = (\delta\mu, a - \gamma\mu).$$

They meet at $\lambda_* = \mu_* = 8h\rho/(3(D+3))$ in

$$(72) \quad J = \left(\frac{2b+a-aD}{D+3}, \frac{b+2a-bD}{D+3} \right).$$

Thus the two public line pieces have the coordinate evaluations

$$\ell_- = \{\Phi(L_-(\lambda)) : \lambda_* \leq \lambda \leq h^{-1}\}, \quad \ell_+ = \{\Phi(L_+(\mu)) : \mu_* \leq \mu \leq h^{-1}\}.$$

The actual adjacent handoffs have local centers

$$Z_5(t) = (1+t, t), \quad 1-a \leq t \leq b, \quad \text{and} \quad Z_3(y) = (y, 1+y), \quad 1-b \leq y \leq a.$$

The restrictions of the two unit-circle equations are

$$(73) \quad \begin{aligned} g_-(\lambda) &= \frac{3}{4}\lambda^2 - 3(\alpha + b\beta)\lambda + 3b^2 - 3b + 2, \\ g_+(\mu) &= \frac{3}{4}\mu^2 - 3(\delta + a\gamma)\mu + 3a^2 - 3a + 2. \end{aligned}$$

Let λ_- and μ_+ be their first roots; explicitly,

$$\begin{aligned} \lambda_- &= 2(\alpha + b\beta) - \frac{2}{3}\sqrt{9(\alpha + b\beta)^2 - 9b^2 + 9b - 6}, \\ \mu_+ &= 2(\delta + a\gamma) - \frac{2}{3}\sqrt{9(\delta + a\gamma)^2 - 9a^2 + 9a - 6}. \end{aligned}$$

Lemma E.4 (Moving adjacent-triangle signs). *Throughout (67),*

$$(74) \quad \lambda_* < \lambda_- < h^{-1}, \quad \mu_* < \mu_+ < h^{-1}.$$

Moreover $J_u, J_v < 1/2$. The point $L_+(\mu_+)$ satisfies

$$(L_+(\mu_+))_u + (L_+(\mu_+))_v < 3 - 2a,$$

and the reflected bound holds for $L_-(\lambda_-)$. Writing $F(P, t) = \mathbf{q}(P - Z_5(t)) - 1$, one has, for every $t \in [1-a, b]$,

$$(75) \quad F(L_+(\mu_+), t) \geq 0, \quad F(J, t) > 0, \quad F(L_-(\lambda_-), t) > 0.$$

Under the reflection $(u, v, a, b) \leftrightarrow (v, u, b, a)$, the analogous signs hold for every $y \in [1-b, a]$: the first-root point $L_-(\lambda_-)$ has nonnegative sign against $Z_3(y)$, and the other two witnesses have strictly positive sign.

Proof. Put

$$U = a + b - 1, \quad z = a - b, \quad D^2 = z^2 + 6U + 3U^2.$$

Then $0 < U < 1/6$ and $z^2 < \Omega := 1 - 6U - 3U^2$. We give the fixed-line calculation for the circle centered at $E_+ = (2-a, 1-a)$; reflection gives the calculation for E_- . Put

$$F(P, t) = \mathbf{q}(P - (1+t, t)) - 1, \quad t_0 = 1-a,$$

so this circle is $F(P, t_0) = 0$. At the junction, exact expansion gives

$$(D+3)^2 F(J, t_0) = 3D(z^2 + Uz + 1 - 3U) + P_J,$$

where

$$P_J = z^4 + 5z^2 + 6Uz^2 + 3U^2z^2 + 9Uz + 3U + 15U^2 > 0;$$

indeed $z^2 + Uz + 1 - 3U \geq 1 - 3U - U^2/4 > 0$, and $5z^2 + 9Uz + 15U^2$ is positive definite. Thus $F(J, t_0) > 0$. Also

$$J_u + J_v = \frac{(1+U)(3-D)}{3+D} < 1.$$

Indeed this inequality is $3U < D(2 + U)$, which follows from $D^2 \geq 3U(2 + U)$ and $(2 + U)^3 > 3U$. Since

$$\frac{\partial F}{\partial t}(P, t) = 1 + 2t - P_u - P_v,$$

$F(J, t)$ is strictly increasing for $t \geq 0$, and therefore is positive throughout $[1 - a, b]$.

On the opposite line L_- , the derivative at the junction, after multiplication by a positive factor, has at $t = 1 - a$ the numerator

$$N_0 = D(9 + 3z^2 + 6Uz - 9U^2) + P_0,$$

where

$$P_0 = 18U^3 + 6U^2z^2 + 63U^2 + 18Uz^2 + 18Uz + 54U \\ + 2z^4 + 9z^2 + 9.$$

The coefficient of D is at least $9 - 6U - 9U^2 > 0$, while $P_0 \geq 9 + 54U - 18U > 0$. Hence $N_0 > 0$. At the other endpoint $t = b$, the corresponding numerator is

$$N_1 = D(9 + 3z^2 + 6U - 3U^2) + P_1,$$

where

$$P_1 = 9U^4 - 3U^3z + 45U^3 + 9U^2z^2 - 6U^2z + 72U^2 \\ - Uz^3 + 21Uz^2 + 9Uz + 45U + 2z^4 + 9z^2 + 9.$$

Its D -coefficient is positive and, using $|z| < 1$,

$$P_1 \geq 9 + 45U - 9U - U - 6U^2 - 3U^3 > 0.$$

The junction derivative is affine in t , so it is positive throughout $[1 - a, b]$. Since the restriction of F to L_- is a convex quadratic, is positive at the junction, and is increasing there, it has no intersection on the opposite line side for any actual adjacent handoff.

On the correct line L_+ , the value at its outer endpoint h^{-1} has, after multiplication by a positive factor, numerator

$$A_* = P_2 - 4DU(1 + U + z),$$

where

$$P_2 = 3U^4 + 6U^3z + 6U^3 + 4U^2z^2 + 12U^2z + 12U^2 \\ + 2Uz^3 + 6Uz^2 + 2Uz + 6U + z^4 + 2z^2 - 3.$$

For fixed U , replace the odd powers of z by $x = |z|$. The resulting polynomial $P_2^+(U, x)$ satisfies

$$\partial_x P_2^+ = 2(3U^3 + 4U^2x + 6U^2 + 3Ux^2 + 6Ux + U + 2x^3 + 2x) > 0,$$

and therefore its supremum for $z^2 < \Omega$ occurs at $x = \sqrt{\Omega}$. There

$$P_2^+(U, \sqrt{\Omega}) = 4U(U + \sqrt{\Omega} - 3) < 0.$$

As $1 + U + z = 2a > 0$, this gives $A_* < 0$. The positive junction value and negative endpoint value force the first root strictly between them, proving the L_+ half of (74); reflection proves the other half.

For the coordinate bounds, the exact identities

$$D^2 - (3U - z)^2 = 6U(1 + z - U), \quad D^2 - (3U + z)^2 = 6U(1 - z - U)$$

give $J_u, J_v < 1/2$. It remains to control the coordinate sum along L_+ . For $E = L_+(h^{-1})$, the sign of $3 - 2a - E_u - E_v$ is the sign of

$$D(6U + 2z + 6) + B(U, z),$$

where

$$B(U, z) = -9U^3 - 9U^2z - 9U^2 - 3Uz^2 - 18Uz + 3U \\ - 3z^3 + 3z^2 - 3z + 3.$$

Here

$$B_z = -3(3z^2 + 2Uz + 6U - 2z + 3U^2 + 1) < 0,$$

because the quadratic in parentheses has discriminant $-32U^2 - 80U - 8 < 0$. Hence its minimum on $z^2 < \Omega$ is at $z = \sqrt{\Omega}$, where $B(U, \sqrt{\Omega}) = 6(1 - 3U - \sqrt{\Omega}) > 0$ because $(1 - 3U)^2 - \Omega = 12U^2$. Thus $E_u + E_v < 3 - 2a$. Also $J_u + J_v < 1 < 3 - 2a$, and the coordinate sum is affine on the line segment, so the same strict bound holds at the first root. Now

$$\frac{\partial F}{\partial t}(P, t) = 1 + 2t - P_u - P_v.$$

Since $F(L_+(\mu_+), 1 - a) = 0$, the coordinate-sum bound makes its t -derivative strictly positive for $t \geq 1 - a$. Together with the no-opposite-line result, this proves (75). Reflection gives both the bound on L_- and the complete $Z_3(y)$ statement. $\square$

Define the local and global witnesses by

$$(76) \quad Q_-^{\text{loc}} = L_-(\lambda_-), \quad Q_0^{\text{loc}} = J, \quad Q_+^{\text{loc}} = L_+(\mu_+),$$

and

$$Q_- := \Phi(Q_-^{\text{loc}}), \quad Q_0 := \Phi(Q_0^{\text{loc}}), \quad Q_+ := \Phi(Q_+^{\text{loc}}).$$

Thus J is the unique meeting point of the two maximal line pieces, and the first-root inequalities in theorem E.4 give exactly

$$\{Q_0\} = \ell_- \cap \ell_+, \quad \{Q_-\} = \ell_- \cap \mathcal{C}_-, \quad \{Q_+\} = \ell_+ \cap \mathcal{C}_+.$$

Lemma E.5 (Direct asymmetric forcing). *Assume that $U_C, U_0, \dots, U_5$ cover H . Then*

$$Q_-, Q_0, Q_+ \in \text{int}(H) \setminus \bigcup_{i=0}^5 U_i,$$

and consequently all three witnesses belong to U_C .

Proof. The root order and positivity of the line coordinates give $0 < u, v < 1$ for all three points. Each belongs to the local representation of $\mathcal{R}_{AB}(a, b)$ obtained from theorem E.1 by interchanging the two cone coordinates, and hence lies in a closed unit triangle containing V_4 , so $\mathbf{q}(u, v) \leq 1$; hence $|u - v| < 1$ and all three points lie in $\text{int}(H)$.

The closure of U_4 is one of the triangles represented by $\mathcal{R}_{AB}(a, b)$, since it contains V_4, X_3, X_4 . All three witnesses lie on its exact non-axis frontier. If one belonged to the open triangle U_4 , a sufficiently small plane ball about it would lie both in U_4 and in the interior of the corner cone, making the witness an interior point of $\mathcal{R}_{AB}(a, b)$. This contradiction excludes U_4 .

For the triangles U_0, U_1, U_2 , their contained vertices have local coordinates

i	0	1	2
V_i	(2, 1)	(2, 2)	(1, 2).

For $P = (u, v)$,

$$\|P - (2, 1)\|^2 - 1 = \mathbf{q}(u, v) + 2 - 3u, \quad \|P - (1, 2)\|^2 - 1 = \mathbf{q}(u, v) + 2 - 3v.$$

The inequalities $J_u, J_v < 1/2$ and the line order give distance greater than one from V_0 for Q_-, Q_0 and from V_2 for Q_0, Q_+ . For the remaining pair, the first-root circle and the sum bound in theorem E.4 give

$$\|Q_+ - V_0\|^2 - 1 = a(3 - a - (Q_+)_u - (Q_+)_v) > a^2,$$

and, by reflection, $\|Q_- - V_2\|^2 - 1 > b^2$. All three points are farther than one from $V_1 = (2, 2)$, because writing $u = 1 - \xi$, $v = 1 - \eta$ gives

$$\|P - V_1\|^2 = 1 + \xi + \eta + \xi^2 + \eta^2 - \xi\eta > 1.$$

Since two points in one open unit equilateral triangle have distance strictly less than one, these inequalities exclude U_0, U_1, U_2 .

The actual handoff $X_5 = Z_5(\xi_5)$ belongs to U_5 , and $1 - a < \xi_5 < b$ by (E.2). The three signs in (75) therefore put every witness at distance at least one from X_5 , excluding U_5 . Similarly, if $y = 1 - \xi_2$, then $1 - b < y < a$, $X_2 = Z_3(y) \in U_3$, and the reflected signs exclude U_3 . Thus no V triangle contains any witness. The assumed cover now forces all three interior witnesses into U_C . $\square$

The nine directly forced points are $P_0^{\text{rad}}, \dots, P_5^{\text{rad}}, Q_-, Q_0, Q_+$. Under a putative cover, theorems E.3 and E.5 therefore force the compact set

$$(77) \quad K_{\text{wit}}(a, b) = \mathbb{B}(0, r_*) \cup \{Q_-, Q_0, Q_+\} \subset U_C.$$

Body consequence. This is the full calculation supplying the body containments equations (11) to (13).

E.3. Newton reduction and four-overlap module.

Exact interface. Retain the public $K_{\text{wit}}(a, b)$ from equation (13) and the exact support cap

$$\text{Cap}(X, r) = \{n \in \mathbb{S}^1 : \langle X, n \rangle + r \geq h\}.$$

Variables and feasible set. Work on the strict AB domain in (67) and its nontrivial sheet $c_* > 2/3$. The Newton parameters are taken on the open segments from the line junction Q_0 to the first circle roots Q_- and Q_+ .

Objective. Replace the two independent first-root radicals by rational inner points and verify the ray order, radius bounds, and four consecutive support-cap intersections needed for the cyclic enclosure argument.

Exact result. The Newton containment is theorem E.6 and equation (78), and the complete four-overlap conclusion is theorem E.7.

Solution. For the nontrivial regime $c_* > 2/3$, we first remove the square radicals in the first-root witnesses. Normalize $\bar{\alpha} = \alpha/h$, and similarly for the other three coefficients, and put

$$\xi = \lambda/h, \quad \zeta = \mu/h, \quad \xi_* = \zeta_* = \frac{8\rho}{3(D+3)}.$$

Let g_-, g_+ be the rescaled forms of (73), and take one Newton step at the junction:

$$\hat{\xi}_- = \xi_* - \frac{g_-(\xi_*)}{g'_-(\xi_*)}, \quad \hat{\zeta}_+ = \zeta_* - \frac{g_+(\zeta_*)}{g'_+(\zeta_*)}.$$

Define, in local coordinates,

$$\begin{aligned} P_- &= \left(b - \frac{3}{4}\widehat{\beta\xi}_-, \frac{3}{4}\widehat{\alpha\xi}_- \right), \\ P_0 &= J, \\ P_+ &= \left(\frac{3}{4}\widehat{\delta\zeta}_+, a - \frac{3}{4}\widehat{\gamma\zeta}_+ \right), \end{aligned}$$

and convert them to global vectors.

Lemma E.6 (Newton inner reduction). *The points above have coordinates rational in a, b, D and satisfy*

$$P_- \in (Q_0, Q_-), \quad P_0 = Q_0, \quad P_+ \in (Q_0, Q_+).$$

Consequently

$$\widehat{K} = \text{conv}(\mathbb{B}(0, r_*) \cup \{P_-, P_0, P_+\}),$$

and one has

$$(78) \quad \widehat{K} \subseteq \text{conv}(K_{\text{wit}}), \quad \Lambda(K_{\text{wit}}) \geq \Lambda(\widehat{K}).$$

Proof. Each quadratic g_-, g_+ is strictly convex, positive, and decreasing at the junction. Its tangent zero therefore lies strictly between the junction and its first root. This gives the two open-segment inclusions. All displayed coefficients and Newton quotients are rational in a, b, D , and convexity gives the asserted containment. $\square$

Put

$$P_* := P_+ + RP_-.$$

For a disk $\mathbb{B}(X, r)$ let

$$\text{Cap}(X, r) := \{n \in \mathbb{S}^1 : \langle X, n \rangle + r \geq h\}.$$

Ray order and overlap verification.

Proposition E.7 (Technical four-overlap verification). *Assume (67) and $c_* > 2/3$. Then*

$$\|P_-\|, \|P_+\| > r_*,$$

the rays P_-, P_0, P_+, P_, RP_- occur in that cyclic order, and the four consecutive cap pairs*

$$\text{Cap}(P_-, 2r_*) \leftrightarrow \text{Cap}(P_0, 2r_*) \leftrightarrow \text{Cap}(P_+, 2r_*) \leftrightarrow \text{Cap}(P_*, r_*) \leftrightarrow \text{Cap}(RP_-, 2r_*)$$

overlap across their intervening short sectors.

Proof. Put $\tau = h - 2r_* = h(2c_* - 1)$. The rays of

$$(79) \quad P_-, P_0, P_+, P_*, RP_-$$

occur in this cyclic order from P_- to RP_- . Indeed the two line distances give

$$\frac{\text{cross}(P_-, P_0)}{\|P_0 - P_-\|} = \alpha(1 - b) + \beta > 0, \quad \frac{\text{cross}(P_0, P_+)}{\|P_+ - P_0\|} = \gamma + \delta(1 - a) > 0.$$

If $P_- = (-X, -Y)$ and $P_+ = (-P, -Q)$ in the centered cone basis, then $0 < X, Y, P, Q < 1$, $X, Q > 1/2$, and

$$\frac{\text{cross}(P_+, RP_-)}{h} = PX + (Q - P)Y > 0.$$

The last sign is immediate unless $Q < P$ and $X < Y$, in which case it is larger than $Q - P/2 > 0$. The ray of $P_* = P_+ + RP_-$ lies between those of P_+ and RP_- . The same coordinates give

$$(80) \quad \|P_-\|, \|P_+\| > h/2 > r_*.$$

For the two equal-radius overlaps it is enough to prove

$$(81) \quad \frac{\text{cross}(P_-, P_0)}{\|P_0 - P_-\|} \geq \tau, \quad \frac{\text{cross}(P_0, P_+)}{\|P_+ - P_0\|} \geq \tau.$$

We now give the exact analytic verification. By reflection take $a \leq b$ and set $m = 1 - b$, $M = 1 - a$, $U = a + b - 1$, $s = m + M$, and $w = M - m$. The smaller of the two normalized left sides in (81) is

$$(82) \quad d = \frac{\mathcal{A} + U(2m - 1) + D(\mathcal{A} + U)}{2\rho}, \quad \mathcal{A} = 1 - m + m^2.$$

On the C_L sheet, write $F_L(z) = z^4 - z^2 + mz - m^2$. This defining polynomial evaluated at $r = 1 - m/2 + m^2/2$ is

$$F_L(r) = \frac{m^2(m-1)}{16}(m^5 - 3m^4 + 11m^3 - 17m^2 + 28m - 12) > 0.$$

Indeed the polynomial in parentheses is increasing on $(0, 1/2)$ and has value $-33/32$ at $1/2$; its derivative is $28 - 34m + m^2(5m^2 - 12m + 33) > 0$. Since $F_L(h) = -(m - h/2)^2 \leq 0$ and the selected root is unique, $2c_* - 1 \leq \mathcal{A}$. Put

$$X_0 = \mathcal{A} + U, \quad Y_0 = 2\rho\mathcal{A} - \mathcal{A} - U(2m - 1).$$

Then

$$d - \mathcal{A} = \frac{D(\mathcal{A} + U) - Y_0}{2\rho},$$

and exact expansion gives

$$\begin{aligned} Y_0 &= 2(m^2 - m + 1)U^2 + (2m^3 - 2m + 3)U \\ &\quad + (m^2 - m + 1)(2m^2 - 2m + 1) > 0. \end{aligned}$$

Furthermore

$$D^2X_0^2 - Y_0^2 = 4m\rho\Phi(U, m),$$

where

$$\begin{aligned} \Phi(U, m) &= (1 - m)(m^2 - m + 2)U^2 \\ &\quad + (2 - m^2)(m^2 - m + 1)U \\ &\quad - m(1 - m)^2(m^2 - m + 1). \end{aligned}$$

In these variables

$$\chi = U^4 - 4U^3 + 5U^2 - (m + 2)U + m(1 - m).$$

Since $U^2(U^2 - 4U + 5) > 0$, the selector $\chi \leq 0$ implies $U > m(1 - m)/(m + 2)$; Φ is increasing in U and at that lower endpoint is

$$\frac{m^2(1 - m)(m^4 - 2m^3 + 6m^2 - 6m + 4)}{(m + 2)^2} > 0.$$

The last quartic equals $(m^2 - m)^2 + 5m^2 - 6m + 4 > 0$. Thus $d > \mathcal{A} \geq 2c_* - 1$.

On the other radial sheet let $E = \sqrt{4s^2 - 3}$. Then $0 \leq w < sE$ and $c_* = (s + w)/(1 + E)$. At fixed U , differentiation of (82) can be checked without an implicit estimate. Namely,

$$d = \frac{1 + D}{2} - UB(D, w), \quad B(D, w) = \frac{(1 + U)(3 + D) + w(1 - D)}{D^2 + 3}.$$

Here $D^2 = w^2 + 6U + 3U^2$, so $0 \leq D' = w/D \leq 1$, and

$$B_w = \frac{1 - D}{D^2 + 3} \geq 0.$$

Writing $N = (1 + U)(3 + D) + w(1 - D)$, direct differentiation gives

$$B_D = \frac{(1 + U - w)(D^2 + 3) - 2DN}{(D^2 + 3)^2} > -\frac{34}{27}.$$

Indeed the first term in the numerator is positive, while $1 + U - w = 2a > 0$, $N < (7/6)4 + 1 = 17/3$, $D < 1$, and $D^2 + 3 \geq 3$. If $B_D < 0$, then $B_D D' \geq B_D$; otherwise $B_D D' \geq 0$. Hence $B' = B_w + B_D D' > -34/27$, and $U < 1/6$ yields

$$d' < \frac{1}{2} + \frac{1}{6} \frac{34}{27} = \frac{115}{162} < 1,$$

whereas $(2c_* - 1)' = 2/(1 + E) \geq 1$. Hence their difference decreases with w . At the boundary $w = sE$, where $\chi = 0$ and $c_* = s$, the preceding sheet applies: this is an admissible point because $h < s < 1$ and $D^2 = 4s^4 - 12s + 9 < 1$. Indeed $s^4 - 3s + 2 = (s - 1)(s^3 + s^2 + s - 2) < 0$ on this interval; the second factor is increasing and already equals $(7h - 5)/4 > 0$ at $s = h$. The preceding sheet gives $d \geq 2s - 1$. Therefore (81) holds on both sheets, and reflection supplies the other line.

It remains to check the two mixed overlaps. The exact rational envelopes, Gram reductions, and global positive-basis identities are recorded once in appendix F. theorem F.3 gives precisely the overlaps for $\mathbb{B}(P_+, 2r_*)$ with $\mathbb{B}(P_*, r_*)$ and for $\mathbb{B}(P_*, r_*)$ with $\mathbb{B}(RP_-, 2r_*)$.

Thus the four consecutive pairs of caps centered on the rays (79) overlap. $\square$

Body consequence. These exact Newton and overlap data supply the nontrivial branch of the support-arc solution below, and hence the body conclusion theorem 5.25.

E.4. Support-arc enclosure module.

Exact interface. The body supplies the compact witness

$$K_{\text{wit}}(a, b) = \mathbb{B}(0, r_*) \cup \{Q_-, Q_0, Q_+\}$$

in equation (13), together with the exact level- h cap

$$\text{Cap}(X, r) = \{n \in \mathbb{S}^1 : \langle X, n \rangle + r \geq h\}.$$

Variables and feasible set. The parameters satisfy $0 < a, b < 1$, $a + b > 1$, and $a^2 + ab + b^2 < 1$. The disk branch is split exactly at $c_* = 2/3$; on the strict branch $c_* > 2/3$, use the Newton points and cap overlaps from the preceding module. All support directions range over the closed unit circle $\mathbb{S}^1$.

Objective. Determine the enclosure gauge of $K_{\text{wit}}(a, b)$ by proving the sharp sufficient bound $\Lambda(K_{\text{wit}}(a, b)) \geq 1$ on the entire feasible domain.

Exact result. The cyclic cap implication is theorem E.8; together with the disk branch and the four-overlap module it yields theorem E.9.

Solution. Let R denote counterclockwise rotation through $2\pi/3$. For $X \in \mathbb{R}^2$ and $r \geq 0$, recall

$$\mathbb{B}(X, r) = \{Y : \|Y - X\| \leq r\}, \quad \text{Cap}(X, r) = \{n \in \mathbb{S}^1 : \langle X, n \rangle + r \geq h\}.$$

Lemma E.8 (Cyclic support-arc covering). *Let K be compact and put*

$$\Sigma_K = K + RK + R^2K.$$

Suppose Σ_K contains the full R -orbits of

$$\mathbb{B}(P_-, 2r_*), \quad \mathbb{B}(P_0, 2r_*), \quad \mathbb{B}(P_+, 2r_*), \quad \mathbb{B}(P_*, r_*), \quad P_* := P_+ + RP_-.$$

Let θ_X denote the argument of the ray through X . Assume the rays

$$P_-, P_0, P_+, P_*, RP_-$$

occur in this cyclic order, every corresponding support arc is connected with half-width less than $\pi/2$, and, for each consecutive pair $(X, r), (Y, s)$ in

$$(P_-, 2r_*), (P_0, 2r_*), (P_+, 2r_*), (P_*, r_*), (RP_-, 2r_*),$$

one has

$$\text{Cap}(X, r) \cap \text{Cap}(Y, s) \cap \{n : \arg n \in [\theta_X, \theta_Y]\} \neq \emptyset,$$

where $[\theta_X, \theta_Y]$ is the intervening short angular interval. Then

$$\Lambda(K) \geq 1.$$

Proof. The four consecutive intersections and the cyclic order make the five support arcs cover the sector from the ray of P_- to the ray of RP_- . Their R -orbits cover $\mathbb{S}^1$. Hence, for every unit vector n , one of the displayed disks has support at least h in direction n , and therefore

$$h_{\Sigma_K}(n) \geq h.$$

Since

$$h_{\Sigma_K}(n) = \sum_{j=0}^2 h_K(R^j n),$$

theorem A.9 gives $\Lambda(K) \geq 1$. $\square$

Theorem E.9 (Zero-gap nine-point optimization). *For every pair (a, b) in the strict domain*

$$0 < a, b < 1, \quad a + b > 1, \quad a^2 + ab + b^2 < 1,$$

the exact public witness satisfies

$$\Lambda(K_{\text{wit}}(a, b)) \geq 1.$$

Proof. The centered disk alone gives

$$\Lambda(K_{\text{wit}}) \geq \frac{3r_*}{h} = 3(1 - c_*),$$

so the assertion follows when $c_* \leq 2/3$. Assume $c_* > 2/3$. By theorem E.6 and equation (78), $\hat{K} \subseteq \text{conv}(K_{\text{wit}})$ and $\Lambda(K_{\text{wit}}) \geq \Lambda(\hat{K})$. The Minkowski sum

$$\hat{K} + R\hat{K} + R^2\hat{K}$$

contains the disk orbits listed in theorem E.8. The ray order, radius bounds, and all four intervening overlaps are theorem E.7; the two mixed overlaps there are certified exactly by theorem F.3. Thus theorem E.8 gives $\Lambda(\hat{K}) \geq 1$, and the Newton containment finishes the proof. $\square$

Body consequence. This solved optimization supplies theorem 5.25. Combined with $K_{\text{wit}}(a, b) \subset U_C$, it gives the compact body obstruction theorem 5.26 and its routing consequence theorem 5.27.

APPENDIX F. EXACT POLYNOMIAL POSITIVITY CERTIFICATE

The two mixed cap overlaps in the finite-enclosure method are certified by exact symbolic arithmetic. This section gives the mathematical reduction and verification algorithms; the repository provenance manifests identify the accompanying code and sparse integer data. No floating-point arithmetic, interval arithmetic, adaptive subdivision, branch-and-bound, or unrecorded numerical tolerance is used.

F.1. Certificate manifest. The certificate has a machine-readable provenance manifest. The file `certificate_provenance.json`, in the numbered 3105X computation directory, identifies the six sparse-data shards and both exact verifiers by full Git blob and records their roles. They decode to one canonical sparse-polynomial transcript with SHA-256 digest

`dc46aaf263655d5159ecd3a81db72ee82477951d06172f4743b248df37209485`

The derivation verifier reconstructs the transcript from the witness formulas and compares every coefficient with the authenticated object. The positivity verifier derives the Bernstein coefficients from that object and checks their signs exactly.

F.2. Rational radial upper envelopes. Retain the notation of appendix E. Thus

$$p = 1 - b, \quad q = 1 - a, \quad s = p + q, \quad m = \min\{p, q\}, \quad M = \max\{p, q\},$$

$$\chi = s^4 - s^2 + mM, \quad h = \frac{\sqrt{3}}{2}, \quad c_* = c_{\max}(p, q).$$

For the certificate formulas only, abbreviate

$$A := P_-, \quad C := P_+, \quad \eta := r_*, \quad I(X, r) := \text{Cap}(X, r).$$

The strict witness domain gives

$$(83) \quad mM > (1 - s)(2 - s), \quad m > 1 - s, \quad s > \frac{2}{3}.$$

Put

$$F_m(x) = x^4 - x^2 + mx - m^2, \quad \kappa = F'_m(s) = 4s^3 - 2s + m.$$

Then

$$\kappa > 4s^3 - 3s + 1 = (s + 1)(2s - 1)^2 > 0.$$

Define

$$(84) \quad \bar{c}_L = s - \frac{\chi}{\kappa} \quad (\chi \leq 0), \quad \bar{c}_T = s - \frac{\chi}{1 - m} \quad (\chi \geq 0).$$

Lemma F.1 (Branchwise radial upper bounds). *On the corresponding radial cells,*

$$c_* \leq \bar{c}_L < 1, \quad c_* \leq \bar{c}_T < 1.$$

At $\chi = 0$ both upper bounds equal $s = c_$.*

Proof. On the L cell, $F_m(s) = \chi \leq 0$ and $F_m(c_*) = 0$. For $x \geq s > 2/3$, $F_m''(x) > 0$, so convexity gives

$$0 \geq F_m(s) + F_m'(s)(c_* - s),$$

and hence $c_* \leq \bar{c}_L$. If $U = 1 - s$, then

$$\chi + U\kappa > U\{m + 1 - 4U + 8U^2 - 3U^3\} > 0$$

by (83); therefore $\bar{c}_L < 1$.

On the T cell put

$$E = \sqrt{4s^2 - 3}, \quad m_- = \frac{s(1 - E)}{2}, \quad m_+ = \frac{s(1 + E)}{2}.$$

Then

$$\chi = (m - m_-)(m_+ - m), \quad s - c_* = \frac{2(m - m_-)}{1 + E}.$$

Consequently

$$\frac{\chi}{s - c_*} = \frac{1 + E}{2}(m_+ - m) \leq 1 - m,$$

which gives $c_* \leq \bar{c}_T$. The strict inequality $\chi > 0$ gives $\bar{c}_T < s < 1$. $\square$

Reflect so that $z = a - b \geq 0$ and put $U = a + b - 1$. Then

$$s = 1 - U, \quad m = \frac{1 - U - z}{2}, \quad D^2 = z^2 + 6U + 3U^2 =: K(U, z),$$

and

$$\begin{aligned} \Xi(U, z) &= 4\chi = 1 - 10U + 21U^2 - 16U^3 + 4U^4 - z^2, \\ G(U, z) &= 2\kappa = 5 - 21U + 24U^2 - 8U^3 - z. \end{aligned}$$

Thus

$$(85) \quad \bar{c}_L = 1 - U - \frac{\Xi(U, z)}{2G(U, z)}, \quad \bar{c}_T = 1 - U - \frac{\Xi(U, z)}{2(1 + U + z)}.$$

F.3. Gram reduction and the geometric implication. On the active cell let $\bar{c}$ denote the corresponding expression in (85), and put

$$\bar{\eta} = h(1 - \bar{c}), \quad e = 1 - \bar{c}, \quad t = 2\bar{c} - 1.$$

theorem F.1 gives $0 < \bar{\eta} \leq \eta = h(1 - c_*)$. It therefore suffices to prove the mixed intersections for the smaller disks with radii $2\bar{\eta}, \bar{\eta}, 2\bar{\eta}$.

Represent a global vector as (x, hy) and put $C' = R^{-1}C$. Define

$$N_A = \|A\|^2, \quad N_C = \|C'\|^2, \quad \nu = \langle A, C' \rangle,$$

$$\Delta_0 = x_A y_{C'} - y_A x_{C'} > 0.$$

Thus the Euclidean cross product is $h\Delta_0$, and the Gram identity is

$$(86) \quad h^2 \Delta_0^2 = N_A N_C - \nu^2.$$

Define

$$(87) \quad Q_A = \Delta_0^2 - \|tA - eC'\|^2, \quad Q_C = \Delta_0^2 - \|tC' - eA\|^2.$$

Lemma F.2 (Residual signs imply the mixed cap intersections). *If $Q_A, Q_C \geq 0$, then*

$$I(C, 2\bar{\eta}) \cap I(C + RA, \bar{\eta}) \neq \emptyset,$$

and

$$I(C + RA, \bar{\eta}) \cap I(RA, 2\bar{\eta}) \neq \emptyset.$$

Consequently the same intersections hold with η in place of $\bar{\eta}$.

Proof. Rotate the first pair by R^{-1} . Its centers become C' and $A + C'$, so their difference is A . An outer common tangent normal n_A is chosen with

$$\langle A, n_A \rangle = \bar{\eta}.$$

Since $\|A\| > \bar{\eta}$ and $\Delta_0 > 0$, choose the tangent side for which

$$\langle C', n_A \rangle = \frac{\bar{\eta}\nu + \sqrt{N_A - \bar{\eta}^2} h \Delta_0}{N_A}.$$

The common support of the two disks is therefore

$$2\bar{\eta} + \frac{\bar{\eta}\nu + \sqrt{N_A - \bar{\eta}^2} h \Delta_0}{N_A}.$$

Let $\bar{\tau} = h - 2\bar{\eta} = ht$. Expanding (87) and using (86) gives

$$(88) \quad \bar{P}_A := (N_A - \bar{\eta}^2)h^2\Delta_0^2 - (\bar{\tau}N_A - \bar{\eta}\nu)^2 = h^2N_AQ_A.$$

Thus $Q_A \geq 0$ implies

$$\sqrt{N_A - \bar{\eta}^2} h \Delta_0 \geq \bar{\tau}N_A - \bar{\eta}\nu,$$

and the common support is at least $2\bar{\eta} + \bar{\tau} = h$. Hence the two level- h caps intersect. The same calculation with A and C' exchanged gives

$$(89) \quad \bar{P}_C := (N_C - \bar{\eta}^2)h^2\Delta_0^2 - (\bar{\tau}N_C - \bar{\eta}\nu)^2 = h^2N_CQ_C,$$

and proves the second intersection. Enlarging the disk radii from $\bar{\eta}$ to η only enlarges the caps. $\square$

F.4. Eight exact integer-polynomial signs. The Newton witnesses A, C are rational functions of U, z, D . Substituting (85) into (87), clearing denominators, and reducing by

$$D^2 - K(U, z) = 0$$

gives, for $B \in \{L, T\}$ and $X \in \{A, C\}$,

$$(90) \quad \text{num}(Q_{B,X}) = 32(K+3)^2\{A_{B,X}(U, z) + DB_{B,X}(U, z)\},$$

where all eight core polynomials lie in $\mathbb{Z}[U, z]$. Every omitted denominator is strictly positive on the strict domain: its factors are positive constants, $(D+3)^4$, $G(U, z)^2$ or $(1+U+z)^2$, and squares of the two Newton-derivative numerators. Here $G = 2\kappa > 0$, and the fixed-line sign theorem proves both Newton derivatives strictly negative.

The derivation verifier constructs A, C , both rational envelopes, and all four residuals directly in the exact field $\mathbb{Q}(U, z, D)$. It performs the reduction modulo $D^2 - K$, removes the common positive factor, primitive-normalizes the eight integer polynomials, and compares their sparse term lists coefficient by coefficient with the authenticated transcript. Thus the data are checked against the geometric formulas rather than trusted as an independent precomputation.

F.5. **Three global positive-basis certificates.** Set

$$\mathcal{G}(U) = 1 - 6U - 3U^2, \quad \mathcal{P}(U) = 1 - 10U + 21U^2 - 16U^3 + 4U^4,$$

$$U_{\Delta} = 1 - \frac{\sqrt{3}}{2}, \quad U_{\max} = \frac{2}{\sqrt{3}} - 1.$$

The complete reflected domain is the union of the three closed cells

$$\begin{aligned} T : & \quad 0 \leq U \leq U_{\Delta}, \quad 0 \leq z^2 \leq \mathcal{P}(U), \\ L_0 : & \quad 0 \leq U \leq U_{\Delta}, \quad \mathcal{P}(U) \leq z^2 \leq \mathcal{G}(U), \\ L_1 : & \quad U_{\Delta} \leq U \leq U_{\max}, \quad 0 \leq z^2 \leq \mathcal{G}(U). \end{aligned}$$

For the T cell use

$$U = \frac{(x-1)(3-x)}{4x}, \quad z = y \frac{9-x^4}{8x^2}, \quad 1 \leq x \leq \sqrt{3}, \quad 0 \leq y \leq 1,$$

and then $x = 1 + (\sqrt{3} - 1)r$. This maps $[0, 1]^2$ onto the complete cell. After multiplication by positive powers of 2 and x , the four transformed core polynomials have global tensor Bernstein expansions with data

	bidegree	coefficient count	zero count
$A_{T,A}$	$(88, 22)$	2047	2
$B_{T,A}$	$(80, 20)$	1701	0
$A_{T,C}$	$(88, 22)$	2047	2
$B_{T,C}$	$(80, 20)$	1701	0.

Every nonzero coefficient is strictly positive.

For an L -cell polynomial write

$$F(U, z) = E_F(U, w) + zO_F(U, w), \quad w = z^2,$$

and

$$R_F(U, w) = E_F(U, w)^2 - wO_F(U, w)^2.$$

The implications $E_F \geq 0$ and $R_F \geq 0$ give $E_F \geq \sqrt{w}|O_F|$, hence $F \geq 0$ for $z \geq 0$. Use the exact charts

$$U = U_{\Delta}r, \quad w = \mathcal{P}(U) + s\{\mathcal{G}(U) - \mathcal{P}(U)\}$$

for L_0 , and

$$U = U_{\Delta} + (U_{\max} - U_{\Delta})r, \quad w = s\mathcal{G}(U)$$

for L_1 . The positivity verifier forms one global Bernstein expansion for each of E_F, R_F on each square. The authenticated transcript records the bidegrees and coefficient counts of all sixteen expansions; every coefficient is strictly positive.

The verifier implements polynomial addition, multiplication, substitution, and power-to-Bernstein conversion over exact integers. Coefficients in $\mathbb{Q}(\sqrt{3})$ are represented as rational pairs and signed by exact rational comparisons. If $P(r, s) = \sum a_{ij}r^i s^j$ has bidegree (m, n) , its exact tensor Bernstein coefficients are

$$b_{k\ell} = \sum_{i \leq k} \sum_{j \leq \ell} a_{ij} \frac{\binom{k}{i}}{\binom{m}{i}} \frac{\binom{\ell}{j}}{\binom{n}{j}}.$$

This is precisely the conversion implemented by the verifier.

Theorem F.3 (Exact mixed-overlap certificate). *On the complete strict witness domain with $c_* > 2/3$, all eight polynomials in (90) are nonnegative. Hence $Q_A, Q_C \geq 0$, and the two mixed cap intersections required by theorem E.7 hold.*

Proof. The T chart proves the four T -cell core signs by nonnegative Bernstein coefficients. On each L chart the positive Bernstein certificates for E_F and R_F prove the four L -cell signs. Since $D \geq 0$, (90) gives $Q_A, Q_C \geq 0$ for $z \geq 0$; reflection supplies $z \leq 0$. theorem F.2 then gives the two cap intersections. $\square$

Remark F.4 (Reproduction). The authenticated data and the two verifiers constitute the exact electronic certificate used by the proof. Reproduce it from its package directory by executing

```
verify_mixed_overlap_core_derivation.py
verify_global_core_positivity.py
```

Both commands use exact arithmetic and must reproduce the transcript digest above and the asserted sign checks.